

AN L^3 BOUND FOR THE TRILINEAR HILBERT TRANSFORM

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ABSTRACT. We prove that the trilinear Hilbert transform with fixed slopes 1, 2, 3 maps $L^3(\mathbb{R}) \times L^3(\mathbb{R}) \times L^3(\mathbb{R})$ to $L^1(\mathbb{R})$, resolving this case of the trilinear Hilbert transform conjecture.

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1. INTRODUCTION

For Schwartz functions $f_1, f_2, f_3 : \mathbb{R} \rightarrow \mathbb{C}$, set

$$T(f_1, f_2, f_3)(x) = \text{p. v.} \int_{\mathbb{R}} f_1(x-t) f_2(x-2t) f_3(x-3t) \frac{dt}{t}.$$

The principal value removes a symmetric interval about the origin. Our main result is the following estimate.

Theorem 1.1. *There is a finite constant C such that*

$$\|T(f_1, f_2, f_3)\|_{L^1(\mathbb{R})} \leq C \prod_{j=1}^3 \|f_j\|_{L^3(\mathbb{R})} \quad (f_1, f_2, f_3 \in \mathcal{S}(\mathbb{R})).$$

Consequently T has a unique bounded trilinear extension from $L^3(\mathbb{R}) \times L^3(\mathbb{R}) \times L^3(\mathbb{R})$ to $L^1(\mathbb{R})$.

Theorem 1.1 resolves affirmatively the $L^3 \times L^3 \times L^3 \rightarrow L^1$ case of the standard trilinear Hilbert transform conjecture for the fixed slopes 1, 2, 3. The full conjecture asks for a broader range of exponents; see [4, Conjecture 1.7]. The result here concerns the displayed fixed operator.

The bilinear predecessor was treated by Lacey and Thiele [6, 7]. The difficulty in the trilinear setting is the interaction of cancellation across scales with quadratic modulation. For example, if

$c = (-1, 3, -3, 1)$, then

$$\sum_{j=0}^3 c_j(x + jt)^a = 0 \quad (a = 0, 1, 2).$$

Thus matched quadratic phases can persist in the associated four-linear form. Quantitative information at one scale must be organized so that this persistence does not produce a loss proportional to the number of scales. Tao [11] obtained cancellation for truncated multilinear Hilbert transforms, improving the elementary logarithmic truncation bound to a sublogarithmic one. That argument established a connection with higher-order uniformity. The straight-line trilinear boundedness problem remained open in the account of Hu and Lie [5, Section 1.2]. Their results for curved transforms [4, 5] address a different source of oscillation.

The inverse theorem of Leng, Sah and Sawhney [8, Theorem 1.2] provides the quantitative structural input used below. We use its degree-two case and prove the continuous chart and localization statements needed for the singular integral. The inverse theorem alone does not sum the scales. The proof of that summation has three main features.

First, private orthogonal coordinates make the coefficients of each selected column recoverable. Lemma 3.4 then controls products of inherited projections uniformly in spatial depth. Its loss is a power of the logarithm of a common budget for column sizes, inverse private-coordinate sizes, and the number of available labels along a spatial path; the total number of labels on disjoint branches is unrestricted.

Second, we localize curvature jointly in all four arguments and retain the separation needed when projection adjoints are evaluated on earlier cells. Transferring complementary filters leaves coupled narrow bands. The comparison in Lemma 7.2 gives a power of block length whose exponent tends to zero as the auxiliary exponent decreases to two. This control is essential to the final absorption.

Third, each matched column receives one fixed mode expansion. Frequency graphs separate errors with square-summable energy from coherent components with hereditary representatives. Lemma 9.1 bounds the total mass of progression tests on scales of small density, with a positive-power saving in the density. Its application uses all four anchor cancellations and polynomial dimension costs. The compression and counting lemmas isolate tools for other progression forms with the same projection and cancellation structures.

Organization and notation. Section 2 reduces the operator estimate to a uniform maximal-function bound for local forms. Sections 3 and 4 develop the projection and quadratic tools. Section 5 produces a pair-small base form with a bounded number of active scales on each path. Section 6 localizes this form by curvature. Section 7 gives the side estimates under a square-energy hypothesis, which is proved in Section 8. Section 9 estimates the remaining core terms. Section 10 chooses the parameters and completes the absorption.

We write $e(t) = \exp(2\pi it)$ and use progression coordinates $y_j = x + jt$, $0 \leq j \leq 3$. Inner products are linear in the first argument. Replacing the shifts $x - jt$ by $x + jt$ in the principal-value integral changes the operator's sign, by the substitution $t \mapsto -t$. Norms and inner products carrying an interval subscript use normalized Lebesgue measure on that interval. Other function norms use ordinary Lebesgue measure. In vector-valued arguments the scalar component entering the form is called the physical component. All depth trees used in the estimates are finite; constants will be uniform in their depth. The polynomial and quasipolynomial budget conventions are specified in Section 3.

2. LOCAL FORMS AND THE OUTER REDUCTION

Fix a dyadic lattice on $\mathbb{R}$. The depth increases by one when the interval length is halved. For an interval I of length r , define

$$H_I(z_0, z_1, z_2, z_3) = r^{-2} \int_{\mathbb{R}^2} w_I(x, t) \prod_{j=0}^3 z_j(x + jt) dx dt.$$

We fix constants $c_0 > 0$ and $C_0 \geq 1$ and call a kernel admissible when it has the following properties:

- (1) Its support consists of points for which all four $x + jt$ lie in I at distance at least $c_0 r$ from its endpoints.
- (2) For every $u \in \mathbb{R}$ and every $j \in \{0, 1, 2, 3\}$,

$$\int_{\mathbb{R}} w_I(u - jt, t) dt = 0.$$

- (3) Every derivative of total order n is bounded in absolute value by $r^{-n}(C_0(n+2))^{C_0(n+2)}$.

Kernels at different intervals may be chosen independently and may be multiplied by unimodular scalars. Constants in the proof may depend on these fixed admissibility parameters. For vector-valued functions the form uses only their physical scalar components.

Lemma 2.1 (Local absolute estimate). *If $p_j \geq 1$ and $\sum_{j=0}^3 p_j^{-1} \leq 2$, then*

$$r^{-2} \int |w_I(x, t)| \prod_{j=0}^3 \|z_j(x + jt)\| dx dt \lesssim \prod_{j=0}^3 \|z_j\|_{p_j, I}.$$

Proof. For distinct indices i, j , the map $(x, t) \mapsto (y_i, y_j)$ has nonzero constant determinant. The estimate follows immediately when $p_i = p_j = 1$ and the other two exponents are infinity. The estimate with all exponents infinity follows from the support and kernel bounds. The endpoints with just one exponent equal to one follow from the two-exponent bounds and normalized norm monotonicity. Interpolation, or log-convexity applied to this positive form, fills the polytope $0 \leq p_j^{-1} \leq 1$, $\sum p_j^{-1} \leq 2$. $\square$

Let M be the uncentered Hardy–Littlewood maximal operator and put $M_q f = (M(|f|^q))^{1/q}$. For $2 < q < 3$, let $C_N = C_N(q)$ be the least constant such that

$$\sum_{I \in \mathcal{I}} |I| |H_I(f_0, f_1, f_2, f_3)| \leq C_N \int_{\mathbb{R}} \prod_{j=0}^3 M_q f_j$$

for every finite collection $\mathcal{I}$ in at most N consecutive depths, every lattice, every admissible kernel family, and compactly supported L^q inputs. Lemma 2.1, with all exponents two, gives $C_N \lesssim N$. Indeed the local L^2 norms are at most the local L^q norms, whose product is at most $\prod_j M_q f_j$ at every point of I ; sum over disjoint intervals at each depth.

Theorem 2.2 (Uniform local estimate). *For some $q \in (2, 3)$, the constants $C_N(q)$ are bounded independently of N .*

It is useful to isolate the normalized assertion that will be proved in Section 10. A rooted subtree contains all ancestors between each of its cells and its root; the summation may use any subset of those cells.

Proposition 2.3 (Normalized absorption estimate). *For a suitable fixed $q \in (2, 3)$ and every sufficiently small fixed $c > 0$, there is an $A < \infty$, independent of N , with the following property. Suppose a finite rooted subtree has root I_* , all summation cells lie in at most N consecutive depths, and*

$$\|f_j\|_{q, I} \leq 1 \quad (0 \leq j \leq 3, I \text{ in the subtree}).$$

Then

$$\sum_{I \text{ a summation cell}} |I| |H_I(f)| \leq (A + cC_N) |I_*|.$$

The rest of the proof establishes Proposition 2.3. At this stage C_N is finite, with the preliminary bound $C_N \lesssim N$. The constant A may depend on the fixed choices of q , c , and the accuracy parameters, but is independent of N . The stopping argument below gives $C_N \leq CA + CcC_N$; choosing c sufficiently small then gives the uniform bound. Thus the long construction needs a small coefficient of the finite constant C_N , which will be absorbed only after Proposition 2.3 is proved.

Lemma 2.4 (Outer reduction). *Proposition 2.3 implies Theorem 2.2, and Theorem 2.2 implies Theorem 1.1.*

Proof. We first remove the normalization. Start with disjoint roots at the largest interval size occurring in the finite collection. At a root J , put $a_{j,J} = \|f_j\|_{q,J}$. A zero root size gives zero contribution below that root. Otherwise stop at the first strict descendants J' for which one average of $|f_j|^q$ exceeds $16a_{j,J}^q$. For each fixed j the union of its first stops has length at most $|J|/16$, so the union of all stops has length at most $|J|/4$. On the intervening tree, division of f_j by $16^{1/q}a_{j,J}$ makes all required local sizes at most one. Proposition 2.3 therefore bounds this tree by

$$16^{4/q}(A + cC_N)|J| \prod_j a_{j,J}.$$

Repeat at the stopped roots. The sets $E_J = J \setminus \bigcup J'$ are disjoint over this recursive family, have measure at least $3|J|/4$, and satisfy $M_q f_j(x) \geq a_{j,J}$ for $x \in E_J$. Hence

$$\sum_J |J| \prod_j a_{j,J} \leq \frac{4}{3} \int \prod_j M_q f_j.$$

Taking the supremum gives $C_N \leq CA + CcC_N$, where C is bounded uniformly for $2 < q < 3$. Choose c so that $Cc < 1/2$ and absorb. This proves Theorem 2.2.

To recover the continuous kernel, take an odd compactly supported Gevrey function ϕ with $\phi^{(3)}(0) \neq 0$, and a compactly supported Gevrey function ω such that $\sum_{m \in \mathbb{Z}} \omega(X - m) = 1$. Set

$$W(X, S) = \prod_{j=0}^3 (\partial_S - j\partial_X)[\omega(X)\phi(S)].$$

Each factor is differentiation in the direction of integration on the corresponding line $X = u - jS$. Compact support therefore makes all four one-point marginals of W vanish. Spatial summation and a change of variables give

$$\sum_{m \in \mathbb{Z}} W(x/a - m, t/a) = \phi^{(4)}(t/a), \quad \int_0^\infty \phi^{(4)}(t/a) \frac{da}{a^2} = -\frac{\phi^{(3)}(0)}{t} \quad (t \neq 0).$$

For the second equality one first takes $t > 0$ and substitutes $s = t/a$; for $t < 0$ use oddness of $\phi^{(4)}$.

Write $a = b2^n$, $1 \leq b < 2$. For each fixed b , a kernel $a^{-1}W(x/a - m, t/a)$ can be put into the form $r^{-1}w_I(x, t)$ with $r = Da$ for a fixed sufficiently large dyadic D . To choose I , use a fixed finite family of shifted lattices. At scale r their boundaries may be placed at offsets $0, r/3, -r/3$, with alternating signs between depths to preserve nesting. The four progression coordinates on the support occupy an interval of length $O(a)$; for large D , one of these cells contains that interval with the fixed interior margin. Each chosen cell receives only a bounded number of translates. The rescaled derivative and support bounds are fixed, so choose c_0, C_0 once to include them. A bounded number of contributions at one cell can be estimated separately.

Theorem 2.2, summed over these lattices, gives a uniform bound for the absolute sum over any finite range of n and the relevant spatial translates. Integrate in b with measure db/b . For compactly supported bounded f_0 and Schwartz f_1, f_2, f_3 , the resulting bound is

$$C \int \prod_{j=0}^3 M_q f_j \leq C_q \|f_0\|_\infty \prod_{j=1}^3 \|f_j\|_3.$$

Here M_q is bounded on L^3 because $q < 3$; use Hölder with the three exponents equal to three. Compact truncations of the other inputs justify use of the defining class for C_N : at a finite range of scales and with compact support in slot zero, every relevant progression coordinate lies in a fixed compact set.

After spatial summation the kernel is odd. Pair positive and negative t near zero; the difference of the two products of Schwartz functions is $O(|t|)$ uniformly on the compact support of f_0 . At large $|t|$ the Schwartz bounds are integrable. The integrated scale kernels have absolute value at most $C/|t|$. Dominated convergence therefore permits the scale range to tend to all positive a and recovers a nonzero constant times the principal-value four-linear form. Reflection of t changes only its sign and gives the operator in Theorem 1.1. Duality against compactly supported bounded f_0 proves the inequality in Theorem 1.1. Density of Schwartz functions, trilinearity, and completeness of L^1 give the stated extension. $\square$

3. BUDGETS, PRIVATE COORDINATES, AND COMPRESSION

We first make precise the quantitative conventions and the projection estimates used below. All trees in this section are finite. Bounds are independent of their number of generations.

3.1. Quantitative conventions and stopping attempts. A rational number has height at most B if it can be written as p/q with $p \in \mathbb{Z}$, $q \in \mathbb{N}$, and $\max(|p|, q) \leq B$. Heights of rational vectors and matrices are entrywise heights. For parameters $H_1, \dots, H_s \geq e$ and $0 < \delta_1, \dots, \delta_t \leq 1$, a quasipolynomial bound means a bound of the form

$$\exp \left(C_q \left(1 + \sum_{a=1}^s \log H_a + \sum_{b=1}^t \log(1/\delta_b) \right)^C \right).$$

A polylogarithmic bound is the same expression without the outer exponential. Here and below the degrees denoted by C can be fixed independently of q , of spatial lengths and frequencies, and of the number of scales. Constants such as C_q may depend on the fixed choice of q and on the fixed kernel class. A bounded number of compositions of quasipolynomial constructions is again quasipolynomial. When the number of constructions is itself a parameter, we keep track of the resulting growth explicitly. Expressions such as $D \leq \exp(k^C)$ allow an increase of the fixed exponent and a sufficiently large lower threshold for k .

We will use elementary spatial packing in the following form. If J is a dyadic interval, $a_I \geq 0$, and

$$\sum_{I \ni y} a_I \leq A \quad \text{for almost every } y \in J,$$

where all the intervals in the sum are contained in J , then Tonelli's theorem gives

$$\sum_I |I| a_I = \int_J \sum_{I \ni y} a_I dy \leq A |J|.$$

This is the passage from a bound along a spatial path to a length-weighted tree bound.

Lemma 3.1 (Packing of attempts). *Suppose a construction on a root J sends exceptional cells and all their descendants to new estimates on disjoint subroots. Suppose that, at every root R , the new roots R' satisfy*

$$\sum_{R' \text{ child root of } R} |R'| \leq \rho |R|, \quad 0 \leq \rho < 1.$$

Then the sum of the lengths of all roots, including J , is at most $|J|/(1 - \rho)$. In particular, if the contribution assigned to an attempt from R is at most $A|R|$, with its hypotheses valid at every new root, the total contribution is at most $A|J|/(1 - \rho)$.

Proof. The sum of the root lengths in generation n is at most $\rho^n |J|$, by induction. Summing this geometric series proves both assertions. $\square$

We refer to one application of the construction at a root as an *attempt*. Several stopping requirements can be imposed together: if their exceptional root families cost fractions $\rho_1, \dots, \rho_s$, take the maximal intervals in their union. These are disjoint, their total length is at most $(\sum_a \rho_a) |R|$, and Lemma 3.1 applies when this sum is smaller than one. A local normalization required by an estimate must hold at each new root before that estimate is restarted there.

A stopping boundary for the summation does not have to be a boundary for an auxiliary projection family. An inherited column list can be continued on all descendants of a terminal decision cell: retain its columns and refit their constant coefficients on each finer cell. This gives genuine nested projections on the full refining partitions needed for an operator estimate. If a later attempt uses a new family, the extension of the old family is still used for the summands assigned to the old attempt. In particular, a projection on a partition refining I is an $L^2(I)$ contraction whether or not the averages of the original input are normalized on the finer cells.

3.2. Columns and inherited projections. Fix a finite dyadic tree with root J and a finite label set $\mathcal{V}$. The cardinality of $\mathcal{V}$ will not be bounded. To each label v assign a birth cell $J_v \subset J$, a scalar function φ_v on J_v , and a number $\epsilon_v > 0$. Work in the value Hilbert space

$$\mathcal{H} = \mathbb{C} \oplus \ell^2(\mathcal{V}),$$

and define the column on its birth cell by

$$\Phi_v(y) = (\varphi_v(y), \sqrt{\epsilon_v} \mathbf{e}_v), \quad y \in J_v.$$

The first coordinate is called the physical coordinate. Every label has its own private direction, even when two scalar functions φ_v coincide. We write $\tilde{\Phi}_v = \mathbf{1}_{J_v} \Phi_v$ for the extension by zero to J .

Throughout the projection estimates, $D \geq 2$ is a common budget such that

$$\|\Phi_v\|_{L^\infty(J_v; \mathcal{H})} \leq D, \quad \sqrt{\epsilon_v} \geq D^{-1}, \quad \sum_{v \in \mathcal{V}} \mathbf{1}_{J_v}(y) \leq D \quad \text{for almost every } y \in J.$$

The last condition says that at most D labels are born along any spatial path. It counts all available labels, including labels not selected in the projection currently under consideration. It allows arbitrarily many labels on disjoint branches.

A mask $\mathcal{V}^i(I)$ on a cell I is a subset of $\{v : I \subset J_v\}$. An inherited mask family satisfies

$$\mathcal{V}^i(I) \subset \mathcal{V}^i(I') \quad \text{whenever } I' \text{ is a child of } I.$$

Let P_I^i be the orthogonal projection in $L^2(I; \mathcal{H})$ onto

$$\left\{ \sum_{v \in \mathcal{V}^i(I)} c_v \Phi_v|_I : c_v \in \mathbb{C} \right\}.$$

The coefficients in this definition are constant on I . We also regard P_I^i as an operator on $L^2(J; \mathcal{H})$ by restricting its input to I and extending its output by zero. For a stopping partition π of a cell, P_π^i denotes the direct sum of these local projections. If π' refines π , inheritance implies

$$\text{range}(P_\pi^i) \subset \text{range}(P_{\pi'}^i).$$

Indeed, the restriction of an old constant-coefficient combination to a smaller cell is still an allowed combination there. Thus these direct-sum projections are nested, even though their coefficients are refitted on every cell. Their successive differences are orthogonal projections with mutually orthogonal ranges.

The masks and birth lists may have been chosen using an input or previous stopping decisions. In each application of an operator estimate we hold the entire resulting family fixed.

Lemma 3.2 (Coefficient and projection bounds). *If $\mathcal{A}$ is any set of columns available on a cell I , then*

$$D^{-1} \left(\sum_{v \in \mathcal{A}} |c_v|^2 \right)^{1/2} \leq \left\| \sum_{v \in \mathcal{A}} c_v \Phi_v(y) \right\|_{\mathcal{H}} \leq D^{3/2} \left(\sum_{v \in \mathcal{A}} |c_v|^2 \right)^{1/2} \quad \text{for almost every } y \in I.$$

In particular, the coefficients are unique. The orthogonal projection $P_{I, \mathcal{A}}$ onto their span satisfies

$$\|P_{I, \mathcal{A}} z\|_{L^\infty(I; \mathcal{H})} \leq D^5 \int_I \|z(y)\|_{\mathcal{H}} dy.$$

For an actual scalar input $F = (f, 0)$, write $P_{I, \mathcal{A}} F = \sum_{v \in \mathcal{A}} c_v \Phi_v$. Then

$$\int_I (f - (P_{I, \mathcal{A}} F)_{\text{phys}}) \overline{\varphi_v} = \epsilon_v c_v \quad (v \in \mathcal{A}).$$

Proof. The private component of the sum has squared norm $\sum_v \epsilon_v |c_v|^2$, giving the lower bound. The upper bound follows from Cauchy–Schwarz, since there are at most D columns and each has norm at most D .

Let $E_y : \ell^2(\mathcal{A}) \rightarrow \mathcal{H}$ be the evaluation map $E_y c = \sum_v c_v \Phi_v(y)$, and let

$$G = \int_I E_y^* E_y dy$$

be the normalized Gram matrix. The bounds just proved give $G \geq D^{-2} \text{Id}$ and $\|E_y\| \leq D^{3/2}$. Therefore the projection has the integral representation

$$P_{I, \mathcal{A}} z(y) = \int_I E_y G^{-1} E_t^* z(t) dt, \quad \|E_y G^{-1} E_t^*\| \leq D^5.$$

This proves the supremum estimate. Finally, orthogonality to Φ_v says that the physical pairing in the last assertion minus $\epsilon_v c_v$ is zero. $\square$

3.3. A binary-prefix estimate. The following elementary Hilbert space fact supplies the dependence on the number of stopping generations. In the compression argument that number will be a power of D , so its logarithm is the appropriate cost. Its proof uses the classical Rademacher–Menshov binary-decomposition mechanism; see [9, Section 3, Lemma 1] for a vector-valued presentation. The nested projection products and Bessel rows are treated below.

Lemma 3.3 (Products of nested prefixes). *Let $\mathcal{X}$ be a Hilbert space, and for $1 \leq i \leq m$ let $P_0^i, \dots, P_T^i$ be nested orthogonal projections on $\mathcal{X}$. Set $Z_n = P_n^m \cdots P_n^1$ and*

$$L_T = 1 + \lceil \log_2(T+1) \rceil.$$

Suppose $B_\nu : \mathcal{X} \rightarrow \mathcal{Y}_\nu$ are finitely many bounded linear operators satisfying $\sum_\nu \|B_\nu h\|^2 \leq \|h\|^2$. For any choices $n(\nu) \in \{0, \dots, T\}$,

$$\left(\sum_\nu \|B_\nu Z_{n(\nu)} g\|^2 \right)^{1/2} \leq L_T^m \|g\|.$$

Moreover,

$$\sum_{n=1}^T \|(Z_n - Z_{n-1})g\|^2 \leq m^2 L_T^{2(m-1)} \|g\|^2 \leq m^2 L_T^{2m} \|g\|^2.$$

For $m = 0$ the product is the identity and the first estimate has constant one.

Proof. Put $\Delta_0^i = P_0^i$ and $\Delta_n^i = P_n^i - P_{n-1}^i$ for $n \geq 1$. Within each family these are pairwise orthogonal projections. A prefix of the index set $\{0, \dots, T\}$ can be partitioned into dyadic index blocks, with at most one block of each length. For a block A put $\Delta_A^i = \sum_{n \in A} \Delta_n^i$. At any fixed block length, the operators Δ_A^i are pairwise orthogonal projections.

Expand each factor of $Z_{n(\nu)}$ using this binary decomposition. Fix a tuple of block lengths, one for each of the m factors. A term is zero if one of those lengths does not occur in the relevant prefix. Otherwise it has the form

$$B_\nu \Delta_{A_m(\nu)}^m \cdots \Delta_{A_1(\nu)}^1 g.$$

Group rows with the same tuple of blocks. The row contraction bounds their sum of squared norms by $\|\Delta_{A_m}^m \cdots \Delta_{A_1}^1 g\|^2$. Summing over the block tuples that occur is bounded by the sum over all tuples of those fixed lengths. Successive Bessel inequalities give

$$\sum_{A_1, \dots, A_m} \|\Delta_{A_m}^m \cdots \Delta_{A_1}^1 g\|^2 \leq \|g\|^2.$$

This argument uses orthogonality within each layer, not commutation between layers. There are at most L_T^m tuples of block lengths. The triangle inequality in the Hilbert direct sum of the row spaces proves the first assertion.

For the second, telescope in the order

$$Z_n - Z_{n-1} = \sum_{i=1}^m P_n^m \cdots P_n^{i+1} \Delta_n^i P_{n-1}^{i-1} \cdots P_{n-1}^1.$$

Discard the left contractions and use Cauchy–Schwarz in the sum over i . For a fixed i , the rows $B_n = \Delta_n^i$, $1 \leq n \leq T$, satisfy the row contraction hypothesis. Apply the first assertion to the $i - 1$ preceding families and the prefix index $n - 1$. The resulting bound is

$$\sum_{n=1}^T \|(Z_n - Z_{n-1})g\|^2 \leq m \sum_{i=1}^m L_T^{2(i-1)} \|g\|^2 \leq m^2 L_T^{2(m-1)} \|g\|^2,$$

as claimed. $\square$

3.4. Compression of local projection products.

Lemma 3.4 (Compression). *Use the column system and budget $D \geq 2$ above on a finite dyadic tree with root J . Let $P^1, \dots, P^m$ be inherited mask families, with their local orthogonal projections. Suppose finitely many bounded linear rows*

$$B_\nu : L^2(J; \mathcal{H}) \longrightarrow \mathcal{Y}_\nu$$

satisfy

$$\sum_\nu \|B_\nu z\|_{\mathcal{Y}_\nu}^2 \leq \|z\|_{L^2(J; \mathcal{H})}^2, \quad B_\nu = B_\nu \mathbf{1}_{u(\nu)},$$

where each $u(\nu)$ is a cell of the tree. Then an absolute constant C satisfies

$$\left(\sum_{\nu} \|B_{\nu} P_{u(\nu)}^m \cdots P_{u(\nu)}^1 z\|^2 \right)^{1/2} \leq (C(m+1)^C \log(2D))^{C(m+1)} \|z\|_{L^2(J; \mathcal{H})}.$$

The estimate is uniform in the row cells, the total number of labels, and the number of spatial depths. Identity factors may be omitted. Products containing complements $1 - P^i$ satisfy the same form of bound after changing C . Integrating such compositions against a fixed finite measure costs, in addition, at most the total variation of that measure.

Proof. For a nonempty product, we first divide the row cells into size segments and, on each segment, replace the input by a bounded function with the same first-layer moments. We then stop when the last-layer coefficient vector moves: nested-prefix estimates control the anchor products, and the pathwise birth-label budget controls the coefficient errors. Attempts are summed using both length and energy contraction. Finally, the size segments are summed using the bounds for their weighted averages and representative energies.

The assertion for an empty product is the row contraction hypothesis. Assume $m \geq 1$, and abbreviate the ordinary $L^2(J; \mathcal{H})$ norm to $\|\cdot\|_2$. Add one final dyadic generation below all row cells, retaining columns and refitting there if necessary. This changes neither the hypothesis nor the quantity to be estimated.

Size segments and a bounded moment representative. For every cell K put

$$a_K = \int_K \|z(y)\|_{\mathcal{H}} dy.$$

Starting from J , stop at the first proper descendant cells for which the average is greater than $2a_J$, and repeat inside each stopping cell. It is enough to run this stopping rule through the last possible row depth. A root with zero average contributes no rows and can be omitted. Denote the resulting stopping roots by $\mathcal{S}$.

Assign a row cell u to the deepest root $K \in \mathcal{S}$ containing it. More generally, let $\mathcal{T}_K$ be the full subtree of cells in K that are not contained in a proper stopping child of K , through the last row depth. Thus every assigned row cell lies in $\mathcal{T}_K$. Cut K at its next stopping roots and at final leaves on all other paths; denote this partition by π_K . Every assigned row cell is partitioned by cells of π_K . On a cut cell Q the average of $\|z\|$ is at most $4a_K$: its dyadic parent has average at most $2a_K$, and a child average is at most twice its parent's average.

For $Q \in \pi_K$, take all first-layer labels used by cells of $\mathcal{T}_K$ containing Q , and project $z|_Q$ onto their restrictions to Q . Let g_K be the direct sum of these projections, extended by zero off K . At most D labels occur on each such path. By Lemma 3.2, with

$$G = 4D^5,$$

we have

$$\|g_K\|_{L^\infty(K; \mathcal{H})} \leq Ga_K, \quad \|g_K\|_2^2 = \langle z, g_K \rangle.$$

The second identity follows from orthogonality on each cut cell. If $u \in \mathcal{T}_K$ and $v \in \mathcal{V}^1(u)$, then v was included on every cut cell inside u . Adding its moment identities over that partition of u gives

$$P_u^1 g_K = P_u^1 z, \quad P_u^m \cdots P_u^1 g_K = P_u^m \cdots P_u^1 z.$$

Let $M_d \|z\|$ be the dyadic maximal function within J . Stopping averages more than double on every path. Consequently

$$\sum_{K \in \mathcal{S}} a_K^2 |K| \leq \frac{4}{3} \|M_d \|z\|\|_2^2 \leq \frac{16}{3} \|z\|_2^2.$$

We also need the stronger bound for the moment representatives

$$S := \sum_{K \in \mathcal{S}} \|g_K\|_2^2 \leq C \log(2D) \|z\|_2^2.$$

Here is a proof that avoids any factor which is a power of D . At a fixed point y , split the roots containing y according as

$$a_K < \frac{M_d \|z\|(y)}{GD^2} \quad \text{or} \quad a_K \geq \frac{M_d \|z\|(y)}{GD^2}.$$

The sum of a_K in the first class is at most twice its displayed upper threshold. Thus their total pointwise contribution to $\sum_K \|g_K(y)\|$ is at most $2D^{-2}M_d \|z\|(y)$. The second class has at most

$$L_D = 2 + \lceil \log_2(GD^2) \rceil \leq C \log(2D)$$

members, since $a_K \leq M_d \|z\|(y)$ and the averages double. Using the moment identity, pointwise Cauchy–Schwarz for the second class, and then Cauchy–Schwarz in y , we obtain

$$S \leq 2D^{-2} \|z\|_2 \|M_d \|z\| \|z\|_2 + \sqrt{L_D} \|z\|_2 \sqrt{S} \leq 4D^{-2} \|z\|_2^2 + \sqrt{L_D} \|z\|_2 \sqrt{S}.$$

Solving this quadratic inequality in $\sqrt{S}$ proves the asserted bound.

Anchors within one size segment. Fix $K \in \mathcal{S}$, write $g = g_K$ and $a = a_K$, and put

$$\tau = D^{-10}, \quad b = 2D(m+1), \quad T = \lceil b^{128(m+1)^2} \rceil.$$

For each cell $u \in \mathcal{T}_K$, let $c(u)$ be the unique last-layer coefficient vector of $P_u^m \cdots P_u^1 g$, extended by zero to all labels.

We describe an attempt from a cell $R \in \mathcal{T}_K$, initially $R = K$. Declare R an anchor. From an anchor A , choose as its next anchors the first proper descendant cells $u \in \mathcal{T}_K$ for which

$$\|c(u) - c(A)\|_{\ell^2(\mathcal{V})} > \tau a.$$

Stop a path at its T th transition and send that cell and its descendants to a new attempt, using the same g and a . Denote the disjoint new roots from R by $\mathcal{R}(R)$.

To apply Lemma 3.3, complete each anchor generation to a stopping partition π_n of R , $0 \leq n \leq T$. The active atoms are the anchors of generation n . The other atoms are terminal cut cells on paths that have ended; leave those atoms unchanged in later partitions. When refining an active anchor, its next anchors together with the terminal cut cells not lying inside a next anchor partition it. Use the original masks on active anchors. On a terminal cut one can use an inherited local refit, for example with its original mask if that mask is already defined, and keep this local projection unchanged afterward. Hence these are refining partitions, and the resulting direct-sum auxiliary projection families $P_n^i = P_{\pi_n}^i$ are nested. Put $Z_n = P_n^m \cdots P_n^1$ on $L^2(R; \mathcal{H})$.

For any $R' \in \mathcal{R}(R)$, its ancestors in the anchor tree have generations $0, \dots, T$. On R' the restriction of $(Z_n - Z_{n-1})(\mathbf{1}_R g)$ is the difference of the constant-coefficient combinations at the corresponding two consecutive anchors. Inheritance makes all the older labels available at the newer anchor. The private coordinate bound in Lemma 3.2 therefore gives

$$\|(Z_n - Z_{n-1})(\mathbf{1}_R g)(y)\|_{\mathcal{H}} > D^{-1} \tau a \quad \text{for almost every } y \in R', \quad 1 \leq n \leq T.$$

Integrating over the disjoint new roots, summing in n , and applying Lemma 3.3 yields

$$TD^{-2} \tau^2 a^2 \sum_{R' \in \mathcal{R}(R)} |R'| \leq m^2 L_T^{2m} \|\mathbf{1}_R g\|_2^2.$$

Since $\|g\|_{\infty} \leq Ga$, this implies both

$$\sum_{R' \in \mathcal{R}(R)} |R'| \leq E |R|, \quad \sum_{R' \in \mathcal{R}(R)} \|\mathbf{1}_{R'} g\|_2^2 \leq E \|\mathbf{1}_R g\|_2^2, \quad E = \frac{16D^{32} m^2 L_T^{2m}}{T}.$$

Our fixed choice of T makes $E < 1/2$. For completeness, $b \geq 8$ and

$$L_T \leq 129(m+1)^2 \log_2 b \leq b^4.$$

The second inequality follows from $m+1 \leq b/4$ and $\log_2 b \leq b/2$. Thus

$$E \leq 16b^{34+8m-128(m+1)^2} \leq 16 \cdot 8^{-470} < \frac{1}{2}.$$

It follows by generation-by-generation summation that all attempt roots inside this size segment, denoted by $\mathcal{R}_K$, satisfy

$$\sum_{R \in \mathcal{R}_K} |R| \leq 2|K|, \quad \sum_{R \in \mathcal{R}_K} \|\mathbf{1}_R g_K\|_2^2 \leq 2\|g_K\|_2^2.$$

Rows assigned to an attempt. Assign a row in the size segment to the deepest attempt root containing its cell. For a row cell u assigned to the attempt from R , let A be its last ancestor anchor in that attempt, and let j be the generation of A . Here $0 \leq j < T$. The cell A is an atom of π_j and contains u . Consequently the restriction of $Z_j(\mathbf{1}_R g_K)$ to u is exactly the product at the anchor, restricted to u . The stopping rule gives

$$\|c(u) - c(A)\|_{\ell^2(\nu)} \leq \tau a_K.$$

The support condition on B_ν now permits the decomposition of its input product into this anchor product and its coefficient error. The rows assigned to R , restricted to $L^2(R; \mathcal{H})$, still satisfy the row contraction hypothesis. Lemma 3.3 shows that their anchor contributions have squared sum at most

$$L_T^{2m} \|\mathbf{1}_R g_K\|_2^2.$$

This is the step in which arbitrary, unequal physical depths of the row cells cause no additional cost: the prefix index is the row's last anchor generation, and its cell lies wholly in that anchor atom.

For the error write $e_{\nu,v} = c(u(\nu))_v - c(A(\nu))_v$. It is supported on labels available on $u(\nu)$ and has squared coefficient sum at most $\tau^2 a_K^2$. Since $u(\nu) \subset R \cap J_v$ whenever this coefficient can be nonzero, locality of B_ν gives

$$B_\nu \left(\mathbf{1}_{u(\nu)} \sum_v e_{\nu,v} \Phi_v \right) = \sum_v e_{\nu,v} B_\nu(\mathbf{1}_R \tilde{\Phi}_v).$$

Cauchy–Schwarz in v , followed by the row contraction for each fixed label, bounds the squared sum of errors in this attempt by

$$\begin{aligned} \tau^2 a_K^2 \sum_v \sum_{\substack{\nu \text{ assigned to } R \\ v \text{ available on } u(\nu)}} \|B_\nu(\mathbf{1}_R \tilde{\Phi}_v)\|^2 \\ \leq \tau^2 a_K^2 \sum_v \|\mathbf{1}_R \tilde{\Phi}_v\|_2^2 \leq \tau^2 a_K^2 D^3 |R|. \end{aligned}$$

The last inequality uses the pathwise budget in its integral form:

$$\sum_v \|\mathbf{1}_R \tilde{\Phi}_v\|_2^2 = \int_R \sum_{v: y \in J_v} \|\Phi_v(y)\|_{\mathcal{H}}^2 dy \leq D^3 |R|.$$

In particular, no estimate of the total number of labels on different branches has been used.

Summation and the final constant. Each row belongs to exactly one size segment and one attempt within it. The triangle inequality in the full row Hilbert space, the two packing bounds, and the estimates for S and the stopping averages give

$$\begin{aligned} \left(\sum_{\nu} \|B_{\nu} P_{u(\nu)}^m \cdots P_{u(\nu)}^1 z\|^2 \right)^{1/2} &\leq L_T^m \left(2 \sum_K \|g_K\|_2^2 \right)^{1/2} + \tau D^{3/2} \left(2 \sum_K a_K^2 |K| \right)^{1/2} \\ &\leq C \left(L_T^m \sqrt{\log(2D)} + D^{-17/2} \right) \|z\|_2. \end{aligned}$$

Finally,

$$L_T \leq C(m+1)^2 \log(2D(m+1)) \leq C(m+1)^3 \log(2D),$$

which is bounded by the stated expression after increasing its absolute constant. A complement is expanded as $1 - P^i$; the at most 2^m resulting ordered products obey the same form of bound, and their number is absorbed in that constant. For a fixed average of compositions, apply the estimate to each member and use the integral triangle inequality in the row Hilbert space. This proves the remaining assertions. $\square$

3.5. Rational heights and occupied scale ranges. The next observation is useful when real frequencies are unrestricted but all coefficients used to combine them have bounded rational height. Small divisors can place the relevant scale ranges far apart; they do not create arbitrarily many such ranges.

Lemma 3.5 (Height bins). *Let $d \geq 1$, $B \geq 2$, and $\xi \in \mathbb{R}^d$. Let*

$$\mathcal{E} = \left\{ |a \cdot \xi| : a \in \mathbb{Q}^d \text{ has height at most } B, a \cdot \xi \neq 0 \right\}.$$

There is an absolute constant C such that the following assertions hold.

- (1) *The set $\{\log_2 v : v \in \mathcal{E}\}$ is contained in a union of at most d intervals, each of length at most $C(d+1)^3 \log(2B)$. In particular, $\mathcal{E}$ meets at most $C(d+1)^4 \log(2B)$ dyadic magnitude bins.*
- (2) *If $r_n = r_0 2^{-n}$ with $r_0 > 0$, $\ell \in \{1, 2\}$, and $0 < \alpha \leq \beta$, then*

$$\#\{n \in \mathbb{Z} : \text{some } v \in \mathcal{E} \text{ satisfies } \alpha \leq r_n^{\ell} v \leq \beta\} \leq C(d+1)^4 (\log(2B) + \log(2\beta/\alpha)).$$

- (3) *For every $\varepsilon > 0$, with the maximum of an empty set defined to be zero,*

$$\sum_{n \in \mathbb{Z}} \max(\{r_n^{\ell} v : v \in \mathcal{E}, r_n^{\ell} v \leq \varepsilon\} \cup \{0\}) \leq C(d+1)^4 \log(2B) \varepsilon.$$

The same conclusions apply to suprema of finitely many such coordinate evaluations. For Euclidean norms of evaluation vectors with at most h coordinates, one can add $Cd \log(2h)$ to the bounds for the number of occupied bins and scales, and to the coefficient of ε in the last estimate. No bound on the size of ξ is required.

Proof. If $\mathcal{E}$ is empty there is nothing to prove. It is otherwise a finite set. We first bound coefficients in a minimal rational dependence. Suppose $a_1, \dots, a_k$ are minimally dependent vectors of the indicated height. Then $k \leq d+1$ and their rank is $k-1$. For each i , clear the denominators in its d entries by a common denominator $q_i \leq B^d$. The integer vector $v_i = q_i a_i$ has each entry bounded by B^d . Select $k-1$ coordinate rows on which the vectors have rank $k-1$. Their signed maximal minors give a null vector $(b_1, \dots, b_k)$ with

$$0 < |b_i| \leq d! B^{d^2}.$$

None of these minors is zero: a zero component of a null vector would give a dependence on a proper subset, contrary to minimality. Returning to the original vectors gives

$$\sum_{i=1}^k c_i a_i = 0, \quad c_i = b_i q_i \in \mathbb{Z} \setminus \{0\}, \quad |c_i| \leq H := d! B^{d^2+d}.$$

Set $R = 4dH$. Suppose coefficient vectors have nonzero evaluation magnitudes

$$0 < s_1 < s_2 < \cdots < s_k, \quad s_{i+1} > R s_i.$$

If these vectors were dependent, apply the preceding construction to a minimal dependent subset, ordered by evaluation magnitude, and let s_* be its largest magnitude. The coefficient of that term has absolute value at least one, whereas all the other terms together have absolute value at most

$$dH \frac{s_*}{R} < s_*.$$

They cannot sum to zero. Thus vectors whose evaluation magnitudes are successively separated by a factor greater than R are linearly independent, and there can be at most d of them.

Choose the smallest element s of $\mathcal{E}$, cover $[s, Rs]$, and repeat with the smallest still uncovered element. The selected elements are successively separated by more than R , so at most d intervals are needed. Their logarithmic lengths are $\log_2 R$, and

$$\log_2 R \leq C(d+1)^2 \log(2dB) \leq C(d+1)^3 \log(2B).$$

Each logarithmic interval meets at most $2 + \log_2 R$ unit bins. This proves the first assertion.

For the second, consider one logarithmic value interval $[s, s + \log_2 R]$. The condition $\alpha \leq r_n^\ell v \leq \beta$ confines n to a real interval of length at most

$$\frac{\log_2 R + \log_2(\beta/\alpha)}{\ell}.$$

That interval has at most two plus its length integer points. Summing over the at most d value intervals and using the preceding bounds proves the claimed scale count. The factor r_0 translates the interval and has no effect on its length.

For the third, partition $\mathcal{E}$ by its occupied bins $[2^j, 2^{j+1})$. For one such bin, a contribution at scale n is possible only if $r_n^\ell 2^j \leq \varepsilon$, and its maximum is at most $r_n^\ell 2^{j+1}$. Let n_0 be the first integer for which the former inequality holds. The sum of these upper bounds for $n \geq n_0$ is at most

$$2\varepsilon \sum_{a=0}^{\infty} 2^{-\ell a} \leq 4\varepsilon.$$

The maximum over all occupied bins is bounded by the sum of their separate maxima. Multiplying by the occupied-bin count proves the assertion.

Finally, the supremum norm of a nonzero evaluation vector is the magnitude of one of its nonzero coordinates, so the same set $\mathcal{E}$ contains all such suprema. If there are at most h coordinates, its Euclidean norm lies between that supremum and $\sqrt{h}$ times the supremum. Enlarging each of the d logarithmic intervals to the right by $\frac{1}{2} \log_2 h$ covers all these Euclidean norms. The preceding bin, scale, and geometric-tail arguments then give exactly the additional costs stated in Lemma 3.5. $\square$

In particular, these estimates permit the rational coefficient vector to be chosen anew at each depth: every such choice still lies in the same bounded-height family. They also apply after adjoining finitely many fixed real parameters to ξ , for example the coordinates of a horizontal frequency vector and their pairwise products.

4. QUADRATIC CHARTS AND LOCAL DETECTION

This section establishes the scalar calculus used in the subsequent projection constructions. The distinction between a chart's complexity and its real oscillation parameters is essential: the latter will never be bounded. All constants in this section are independent of those parameters and of the interval on which an atom is tested.

We first combine a scale-wise Weyl alternative and Fourier expansion with an algebraic separation identity along the progression, obtaining separated moments at one scale. We then transfer the discrete degree-two inverse theorem to bounded continuous witnesses and use Hahn–Banach to decompose bounded inputs. Applying that decomposition to two opposing inputs reduces local detection to the separated moments already established.

4.1. Atoms and quantitative conventions. Fix once and for all the Gevrey order used for the smooth cutoffs. Enlarging its fixed constants does not change any of the estimates below. For a rational number, its height is the maximum of the absolute numerator and the positive denominator in reduced form. The height of a rational vector or matrix is the maximum of the heights of its entries. We use Euclidean norms for real vectors and the associated matrix norms; changing between these norms and coordinate maximum norms only changes the quantitative constants.

Definition 4.1 (Quadratic chart atom). A quadratic chart atom is a function

$$\varphi(y) = e(\lambda y^2 + \beta y) \sum_{n \in \mathbb{Z}^d} \chi(\theta y - n) e((\theta y)^T C(\theta y - n)). \quad (1)$$

Here $\theta \in \mathbb{R}^d$, $\lambda, \beta \in \mathbb{R}$, C is a rational d -by- d matrix, and χ is smooth and compactly supported. The case $d = 0$ includes ordinary quadratic phases. An overall unimodular constant factor is allowed. Every non-unimodular scalar multiplying an atom is either absorbed into χ , and is then subject to its stated complexity bounds, or is written as an external coefficient.

An atom has complexity at most H , where $H \geq e^2$, if $d \leq \log H$, the height of C is at most H , $\text{supp } \chi \subset [-H, H]^d$, and, for every multi-index α ,

$$\|\partial^\alpha \chi\|_\infty \leq H^{c_0(|\alpha|+1)} (|\alpha| + 1)^{c_0(|\alpha|+1)}.$$

The constant c_0 is fixed throughout. The parameters λ, β, θ are unrestricted. A bounded atom means an atom normalized to have supremum norm at most one.

A quantity described below as quasipolynomial in $H_1, \dots, H_m$ is bounded by

$$\exp \left(C \left(1 + \sum_{\nu=1}^m \log H_\nu \right)^C \right), \quad H_\nu \geq 2,$$

with an absolute C ; a fixed number of compositions of such bounds is again of this form. When $q > 2$ occurs, the constant multiplying the logarithms may depend on q , but the exponent C can be chosen independently of q . Constants attached to a fixed requested Fourier-tail power may also depend on that power.

There are at most $(2H + 3)^d$ nonzero summands at a point in (1). Consequently its supremum norm is quasipolynomial in H . Dividing an atom by such a bound normalizes it; the factor is then included in the coefficient of the expansion in which it occurs.

Lemma 4.2 (Elementary chart operations). *Complex conjugation, a product of a bounded number of atoms, and translation or affine rescaling of the real argument preserve the class (1), with a quasipolynomial change of complexity. The same statement holds for the multivariable charts introduced below. Only the unrestricted real oscillation parameters can depend on the size of the translation or rescaling.*

Proof. For conjugation replace C, λ, β by their negatives and conjugate χ . For products use the direct sum of the horizontal coordinates and the block diagonal sum of the matrices. The product rule for derivatives gives the required Gevrey bounds.

It remains to check a translated horizontal argument. Let that argument be $\theta y + a$. Choose a positive integer P such that every entry of PC is integral. The product of the denominators of the d^2 entries is an admissible choice, so $P \leq H^{d^2}$. Write $a = m_0 + a_0$, with $m_0 \in P\mathbb{Z}^d$ and $\|a_0\|_\infty \leq P$, and replace the summation index by $n + m_0$. With $w = \theta y - n$, the new phase is

$$(\theta y + m_0 + a_0)^T C(w + a_0) = (\theta y)^T Cw + (\theta y)^T Ca_0 + m_0^T C\theta y + a_0^T C(w + a_0) + m_0^T Ca_0 \pmod{\mathbb{Z}}.$$

The two middle terms linear in θy change the unrestricted linear phase. The last term is constant. The term $a_0^T C(w + a_0)$ can be included in the smooth amplitude $\chi(w + a_0)$. The support and derivatives of that amplitude have quasipolynomial bounds because a_0 , unlike m_0 , is bounded by P . Translating an ordinary quadratic phase produces only a quadratic phase, a linear phase, and a constant. Rescaling simply rescales θ, λ, β . All these calculations hold with θy replaced by Vz . $\square$

4.2. A quantitative Weyl alternative. For $z \in \mathbb{R}^k$, where $1 \leq k \leq 3$, consider

$$Q(z) = e(z^T \Lambda z + b^T z) \sum_{n \in \mathbb{Z}^d} \chi(Vz - n) e((Vz)^T C(Vz - n)), \quad A = C - C^T.$$

The matrix Λ is real and symmetric and V is real. The complexity conditions apply to d, C, χ as in Definition 4.1. We write

$$C_{\text{sy}} = \frac{C + C^T}{2}, \quad C_{\text{sk}} = \frac{C - C^T}{2}, \quad \text{sym } B = \frac{B + B^T}{2}.$$

Lemma 4.3 (Scale-wise Weyl alternative). *Let ϑ be supported in a fixed box in $\mathbb{R}^k$, with supremum norm and Lipschitz norm at most H . Suppose $0 < \delta < 1/2$ and*

$$\sup_{\xi \in \mathbb{R}^k} \left| \int_{\mathbb{R}^k} \vartheta(z) Q(z) e(-\xi^T z) dz \right| \geq \delta.$$

There is a rational subspace $U \subset \mathbb{R}^d$, with orthogonal projections P_f onto U and $P_s = 1 - P_f$, such that

$$\begin{aligned} A|_{U \times U} &= 0, & \|V_s\| &\leq H', & \|\Lambda_*\| &\leq H', \\ V_f &= P_f V, & V_s &= P_s V, & \Lambda_* &= \Lambda + \frac{1}{2} V^T C_{\text{sy}} V + \text{sym}(V_f^T C_{\text{sk}} V_s). \end{aligned} \tag{2}$$

The heights of P_f, P_s are at most H' , where

$$H' \leq \exp\left(C(1 + \log H + \log(1/\delta))^C\right).$$

Proof. We first show that every sufficiently small return has a bounded frequency increment. The lattice directions generated by these returns form a rational isotropic subspace with a bounded-height projection and bounded slow horizontal matrix. Returns in each coordinate direction then bound the remaining effective quadratic matrix. Let P clear the denominators of C , as in Lemma 4.2. Put $F = \vartheta Q$. The support of F , its supremum norm, its L^2 norm, and $\int (1 + \|z\|) |F(z)| dz$ are bounded by a quantity D quasipolynomial in H . The value of D can be enlarged finitely many times below.

Suppose

$$Vh = m + d_h, \quad m \in P\mathbb{Z}^d.$$

Translate z by h and replace the lattice index by $n + m$. On the support of the translated amplitude, $Vz - n$ has norm bounded in terms of the chart complexity. Expanding the phase therefore gives

$$F(z + h) = e(c(h, m)) e(\Omega(h, m)^T z) F(z) + E_{h, m}(z), \quad \Omega(h, m) = (2\Lambda + V^T C V)h - V^T A m,$$

where $c(h, m)$ is constant in z , and

$$\|E_{h,m}\|_{L^1} + \|E_{h,m}\|_{\infty} \leq D(\|h\| + \|d_h\|)$$

provided $\|h\| + \|d_h\| \leq 1$. To verify that the error contains no unbounded coefficient, put $u = Vz$, $w = u - n$, and $d = d_h$. The nonconstant increment of the chart phase is

$$u^T C d + m^T C u + d^T C w.$$

The first two terms, together with the ordinary quadratic increment, are exactly $\Omega(h, m)^T z$; the term $m^T C n$ discarded in this calculation is integral. The remaining $d^T C w$ is bounded by $D\|d\|$. Translating the amplitude and ϑ costs $D(\|h\| + \|d\|)$. This proves the claimed estimate.

The Fourier transform of F is D -Lipschitz. Choose a point ξ_0 where its absolute value is at least $3\delta/4$, and put $\rho = \delta/(100D)$. Choose an integer

$$M \geq 100D^2\delta^{-2}\rho^{-k}.$$

If necessary, increase D to cover the fixed volume constant in Plancherel's theorem. Choose $a > 0$ so small that

$$a \leq \frac{\delta}{100DM}, \quad a \leq \frac{1}{100(1 + \|A\|)}.$$

All of M, a^{-1}, ρ^{-1} are quasipolynomial in H, δ^{-1} . For a return with $\|h\| + \|d_h\| \leq a$, the same translation calculation applies to $\ell(h, m)$, $1 \leq \ell \leq M$. It produces Fourier values of absolute value at least $\delta/2$ at $\xi_0 - \ell\Omega(h, m)$. If $\|\Omega(h, m)\| \geq 1$, the balls of radius ρ about those points are disjoint, and on each such ball the Fourier transform has magnitude at least $\delta/3$. Their contribution to its squared L^2 norm exceeds D^2 , by the choice of M , contradicting Plancherel. Hence

$$\|\Omega(h, m)\| < 1 \quad \text{whenever} \quad \|h\| + \|d_h\| \leq a.$$

For two returns satisfying this condition, direct antisymmetrization gives

$$h^T \Omega(h, m) - h'^T \Omega(h', m') = d_h^T A d_h - m'^T A m.$$

The integer $m'^T A m$ has absolute value less than one after decreasing a by a fixed factor. It is therefore zero.

We next find all directions generated by short returns. For $d \geq 1$ let

$$\mathcal{K} = P^{-1}V[-1, 1]^k + [-1, 1]^d.$$

This is a symmetric convex body with nonempty interior. Write $\lambda_1 \leq \dots \leq \lambda_d$ for its successive minima for $\mathbb{Z}^d$, and $\mu_1 \leq \dots \leq \mu_d$ for the successive minima of its polar body. We need the elementary transference bounds

$$1 \leq \lambda_j \mu_{d+1-j} \leq (d!)^2.$$

For the lower bound, independent lattice vectors in subspaces of dimensions j and $d+1-j$ cannot have all cross pairings zero. One of their integer pairings has absolute value at least one. For the upper bound, Minkowski's second theorem [3, Theorem 1.3] gives

$$\prod_{j=1}^d \lambda_j \mu_j \leq \frac{4^d}{\text{vol}(\mathcal{K}) \text{vol}(\mathcal{K}^\circ)} \leq (d!)^2.$$

Here the last volume-product bound has a short direct proof. Choose maximal-determinant columns $v_1, \dots, v_d$ in $\mathcal{K}$. Their cross-polytope is contained in $\mathcal{K}$, and maximality places $\mathcal{K}$ in their coordinate parallelotope. Taking polars gives an inscribed cross-polytope in $\mathcal{K}^\circ$ with reciprocal determinant. The product of the two cross-polytope volumes is $4^d/(d!)^2$. Since every opposite-minimum product is at least one, their product upper bound proves each individual upper bound.

Set $L = 16(d!)^2$, and choose $b_{\max} \leq a/(100P(d+k+1))$. The $d+1$ disjoint intervals

$$(b_{\max}L^{-j-1}, b_{\max}L^{-j}], \quad 0 \leq j \leq d,$$

cannot all contain a successive minimum. Select one containing none, write its left endpoint as b_0 , and put

$$U = \text{span}(\mathbb{Z}^d \cap b_0\mathcal{K}).$$

The reciprocal of b_0 is quasipolynomial in the required parameters. Every $n \in \mathbb{Z}^d \cap b_0\mathcal{K}$ has a representation $n = P^{-1}Vh + w$, with $\|h\|_\infty, \|w\|_\infty \leq b_0$. Thus $m = Pn$, $d_h = -Pw$ form a return satisfying the threshold a . The integrality argument above, applied to any two such n , shows that A vanishes on $U \times U$.

Let $r_0 = \dim U$. If $r_0 < d$, then $\lambda_{r_0+1} > Lb_0$; transference supplies $d - r_0$ independent integer vectors q_ν in

$$\frac{(d!)^2}{Lb_0}\mathcal{K}^\circ.$$

Each has zero pairing with $\mathbb{Z}^d \cap b_0\mathcal{K}$, since that pairing is integral and has absolute value at most $(d!)^2/L < 1$. They consequently span $U^\perp$. The inclusion $[-1, 1]^d \subset \mathcal{K}$ bounds their coordinates, while $P^{-1}V[-1, 1]^k \subset \mathcal{K}$ bounds each $q_\nu^T V$, by quasipolynomial quantities. If S has the q_ν^T as rows,

$$P_s = S^T(SS^T)^{-1}S.$$

Cramer's rule, and the positive integral determinant of SS^T , give quasipolynomial rational height and norm bounds for this matrix and for $P_s V$. When $r_0 = d$, take $P_s = 0$; the same conclusions hold. Put $P_f = 1 - P_s$.

Finally we bound the effective ordinary quadratic part. For each coordinate vector $e_j \in \mathbb{R}^k$, sample $P^{-1}V(te_j) \bmod \mathbb{Z}^d$ at equally spaced points $t \in [0, b_0]$. Partition the horizontal torus into cubes of side at most $b_0/4$. More than $(1 + 4/b_0)^d$ samples give two in the same cube. Their difference yields a return

$$h = t_j e_j, \quad b_0/M_0 \leq t_j \leq b_0, \quad \|d_h\|_\infty \leq Pb_0/4,$$

where $M_0 \leq 2(1 + 4/b_0)^d$. Its lattice vector satisfies $P^{-1}m \in b_0\mathcal{K}$, so $m \in U$. In particular $m - V_f h = -P_f d_h \in U$. Isotropy gives

$$V^T A(m - V_f h) = V_s^T A(m - V_f h),$$

which is quasipolynomially bounded. Direct matrix algebra gives

$$2\Lambda + V^T C V - V^T A V_f = 2\Lambda_* + V_s^T C_{\text{sk}} V_s.$$

The return bound for Ω therefore bounds $2\Lambda_* t_j e_j$ quasipolynomially. Since t_j^{-1} is also quasipolynomial, this bounds every column of Λ_* . This proves (2).

For $d = 0$, the initial translation argument applies to every sufficiently small h , with $\Omega = 2\Lambda h$; take $h = ae_j/2$ and $U = \{0\}$. This also proves the assertion in that case. $\square$

4.3. Fourier expansion after the Weyl splitting.

Lemma 4.4 (Chart expansion and uniform tails). *Suppose a rational orthogonal splitting satisfies (2), with heights and the indicated sizes at most B . On a fixed box in $\mathbb{R}^k$ there is an exact absolutely convergent expansion*

$$Q(z) = e(z^T \Lambda_* z + b^T z) \sum_{\ell} a_\ell e(\sigma_\ell^T V z). \quad (3)$$

It consists of quasipolynomially many families of Fourier series in at most $2d$ integer indices. Their rational frequencies have height at most $D(1 + \|\ell\|)$, where D is quasipolynomial in H, B . The total absolute coefficient sum is bounded by such a D .

More quantitatively, for any fixed integer $J \geq 1$, one can choose a quasipolynomial D_J such that every dyadic shell of indices $T \leq 1 + \|\ell\| < 2T$ has absolute coefficient sum at most

$$D_J T^{-J(d+1)^2}.$$

For fixed underlying oscillation parameters, the same estimate, after enlarging D_J , holds for the coefficient envelope obtained by pooling all rational projection choices of height at most B , separately for each resulting frequency and effective quadratic part. The coefficient envelopes are uniform over amplitudes with the stated complexity bounds and over the bounded slow values entering their construction. In particular the pooled tail can be made at most ε by truncation at a size quasipolynomial in H, B, ε^{-1} .

Proof. Put $u = Vz$, $u_f = P_f u$, $u_s = P_s u$, and $w = u - n$. For the symmetric part,

$$u^T C_{\text{sy}}(u - n) = \frac{1}{2} u^T C_{\text{sy}} u + \frac{1}{2} w^T C_{\text{sy}} w - \frac{1}{2} n^T C_{\text{sy}} n.$$

For the skew part, isotropy on U gives

$$u^T C_{\text{sk}} w = u_f^T C_{\text{sk}} u_s - u_f^T C_{\text{sk}} P_s n + u_s^T C_{\text{sk}} w.$$

The first term is precisely the extra quadratic included in Λ_* .

Choose an even integer P_0 clearing the denominators of C_{sy} . Then $-n^T C_{\text{sy}} n / 2 \bmod \mathbb{Z}$ is determined by $n \bmod P_0 \mathbb{Z}^d$. Also choose a common positive denominator q_0 for P_s . On the box under consideration, $u_s = V_s z$ is bounded, and w belongs to the support of χ . Thus $P_s n = u_s - P_s w$ ranges over a bounded subset of $q_0^{-1} \mathbb{Z}^d$. The size of this set, P_0^d , and q_0 are all quasipolynomial in H, B .

Split the sum over n according to

$$P_s n = N, \quad n \equiv a \pmod{P_0 \mathbb{Z}^d}.$$

For each pair (N, a) , remove the fixed residue phase and the rational linear phase

$$-u_f^T C_{\text{sk}} N.$$

What remains is a smooth amplitude in the two vector variables (u_s, w) . To see explicitly that it admits a periodic Fourier expansion, let $\eta(s)$ be a smooth cutoff equal to one on all values of $V_s z$ in the box. Choose a smooth equality cutoff E with $E(0) = 1$, supported in the coordinate cube of radius $1/(3q_0)$. Then the function

$$\eta(s) \sum_{n \in a + P_0 \mathbb{Z}^d} \chi(u - n) E(s - P_s(u - n) - N) e\left(\frac{1}{2}(u - n)^T C_{\text{sy}}(u - n) + s^T C_{\text{sk}}(u - n)\right)$$

has the required value when $s = P_s u$. Indeed its equality-cutoff argument then equals $P_s n - N$; the lattice spacing makes the cutoff exactly the indicator of the specified equality. It is periodic in u with period P_0 in each coordinate. By taking the support of η strictly inside a larger integer-period box, it extends smoothly and periodically in s , with a period P_1 quasipolynomial in H, B .

The resulting Fourier modes have the form

$$e(p^T s / P_1 + \ell^T u / P_0), \quad p, \ell \in \mathbb{Z}^d.$$

Substitution of $s = P_s u$, and restoration of the rational linear phase for N , gives

$$\sigma = \sigma_N + P_s^T p / P_1 + \ell / P_0, \quad \sigma_N^T u = -(P_f u)^T C_{\text{sk}} N.$$

These frequencies have the height asserted in the lemma. The sum of the two ordinary quadratic terms is $z^T \Lambda_* z$, proving the exact identity (3).

Here are the quantitative convergence details. On each of the bounded supports above, all coefficients in the smooth phases, cutoff scales, periods, and lattice-point counts are bounded by a quasipolynomial quantity D_0 . The product rule and the chain rule imply, at order m , derivative bounds of the form

$$D_0^{C(m+d+1)} (m+1)^{C(m+1)}.$$

This estimate also bounds derivatives after rescaling the periodic variables to unit periods. It follows either by differentiating the quadratic exponential, whose m -th derivative is a polynomial

of degree at most m times that exponential, or directly by induction on m . The cutoffs satisfy the same fixed-order Gevrey estimate.

For a nonzero integer Fourier index $v = (p, \ell)$, select a coordinate with $|v_j| \geq \|v\|/(2d)^{1/2}$ and integrate by parts m times in that coordinate. Thus

$$|\widehat{A}(v)| \leq D_0^{C(m+d+1)} (m+1)^{C(m+1)} (2d)^{m/2} (1 + \|v\|)^{-m},$$

after increasing the constant to include the zero index. A dyadic shell has at most $(CT)^{2d}$ indices. Taking $m \geq J(d+1)^2 + 4d + 4$ gives the displayed shell bound. Since $d \leq \log H$, the prefactor for this derivative order is quasipolynomial in H, B . Summing the shells proves absolute convergence and the bound for the full coefficient sum.

There are at most $B^{C(d+1)^2}$ rational projection matrices of height at most B . For each such matrix there are quasipolynomially many choices (N, a) . All derivative estimates are uniform in the bounded slow parameters and in amplitudes with the stated complexity bounds. Consequently we may replace every coefficient by its uniform envelope before summing over this finite list. If different indices give the same frequency, use the sum of their envelopes at that frequency. Choosing the derivative order larger by $C(d+1)^2$ absorbs, as well, the count $(D_0 T)^{C(d+1)}$ of bounded-height rational frequencies at an index scale T . This proves the pooling assertion in the form stated. Finally, the sum of the shells above a cutoff T_0 is at most $CD_J T_0^{-J(d+1)^2}$; choosing this to be at most ε gives the required quasipolynomial truncation. $\square$

4.4. Separating two atoms along a progression. Recall that $y_j = x + jt$, $0 \leq j \leq 3$.

Lemma 4.5 (Progression separation). *Let φ_i, φ_k be bounded atoms of complexity at most H in two distinct slots i, k , and let a, b denote the remaining slots. Their product has an exact absolutely convergent expansion in terms*

$$c \psi_a(y_a) \psi_b(y_b) Q(t). \quad (4)$$

All three factors are bounded atoms independent of the interval scale. For some D quasipolynomial in H , the terms can be arranged in dyadic shells $T \geq 1$ with total coefficient mass at most DT^{-100} . In a shell T , each factor has dimension at most $\log D$, and complexity at most $(DT)^C$; unrestricted real linear phases are permitted as in Definition 4.1.

Proof. We first separate only φ_i . Temporarily suppress its linear phase, put $\alpha_i = 1$, $\alpha_k = 0$, and choose α_a, α_b so that

$$\sum_{j=0}^3 \alpha_j = 0, \quad \sum_{j=0}^3 j \alpha_j = 0.$$

The four indices lie in a fixed set, so these coefficients are fixed rationals of bounded height.

Choose a nonnegative, compactly supported Gevrey function ρ on $\mathbb{R}^d$ whose integer translates sum to one. Insert its partitions of unity at $\theta y_a, \theta y_b, \theta t$. Write the corresponding lifts as

$$d_a = \theta y_a - m_a, \quad d_b = \theta y_b - m_b, \quad d = \theta t - m.$$

For the original summation index m_i , put

$$d_i = \theta y_i - m_i, \quad k_j = d_j - d_a - (j-a)d \quad (j = i, b), \quad k_a = 0.$$

Each k_j is integral because $y_j = y_a + (j-a)t$. The support restrictions bound the possible k_i, k_b quasipolynomially in H . Conversely, fixing these integers determines m_i from m_a, m , and the condition for k_b is enforced by a smooth equality cutoff in $d_b - d_a - (b-a)d - k_b$. Such a cutoff can equal one at zero and vanish at every other integer vector, so this insertion is exact.

Put $S = \sum_j j^2 \alpha_j$. The two relations defining the α_j 's imply the exact identity

$$\sum_j \alpha_j (\lambda y_j^2 + (\theta y_j)^T C d_j) = S(\lambda t^2 + (\theta t)^T C d) + \sum_j \alpha_j (\theta y_j)^T C k_j.$$

Indeed the coefficients of the common lift d_a and of x cancel, and the coefficient of t in $\sum_j (j-a)\alpha_j y_j$ is S . Solving this identity for the i -phase leaves the a - and b -phases with coefficients $-\alpha_a, -\alpha_b$, and the t -phase with coefficient S . The last displayed sum is linear in x, t , and hence is a sum of a linear function of y_a and a linear function of y_b . Restore the suppressed βy_i in the same way.

The remaining coupled amplitude is a smooth function of (d_a, d_b, d) on a bounded box. It is a product of the partition bumps, the equality cutoff, and $\chi(d_a + (i-a)d + k_i)$. Its derivative bounds are quasipolynomial versions of the fixed Gevrey bounds. Choose slightly larger product bumps equal to one on its support, extend the amplitude periodically inside that larger box, and take its Fourier series. Every Fourier term factors into a function of d_a , a function of d_b , and a function of d . After the independent lattice sums, these are precisely three atoms of the form (1), with rational chart matrices $-\alpha_a C, -\alpha_b C, SC$. A Fourier index of size T contributes only derivatives of size T^m at order m to their smooth amplitudes, so its complexity is bounded by $(DT)^C$.

Apply the same construction to φ_k , now setting the coefficient of the i -slot to zero, and combine the two triples by direct sums of horizontal coordinates. The number of integral choices and the dimensions are quasipolynomially bounded in the original complexity. The integration-by-parts calculation in the proof of Lemma 4.4, with derivative order larger than the Fourier dimension by at least 110, bounds shell coefficient sums by DT^{-100} . The number of Fourier indices and integral choices in each bounded shell is also quasipolynomial. Normalize the three factors to have supremum norm one and put their bounded normalization factors into c . The bounds remain of the stated form. Absolute convergence justifies all rearrangements and proves the identity for every (x, t) . $\square$

Lemma 4.6 (Single-scale separation). *Fix a term in (4), without its coefficient, of complexity at most H . Let ζ be a fixed nonnegative smooth function, equal to one on the range of t/r allowed by a local kernel w_I , and supported on a larger fixed interval. Put*

$$A_r = \sup_{\xi \in \mathbb{R}} \left| \int_{\mathbb{R}} Q(ru) \zeta(u) e(-\xi u) du \right|, \quad r = |I|.$$

Its bilinear form norm in the remaining two $L^2(I)$ slots, with normalized interval norms, is at most CA_r .

If $A_r \geq \delta > 0$, then, for every $0 < \varepsilon < 1/2$, the kernel of this bilinear form, extended by zero off its support, can be approximated uniformly on $I \times I$, to error ε , by a finite separated sum $\sum_\nu c_\nu \phi_{\nu,a}(y_a) \phi_{\nu,b}(y_b)$. The $\phi_{\nu,e}$ are bounded atoms, and their complexities, their number, and $\sum_\nu |c_\nu|$ are quasipolynomial in $H, \delta^{-1}, \varepsilon^{-1}$.

Before separating the smooth kernel, the retained terms have phases $\lambda_ t^2 + \gamma t$, where $|\lambda_*| r^2$ is quasipolynomially bounded. The quantities λ_* are bounded-height rational combinations of λ and the entries of $\theta \otimes \theta$; each $\gamma - \beta$ is a bounded-height rational combination of θ . The height bounds are quasipolynomial in the same parameters.*

Proof. Use (y_a, y_b) as coordinates; their Jacobian is $|b - a|$ and $t = (y_b - y_a)/(b - a)$. After normalization by r , the smooth kernel has a Fourier expansion on a fixed two-dimensional box whose coefficient sum is uniformly bounded. Each Fourier term merely modulates the two inputs. Insert $\zeta(t/r)$, which is one on the relevant support. The remaining operator is convolution with $Q(t)\zeta(t/r)$, after the fixed change of scale $b - a$. Its Fourier multiplier is bounded by CrA_r . Plancherel, the factor r^{-2} in the definition of H_I , and $\|f\|_{L^2(I)} = r^{1/2}\|f\|_{2,I}$ prove the first assertion. The factors ψ_a, ψ_b are bounded multipliers and do not increase the two L^2 norms.

For the second assertion apply Lemma 4.3 to $Q(r \cdot)$, with $V = r\theta$ and $\Lambda = r^2\lambda$. Lemma 4.4, truncated to a sufficiently small uniform error, then gives terms

$$e(\lambda_* t^2 + \gamma t), \quad \gamma = \beta + \sigma^T \theta, \quad |\lambda_*| r^2 \leq D.$$

The formula for Λ_* in (2) gives exactly the asserted rational dependence of λ_* . Split the linear phase by

$$e(\gamma t) = e(-\gamma y_a/(b-a))e(\gamma y_b/(b-a)).$$

The remaining smooth two-variable kernel, $w_I(x, t)e(\lambda_* t^2)$, has derivatives in normalized coordinates bounded by $D^{C(m+1)}(m+1)^{C(m+1)}$ at order m . Its Fourier expansion, with fixed product cutoffs, has quasipolynomial coefficient sum and a quasipolynomial truncation giving the prescribed error. The resulting factors are a bounded one-variable smooth bump at scale r , a linear phase, and ψ_e .

For completeness, a smooth bump on I , restricted to I , is itself represented by a chart with zero matrix and one additional slow horizontal coordinate. Map I to an interval strictly inside one lattice cell, put the bump in the corresponding amplitude, and support that amplitude away from every other lattice translate. On I only the chosen translate contributes. Lemma 4.2 handles the affine offset and the product with ψ_e . Outside I a bounded periodic extension is harmless. This realizes all separated factors as bounded atoms. Choose the successive truncation errors divided by the already bounded coefficient sums. Only a fixed number of quasipolynomial budgets is composed, giving all the stated estimates. $\square$

4.5. A bounded continuous inverse theorem. We use the following external input, only at degree two.

Theorem 4.7 (Leng–Sah–Sawhney). *For a 1-bounded function $f : [L] \rightarrow \mathbb{C}$ with $\|f\|_{U^3[L]} \geq \delta$, there are a degree-two filtered nilmanifold G/Γ , a polynomial sequence $g : \mathbb{Z} \rightarrow G$ adapted to its filtration, and a function $F : G/\Gamma \rightarrow \mathbb{C}$ such that*

$$\left| \frac{1}{L} \sum_{n=1}^L f(n) \overline{F(g(n)\Gamma)} \right| \geq \varepsilon.$$

The dimension is at most $C(1 + \log(1/\delta))^C$, and the complexity of G/Γ , the Lipschitz norm of F , and ε^{-1} are at most $\exp(C(1 + \log(1/\delta))^C)$. The nilmanifold has a rational Malcev basis adapted to the filtration, with integer second-kind coordinates for Γ . The Lipschitz norm includes the supremum norm.

This is the $s = 2$ instance of [8, Theorem 1.2]; the coordinate and complexity conventions are [8, Definitions 2.3–2.4 and 3.3–3.5]. In particular no mean-zero hypothesis is imposed.

For the continuous version, first identify an interval with $[0, 1)$. Fix the circle $\mathbb{T}_{16} = \mathbb{R}/(16\mathbb{Z})$, with its probability Haar measure, and extend functions on $[0, 1)$ by zero to this circle. Define

$$\|g\|_{\mathcal{U}}^8 = \int_{\mathbb{T}_{16}^4} \prod_{\omega \in \{0,1\}^3} \mathcal{C}^{|\omega|} g(x + \omega \cdot h) d\mu(x) d\mu(h_1) d\mu(h_2) d\mu(h_3),$$

where $\mathcal{C}$ denotes complex conjugation. On an arbitrary interval I , $\mathcal{U}(I)$ denotes this norm after the affine normalization to $[0, 1)$. All constants caused by this fixed circle length are absolute.

Lemma 4.8 (Elementary cube bounds). *The quantity $\|\cdot\|_{\mathcal{U}}$ is a seminorm. The mixed cube form with possibly different functions g_ω at its eight vertices satisfies*

$$\left| \int_{\mathbb{T}_{16}^4} \prod_{\omega} \mathcal{C}^{|\omega|} g_\omega(x + \omega \cdot h) d\mu(x) d\mu(h) \right| \leq \prod_{\omega} \|g_\omega\|_{L^2(\mathbb{T}_{16})}.$$

Consequently cube forms are continuous in L^2 , and $\|g\|_{\mathcal{U}} \leq C\|g\|_{2,[0,1)}$ for supported functions.

Proof. The usual cube Cauchy–Schwarz inequality is valid on this circle. One direct derivation writes a vertex as $x_1^{\omega_1} + x_2^{\omega_2} + x_3^{\omega_3}$, using three independent pairs of Haar variables. At the first

Cauchy–Schwarz step the factors with the first coordinate equal to zero and one are put on opposite sides; repeating for the second and third coordinates gives

$$|\Lambda_{\text{cube}}(g\omega)| \leq \prod_{\omega} \|g\omega\|_{\mathcal{U}}.$$

The same iterations for a single function express its eighth power as an integral of squared absolute values, proving nonnegativity. Expanding the cube of $f + g$, applying the displayed inequality to each mixed term, and summing the binomial coefficients gives the triangle inequality.

For the stated L^2 estimate, split the vertices into

$$\{000, 100, 010, 001\} \quad \text{and} \quad \{111, 011, 101, 110\}.$$

Each four-tuple of corresponding linear forms is an integer linear automorphism of the four Haar variables, with determinant of absolute value one. Apply Cauchy–Schwarz to the product of these two four-factor groups, then change variables in each squared integral. Each squared integral factors into the four squared L^2 norms. This proves the estimate. Expanding the difference of two mixed cube forms proves their continuity, first for bounded functions and then for all L^2 functions by truncation. The fixed normalization between circle measure and interval measure proves the final assertion. $\square$

Lemma 4.9 (Continuous bounded inverse). *For $0 < \delta < 1/2$, let g be supported on an interval I , with $\|g\|_{\infty} \leq 1$ and $\|g\|_{\mathcal{U}(I)} \geq \delta$. There is a bounded atom φ of complexity at most*

$$H_{\delta} = \exp\left(C(1 + \log(1/\delta))^C\right)$$

such that

$$\left| \int_I g(y) \overline{\varphi(y)} dy \right| \geq H_{\delta}^{-1}.$$

Proof. Normalize $I = [0, 1)$. For $L \geq 1$, define the cell averages

$$g_L(j) = L \int_{j/L}^{(j+1)/L} g(y) dy, \quad 0 \leq j < L,$$

and extend them by zero to $\mathbb{Z}/(16L\mathbb{Z})$. Their discrete raw cube averages converge to the continuous raw cube average above. Here is a justification that needs no regularity of g . For a continuous bounded function the assertion is Riemann-sum convergence, because its cell averages differ from endpoint values by its modulus of continuity. For two functions bounded by one, telescoping the eight-factor product bounds the difference of their continuous cube averages by $8\|g - b\|_{L^1(\mathbb{T}_{16})}$. The identical discrete bound holds for their cell averages, since

$$\frac{1}{L} \sum_{j=0}^{L-1} |g_L(j) - b_L(j)| \leq \|g - b\|_{L^1([0,1))}.$$

Approximation in L^1 by continuous functions bounded by one proves the assertion.

There is no wraparound contribution hidden in this comparison. For a nonzero supported cube choose $x \in [0, 1)$, and lift h_j as the difference between $x + h_j \in [0, 1)$ and x . Then $|h_j| \leq 1$, and every real vertex belongs to $[-3, 4]$. Within that range, reduction modulo 16 lies in $[0, 1)$ exactly when the real vertex does. Thus the circle and interval zero-extension cube averages differ only by their fixed normalizing factors. The denominator defining $U^3[L]$, namely the cube norm of the interval indicator, likewise converges to a fixed positive constant. For all sufficiently large L , therefore,

$$\|g_L\|_{U^3[L]} \geq c\delta,$$

after shifting the integer labels by one if needed. No bound on this choice of L is required.

Apply Theorem 4.7. Write its dimension as d_0 , its complexity as M_0 , its Lipschitz bound as K_0 , and its correlation as ε_0 . All these quantities have the bounds asserted there. We next convert its nilsequence into chart atoms, tracking only quantities independent of the real orbit coefficients.

In second-kind Malcev coordinates adapted to the central filtration subgroup G_2 , write a group element as (z, s) , with $z \in \mathbb{R}^p$, $s \in \mathbb{R}^{d_0-p}$. Since $[G, G] \subset G_2$ and $[G, G_2] = 1$, commuting the one-parameter coordinate factors into their prescribed order gives the group law

$$(z, s)(z', s') = (z + z', s + s' + B(z, z')).$$

The coefficients of the bilinear map B are rational and have height bounded quasipolynomially in M_0 and e^{d_0} . Indeed, in step two every interchange contributes just its central commutator with coefficient $z_i z'_j$; all further commutators vanish. The rational structure constants in the Malcev basis are part of the nilmanifold complexity. Integer coordinate tuples represent exactly the lattice Γ .

Write the polynomial sequence in these coordinates as $(z(n), s(n))$. Its second group differences have no horizontal part, so $z(n) = z_0 + n\theta$. For the unit right difference $Dg(n) = g(n+1)g(n)^{-1}$, the group law gives

$$Dg(n) = (\theta, \Delta s(n) - B(\theta, z(n))), \quad D^2g(n) = (0, \Delta^2 s(n) - B(\theta, \theta)), \quad D^3g(n) = (0, \Delta^3 s(n)).$$

The degree-two filtration makes $D^3g = 1$. Thus $s(n)$ is an ordinary vector polynomial of degree at most two. No bound is imposed on the coefficients of $z(n), s(n)$.

Choose ρ as the product of fixed one-dimensional nonnegative Gevrey partitions supported in $(-1, 1)$, so that $\sum_{m \in \mathbb{Z}^p} \rho(z - m) = 1$. Right multiplication by the lattice element $(-m, 0)$ gives

$$F(g(n)\Gamma) = \sum_{m \in \mathbb{Z}^p} \rho(z(n) - m) \tilde{F}(z(n) - m, s(n) - B(z(n), m)), \quad \tilde{F}(w, u) = F((w, u)\Gamma).$$

The function $\tilde{F}$ is periodic in every central coordinate. On a bounded horizontal box its coordinate Lipschitz constant is quasipolynomially bounded in K_0, M_0, e^{d_0} . To check this directly, the coordinate difference of two nearby group elements is

$$(w, u)(w', u')^{-1} = (w - w', u - u' - B(w - w', w'));$$

on a bounded box its coordinates are bounded by a controlled multiple of the coordinate displacement. The Malcev metric is bounded by the length of this coordinate displacement path. This proves the claimed coordinate Lipschitz estimate.

Multiply by a fixed horizontal cutoff equal to one on $\text{supp } \rho$ and supported strictly inside a larger period box. Periodize horizontally, and convolve in all coordinates with product Fejér kernels of order N_0 . For a Lipschitz function on a circle the error is bounded by its Lipschitz constant times $C \log(2N_0)/N_0$, since this is an upper bound for the kernel's first moment. Telescoping over the d_0 coordinates gives the same estimate multiplied by d_0 . Choose N_0 so that this error is at most $\varepsilon_0/4$. It may be chosen quasipolynomially in $K_0, M_0, e^{d_0}, \varepsilon_0^{-1}$. The resulting trigonometric polynomial has the form

$$\sum_{\ell, k} a_{\ell, k} e(\ell^T w / P_2 + k^T u), \quad \|\ell\|_\infty, \|k\|_\infty < N_0,$$

where P_2 is a fixed integer horizontal period. Each coefficient has absolute value at most a controlled multiple of K_0 , so its absolute coefficient sum is at most $CK_0(2N_0 + 1)^{d_0}$. Because the partition ρ is nonnegative and sums to one, the approximation error remains at most $\varepsilon_0/4$ after summing over m .

For a fixed central frequency k , define C_k by $z^T C_k w = k^T B(z, w)$. The associated summand of this approximation is

$$e(k^T s(n) - z(n)^T C_k z(n)) \sum_{m \in \mathbb{Z}^p} \rho(z(n) - m) e(\ell^T (z(n) - m) / P_2) e(z(n)^T C_k (z(n) - m)).$$

Its outside phase is an ordinary quadratic in n . The matrix C_k is rational with controlled height, while the horizontal orbit is affine. Lemma 4.2 removes its constant offset, without putting an unbounded parameter in the amplitude. Thus each summand is an atom (1). The positivity of ρ shows that it has supremum at most one before the harmless normalization of the coefficients. Its complexity and the total absolute coefficient sum are quasipolynomial in δ^{-1} . It follows that one such discrete atom $a(n)$ satisfies

$$\left| \frac{1}{L} \sum_{j=0}^{L-1} g_L(j) \overline{a(j)} \right| \geq \kappa, \quad \kappa^{-1} \leq \exp\left(C(1 + \log(1/\delta))^C\right).$$

Here and below a shift of the polynomial argument incorporates the choice of integer labels.

We finish with an exact floor interpolation, which is necessary because the chosen atom may oscillate on the sampling scale. Write its parameters as $(\lambda, \beta, \theta, C, \chi)$. Let P clear the denominators of C , and write $\theta = \theta_0 + Pq$, with $q \in \mathbb{Z}^d$ and $0 \leq (\theta_0)_j < P$. At integer arguments n , reindexing the chart sum by Pqn replaces the ordinary quadratic coefficient by $\lambda + (Pq)^T C \theta_0$; the other extra term is integral. Reduce this new coefficient and β modulo one. The same sampled atom consequently has a representation with

$$0 \leq \lambda, \beta < 1, \quad 0 \leq \theta_j < P.$$

Only this equality at integer arguments is needed.

Put $\alpha = 2\lambda + \theta^T C \theta$, and define the enlarged horizontal vector and matrix

$$\Theta = \begin{pmatrix} \theta \\ 1 \\ \alpha \end{pmatrix}, \quad \tilde{C} = \begin{pmatrix} C & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}.$$

Choose $0 \leq \eta \leq 1$, Gevrey of the fixed order at cutoff scale ϵ , supported in $(0, 1)$, and equal to one outside endpoint strips of total length at most 2ϵ . Choose a one-dimensional Gevrey partition $\sum_{\ell \in \mathbb{Z}} \rho_0(r - \ell) = 1$. For lift variables (w, h, r) , define

$$\tilde{\chi}(w, h, r) = \chi(w - \theta h) \eta(h) \rho_0(r) e\left(-h \theta^T C w + (\lambda + \theta^T C \theta) h^2 - \beta h\right).$$

The atom

$$A(v) = e(\lambda v^2 + \beta v) \sum_{n, m, \ell} \tilde{\chi}(\theta v - n, v - m, \alpha v - \ell) e\left((\Theta v)^T \tilde{C} (\Theta v - (n, m, \ell))\right)$$

satisfies the exact identity

$$A(v) = \eta(\{v\}) a(\lfloor v \rfloor) \quad (v \in \mathbb{R}),$$

with value zero at integer v . Indeed the support of η forces $m = \lfloor v \rfloor$; the ℓ -sum is one. With $h = v - m$ and $w = \theta v - n$, the phase in the remaining summand is

$$\lambda v^2 + \beta v + (\theta v)^T C w - (2\lambda + \theta^T C \theta) v h - h \theta^T C w + (\lambda + \theta^T C \theta) h^2 - \beta h,$$

which equals $\lambda m^2 + \beta m + (\theta m)^T C (\theta m - n)$. This proves the identity.

All parameters occurring in the new amplitude are now bounded: the reductions above bound λ, β, θ and hence α , while its support bounds $h, r, w - \theta h$. The cutoff η has Gevrey derivative bounds with scale ϵ^{-1} . Thus the complexity of A is quasipolynomial in the old complexity and ϵ^{-1} ; its two extra horizontal coordinates and its new matrix entry -1 introduce no uncontrolled height. Moreover $|A| \leq 1$, because its values are a cutoff times sampled values of the bounded atom.

Take $v = Ly$. This changes only unrestricted real oscillation parameters. Exact cell averaging gives

$$\int_0^1 g(y) \overline{a(\lfloor Ly \rfloor)} dy = \frac{1}{L} \sum_{j=0}^{L-1} g_L(j) \overline{a(j)}.$$

Replacing the sampled atom by $A(Ly)$ changes this integral by at most 2ϵ , since the total measure of the removed cell endpoint strips is at most 2ϵ . Choose $\epsilon \leq \kappa/8$. The resulting correlation is at least $3\kappa/4$, and its complexity is quasipolynomial in δ^{-1} , independently of L . This proves the normalized-interval assertion. Lemma 4.2 transfers it to the original interval. $\square$

4.6. A bounded-input decomposition. The argument below uses the inverse-to-three-part decomposition method of Gowers–Wolf [2, Proposition 3.3 and Theorem 3.4], adapted here to the local interval and the L^2 -closure step. The quantitative inverse input remains the Leng–Sah–Sawhney theorem [8] used in Lemma 4.9.

Lemma 4.10 (Structured, uniform, and L^1 decomposition). *For $0 < \tau < 1/2$ and a function f on I with $\|f\|_\infty \leq 1$, there is a finite decomposition*

$$f = \sum_{\nu} c_{\nu} \varphi_{\nu} + u + v \quad \text{on } I$$

such that the φ_{ν} are bounded atoms of complexity at most D_{τ} ,

$$\sum_{\nu} |c_{\nu}| \leq D_{\tau}, \quad \|u\|_{\mathcal{U}(I)} \leq \tau, \quad \|v\|_{1,I} \leq \tau,$$

and $u, v \in L^2(I)$. Here $D_{\tau} \leq \exp(C(1 + \log(1/\tau))^C)$.

Proof. All functions and inner products in this proof use normalized measure on I , and all functions are extended by zero for their $\mathcal{U}(I)$ norms. Put $\sigma = \tau/2$. Let $\mathcal{A}$ be the bounded atoms of the complexity supplied by Lemma 4.9 at threshold σ^2 , restricted to I , and let $\kappa > 0$ be the corresponding correlation bound. Choose $M \geq \max(1, 2\sigma/\kappa)$, quasipolynomial in σ^{-1} . Consider the convex balanced set

$$\mathcal{B} = M \operatorname{absconv}(\mathcal{A}) + \{u \in L^2(I) : \|u\|_{\mathcal{U}(I)} \leq \sigma\} + \{v \in L^2(I) : \|v\|_{1,I} \leq \sigma\}.$$

Here the absolute convex hull consists of finite combinations. This set contains an L^2 ball about zero.

Suppose f were outside the L^2 closure of $\mathcal{B}$. The real Hahn–Banach separation theorem, and the complex Riesz representation of a real functional as a real part, give $h \in L^2(I)$, normalized so that

$$\operatorname{Re}\langle f, h \rangle > 1, \quad \sup_{z \in \mathcal{B}} \operatorname{Re}\langle z, h \rangle \leq 1.$$

Balancedness of each summand then implies

$$\|h\|_{\infty, I} \leq \sigma^{-1}, \quad |\langle u, h \rangle| \leq \sigma^{-1} \|u\|_{\mathcal{U}(I)} \quad (u \in L^2(I)), \quad \sup_{\varphi \in \mathcal{A}} |\langle h, \varphi \rangle| \leq M^{-1}.$$

The first inequality follows by testing bounded simple functions in the L^1 ball; no assumption that the L^1 dual is represented by an arbitrary measure is needed. Since $\|f\|_{2,I} \leq 1$, the first strict pairing inequality gives $\|h\|_{2,I} > 1$. Testing the second bound with $u = h$ yields

$$\|h\|_{\mathcal{U}(I)} \geq \sigma \|h\|_{2,I}^2 > \sigma.$$

Thus σh is bounded by one and has $\mathcal{U}(I)$ norm greater than σ^2 . Lemma 4.9 supplies $\varphi \in \mathcal{A}$ with $|\langle h, \varphi \rangle| \geq \kappa/\sigma > M^{-1}$, a contradiction.

Therefore f belongs to the closure of $\mathcal{B}$. Choose an element of $\mathcal{B}$ within $\tau/2$ in $L^2(I)$. Its finite structured combination has coefficient sum at most M . Put the approximation error into its L^1 error term, using $\|\cdot\|_{1,I} \leq \|\cdot\|_{2,I}$. The L^1 error is then at most τ , and the uniformity error is at most $\tau/2$. All components lie in $L^2(I)$, as required. $\square$

4.7. Detection from a local progression form. We first state explicitly the form estimate used with the uniform part of the preceding decomposition.

Lemma 4.11 (A generalized von Neumann bound with one L^2 input). *For four distinct slots occupied by R, g, b, c , one has*

$$|H_I(R, g, b, c)| \leq C \|R\|_{2,I} \|g\|_{\mathcal{U}(I)} \|b\|_{\infty,I} \|c\|_{\infty,I}.$$

One also has

$$|H_I(R, g, b, c)| \leq C \|R\|_{1,I} \|g\|_{1,I} \|b\|_{\infty,I} \|c\|_{\infty,I}$$

for $g \in L^1(I)$, with the first estimate interpreted for $R, g \in L^2(I)$.

Proof. Normalize $I = [0, 1)$. Extend the kernel smoothly to a sufficiently large fixed periodic (x, t) box, supported where its real arguments lie in I . The interior support margin permits this extension without changing the kernel on its support. Its Fourier coefficient sum is bounded by a constant determined by the fixed kernel bounds. Thus it suffices to prove the first estimate for each character-weighted progression average on the fixed circle.

Let R occupy slot j , g slot i , and the two bounded inputs slots k, ℓ . Disintegrate Haar measure along $y_j = x + jt$ and apply Cauchy–Schwarz in that variable. The first cost is $\|R\|_2^2$, and translation along the free direction $(-j, 1)$ introduces an independent Haar difference parameter h_1 in every remaining factor. Next disintegrate along y_k , keeping h_1 on the base. The repeated b factors are unchanged under translations in $(-k, 1)$; Cauchy–Schwarz therefore removes them at cost $\|b\|_\infty^4$ and introduces h_2 . The corresponding step at slot ℓ removes the four repeated c factors at cost $\|c\|_\infty^8$ and introduces h_3 . After these three steps the eighth power of the original average is bounded by

$$\|R\|_2^8 \|b\|_\infty^8 \|c\|_\infty^8 \int \prod_{\omega \in \{0,1\}^3} C^{|\omega|} g(y_i + \omega_1(i-j)h_1 + \omega_2(i-k)h_2 + \omega_3(i-\ell)h_3).$$

Every coefficient $i-j, i-k, i-\ell$ is a nonzero integer. Multiplication by it preserves Haar probability measure, and the three parameters remain independent. The last integral is exactly $\|g\|_{\mathcal{U}}^8$.

A Fourier character of (x, t) may be distributed among slots $0, 1$, since $x = y_0$, $t = y_1 - y_0$. Multiplication by a circle character preserves the L^2 and supremum norms and the U^3 norm, so the same estimate applies to every Fourier term. The argument starts with bounded inputs; the mixed-cube estimate in Lemma 4.8 and L^2 approximation extend it to the stated class. Summing the kernel’s absolutely convergent Fourier series proves the first bound.

For the second bound, use the two distinct coordinates occupied by R and g . Their Jacobian is a fixed nonzero integer. Bound the other two inputs and the kernel in supremum norm. The double integral then factors into the two L^1 norms, giving the assertion after normalization. $\square$

Lemma 4.12 (Local detection). *Fix $q > 2$ and $C_1 \geq 1$. Suppose that R, h, u, v occupy the four slots in any order and satisfy*

$$\|R\|_{2,I} \leq C_1, \quad \|h\|_{q,I} \leq 1, \quad \|u\|_{2,I}, \|v\|_{2,I} \leq 1, \quad |H_I(R, h, u, v)| > \delta,$$

where $0 < \delta < 1/2$. Then R has a normalized moment against a bounded atom φ of magnitude at least D^{-1} , where the complexity of φ is at most D and

$$D \leq \exp\left(C_q(1 + \log C_1 + \log(1/\delta))^C\right).$$

The absolute exponent C is independent of q . The atom can be conjugated if the desired moment convention is bilinear rather than sesquilinear.

Proof. We keep track of the finite sequence of tolerances in the argument. Write $q' = q/(q-1) < 2$, and truncate each opposing input at absolute value $B \geq 1$, by putting its tail equal to zero. For the h tail,

$$\|h \mathbf{1}_{\{|h|>B\}}\|_{2,I} \leq B^{-(q-2)/2}.$$

For the u, v tails, their normalized L^2 bounds give

$$\|u \mathbf{1}_{\{|u|>B\}}\|_{q',I}, \|v \mathbf{1}_{\{|v|>B\}}\|_{q',I} \leq B^{-(q-2)/q}.$$

Use the local absolute form estimate with exponents $(2, 2, 2, 2)$ for the first tail and $(2, q, q', 2)$ for either of the other tails. In each case the reciprocal exponent sum is at most two. It follows, by replacing the inputs successively, that the total clipping error is at most $CC_1 B^{-(q-2)/q}$. Choose

$$B = \max\left(2, (100CC_1/\delta)^{q/(q-2)}\right).$$

The clipped form has magnitude at least $3\delta/4$. All clipped inputs have supremum norm at most B , and retain their original L^q or L^2 bounds.

Apply Lemma 4.10 to the clipped h/B with

$$\tau_1 = \frac{\delta}{100CC_1 B^3}.$$

On multiplying back by B , write its structured part as $s_1 = \sum_\nu a_\nu \varphi_\nu$, with

$$\sum_\nu |a_\nu| \leq A_1, \quad A_1 = BD_{\tau_1}.$$

Lemma 4.11 bounds the two error contributions by $CC_1 B^3 \tau_1$ each: the other two opposing inputs are bounded by B , and $\|R\|_{1,I} \leq C_1$. The resulting form with s_1 in this slot still has magnitude greater than $\delta/2$.

Next apply Lemma 4.10 to the clipped u/B with

$$\tau_2 = \frac{\delta}{100CC_1 A_1 B^2}.$$

Its structured part $s_2 = \sum_\mu b_\mu \psi_\mu$ has coefficient sum

$$\sum_\mu |b_\mu| \leq A_2, \quad A_2 = BD_{\tau_2}.$$

Sum the error estimates separately over the atoms of s_1 . The two new error contributions are each at most $CC_1 A_1 B^2 \tau_2$. Therefore, denoting the remaining clipped input by v_B ,

$$|H_I(R, s_1, s_2, v_B)| > \delta/4.$$

By the absolute coefficient bounds, some bounded atom pair φ_i, φ_k satisfies

$$|H_I(R, \varphi_i, \varphi_k, v_B)| > \delta_0, \quad \delta_0 = \frac{\delta}{4A_1 A_2}.$$

The complexities of this pair and δ_0^{-1} are quasipolynomial in C_1, δ^{-1} , with a q -dependent constant but an absolute log exponent. Notice also that $\|v_B\|_{2,I} \leq 1$.

Apply Lemma 4.5 to this pair. Let D_0 bound its absolute coefficient sum and its shell constants. For each term without its coefficient the absolute local form is at most CC_1 , since its three atom factors are bounded and the two free inputs have the indicated L^2 sizes. Discard shells above T_0 , with T_0 chosen so that $CC_1 D_0 T_0^{-100} < \delta_0/4$. The retained terms have quasipolynomial complexity. At least one such term has form magnitude at least $\delta_1 = \delta_0/(2D_0)$. The first part of Lemma 4.6 forces its Fourier supremum A_r to be at least $\delta_1/(CC_1)$.

Use the second part of that lemma with a uniform kernel error at most $\delta_1/(10CC_1)$. Pair independence bounds the resulting form error by $\delta_1/10$. The surviving form is a finite sum

$$\sum_\nu c_\nu \left(\int_I R(y) \phi_{\nu,a}(y) dy \right) \left(\int_I v_B(y) \phi_{\nu,b}(y) dy \right),$$

with a quasipolynomial bound D_1 for $\sum_\nu |c_\nu|$ and for all atom complexities. Fixed coordinate Jacobians have been included in the coefficients. Every second moment has magnitude at most one,

by the $L^2(I)$ bound on v_B and the boundedness of its atom. Consequently one of the first moments has magnitude at least $\delta_1/(2D_1)$. Conjugating its atom if needed gives the stated normalized moment.

There are exactly two bounded-input decompositions, followed by the progression separation and the single-scale separation. Thus all budgets above involve a fixed number of quasipolynomial compositions. The only q -dependent exponent used before these compositions is the power defining B , whose logarithm is a q -dependent constant times $1 + \log C_1 + \log(1/\delta)$. This proves the final quantitative assertion. $\square$

5. TWO ACCURACIES, DELAYED PROJECTIONS, AND THE BASE REDUCTION

The aim of this section is the active-cell minimum in Equation (11). We first show that two fixed structured arguments give an absolutely summable scale approximation, and use its finite multipliers as predictor columns. Their moments justify evaluating the other projections on a finer partition. Staggered approximation accuracies then make every full pair small; the delayed projections split into current projections and rolling differences with square-energy bounds. Estimating the term with two rolling differences and removing inactive scales leaves the stated minimum.

Throughout this section, the four argument positions are distinct elements of $\{0, 1, 2, 3\}$. We work on finite dyadic trees; an inherited projection family is extended past the last decision by restriction and local refitting of its existing columns. This convention is useful when a projection evaluated on a descendant partition occurs in a form associated with an ancestor interval. No new column is required for such an extension. All functions carrying private coordinates are interpreted in the augmented Hilbert space from Section 3; the form H_I uses only their physical coordinates.

Here and below, a polynomial has an absolute degree. Its coefficients may depend on the fixed $q > 2$ and on the fixed kernel class. In particular, a bound $\exp(k^{O(1)})$ permits these dependencies in its constant and in the lower threshold for k , but its exponent is chosen independently of q .

5.1. Absolute scale sums with two fixed structured arguments. We first formulate precisely the class of inputs allowed in the two remaining positions. Fix a dyadic interval J and $F \in L^2(J; \mathcal{H})$. An admissible sequence on a collection of descendants of J is one of the following:

- (1) the fixed input F ;
- (2) $P_I F$, where P_I belongs to a nested, cell-local column projection family with path count, column size, and inverse private weight bounded by D_h ;
- (3) a sequence Z_I , supported on I , for which

$$\sum_{I \subseteq J} |I| \|Z_I\|_{2,I}^2 \leq L^2 \|1_J F\|_2^2.$$

A sum of a fixed number of such sequences is also admissible. The number of summands is absorbed into absolute constants. Here the index I may range over an arbitrary subcollection; projections on the missing cells are completed by inheritance. The estimates below remain valid when the projection lists are enlarged: the additional dependence is through $\log(2D_h)$, rather than a power of D_h .

Fix two atoms of Lemma 4.5, in positions i and j , on their common birth subtree J . Let a, b be the other positions. One term of the separation identity, with its scalar coefficient removed, has the form

$$\psi_a(y_a) \psi_b(y_b) Q(t),$$

where the three atoms are fixed on J , have supremum norm at most one, and have complexity at most H . In accordance with the chart calculus, unrestricted fixed linear modulations do not count toward H . Put

$$A_r = \sup_{\xi \in \mathbb{R}} \left| \int_{\mathbb{R}} Q(ru) \zeta(u) e(-\xi u) du \right|,$$

where ζ is the fixed nonnegative cutoff used in the separation lemma. Enlarging fixed constants in the following bands has no effect on the assertion.

Lemma 5.1 (A scale sum for one separated term). *Suppose the inputs in positions a, b are admissible sequences associated with F_a, F_b , with parameters D_h, L . For $m \geq 0$, let $\mathcal{I}_m$ consist of intervals on which $e^{-m-1} < A_{|I|} \leq e^{-m}$. Then*

$$\begin{aligned} \sum_{I \in \mathcal{I}_m} |I| |H_I(\psi_a z_{a,I}, \psi_b z_{b,I}; Q)| \\ \leq e^{-m} \text{poly}(1 + m, \log(2H), \log(2D_h), 1 + L) \|\mathbf{1}_J F_a\|_2 \|\mathbf{1}_J F_b\|_2. \end{aligned} \quad (5)$$

The notation on the left denotes the two-input integral whose kernel is $w_I(x, t)Q(t)$, with the displayed multipliers in their respective positions. All intervals are contained in J . If $A_{|I|} = 0$, the corresponding two-input integral vanishes.

Proof. Set $\mathcal{K} = 1 + m + \log(2H)$. Throughout this proof, $D_1, D_2, \dots$ denote quantities with logarithms bounded by a polynomial in $\mathcal{K}$. The single-scale part of Lemma 4.6 gives the bilinear bound CA_r before any expansion. When $A_r > 0$, Lemmas 4.3 and 4.4 give an expansion into terms

$$e(\lambda_* t^2 + \gamma t), \quad |\lambda_*| r^2 \leq D_1.$$

The possible unscaled λ_* and γ are fixed rational-height evaluations of the original parameters. In particular, they do not change their underlying parameter vector when r changes.

We first justify a finite truncation in a norm appropriate for a scale sum. Use the pooled tail conclusion of Lemma 4.4 to make the sum of the worst-case absolute tail coefficients, over all the effective quadratic and linear choices, smaller than $e^{-2m}(1 + D_1)^{-2}$, with any additional fixed polynomial reserve needed below. For each fixed tail pair (λ_*, γ) , write $e(\lambda_* t^2) = 1 + (e(\lambda_* t^2) - 1)$. The linear part is estimated by the mean-zero square functions proved below, with a fixed modulation γ . The second part has smooth-kernel norm at most $C|\lambda_*| r^2 \text{poly}(D_1)$; increasing the tail reserve handles this polynomial. At a fixed λ_* ,

$$\sum_{r: |\lambda_*| r^2 \leq D_1} |\lambda_*| r^2 \leq CD_1.$$

At each depth, contractions and Cauchy–Schwarz give the requisite product of ordinary L^2 norms; for an energy sequence the same argument costs at most $1 + L$. Thus the tail is acceptable in the absolute scale sum, not merely at an individual interval.

There is now a finite main catalog, with absolute coefficient norm and rational heights bounded by D_2 . Choose a sufficiently small $\epsilon > 0$ after that catalog, with $\log(1/\epsilon)$ polynomial in $\mathcal{K}$. Mark as exceptional every scale where a possible nonzero $|\lambda_*| r^2$, or a possible nonzero $r|\gamma - \gamma'|$, belongs to a prescribed intermediate annulus between ϵ and ϵ^{-1} . Enlarge the annulus by fixed powers of the catalog bounds when necessary. The height-bin observation from Section 3 bounds the number of exceptional scales by a polynomial in $\mathcal{K}$. At these scales use the original single-scale bound $CA_r \leq Ce^{-m}$, before summing coefficients. This gives the desired estimate for the exceptional scales.

At each remaining scale the effective quadratic is negligible, and the linear centers partition into clumps of diameter at most ϵ/r , separated by more than ϵ^{-1}/r . One way to arrange the stated diameter is to use a smaller tiny threshold than ϵ by the main catalog count, and then take connected components of the tiny-distance relation. The enlarged intermediate annulus excludes all intermediate distances. Replace every effective quadratic by zero and every center in a clump by one representative. The height-bin observation gives

$$\sum_r \max\{r^2 |\lambda_*| : r^2 |\lambda_*| \text{ is tiny at } r\} + \sum_r \max\{r|\gamma - \gamma'| : r|\gamma - \gamma'| \text{ is tiny at } r\} \leq \epsilon \text{poly}(\mathcal{K}),$$

after readjusting ϵ by catalog factors. Multiplication by the absolute coefficient norm therefore makes the total replacement error smaller than the right side of (5).

Every combined clump coefficient is $O(e^{-m})$. Indeed, test the original expansion of $Q(r \cdot)$ against $\zeta(u)e(-r\gamma_{\mathcal{C}}u)$, where $\gamma_{\mathcal{C}}$ is the chosen representative. The coefficient of its own clump is multiplied by the nonzero constant $\int \zeta$. All other clumps give rapidly decreasing Fourier tails, and the quadratic, truncation, and replacement errors have already been made smaller than e^{-m} by the choice of ϵ and the tail reserve. The defining Fourier supremum is at most e^{-m} .

For completeness, we prove the square-function estimate used for these linear clumps. Make the fixed change of coordinates $(x, t) \mapsto (y_a, y_b)$. In these coordinates, w_I has zero integral in either variable while the other is fixed. Its double primitive has compact support in an interior rectangle of $I \times I$. Expand that primitive in a smooth Fourier series on a fixed larger rectangle, multiply by interior cutoffs equal to one on its support, and differentiate once in each variable. This expresses the kernel as a sum of tensor products of smooth, interior, mean-zero bumps. Their coefficients decrease faster than every fixed polynomial in the tensor indices, while their fixed-order derivative bounds grow at most polynomially. It is consequently enough to consider rows

$$\mathcal{W}_{I,\nu}F = |I|^{-1/2} \int_I e(\nu y) \omega_I(y) F(y) dy, \quad \nu = \pm \gamma/(b-a),$$

where ω_I is such a bump. The fixed bounded multiplier ψ_a or ψ_b can be incorporated into each row.

These rows are Bessel with squared bound polynomial in $\mathcal{K}$, jointly over the retained scales, intervals, and clump representatives. Here are explicit frequency estimates. Decompose the input for a row into the demodulated shells $|\xi + \nu| \asymp 2^\ell/|I|$. At fixed scale and center, mean zero and the L^2 Poincaré inequality give row-operator norm $O(2^\ell)$ when $\ell < 0$. When $\ell \geq 0$, two frequency antiderivatives and integration by parts give $O(2^{-2\ell})$. These estimates are summed over the disjoint intervals at that scale. For a fixed Fourier variable ξ and shell index ℓ , the height-bin observation, applied to the original horizontal parameters, the fixed modulation, and ξ , bounds the number of possible scales by $\text{poly}(\mathcal{K})$. At one scale, separation of the centers bounds their shell multiplicity by $O(1 + 2^\ell)$. Thus the total frequency multiplicity is $\text{poly}(\mathcal{K})(1 + 2^\ell)$. Taking its square root and summing the preceding row norms over ℓ proves the Bessel estimate. Exact center frequencies form a null set and cause no additional term. A fixed bounded multiplier preserves the estimate.

For a projection sequence, apply Lemma 3.4 to these rows. Its cost is polynomial in $\log(2D_h)$. For an energy sequence, use instead the per-interval Bessel estimate over separated centers and then sum its given energy bound; no composition lemma is needed. Cauchy–Schwarz over the two sets of rows, together with the $O(e^{-m})$ bound for each combined coefficient, proves (5). If $A_r = 0$, the cutoff Fourier transform is zero, so Q vanishes on the part of the t -axis relevant to the kernel. $\square$

Corollary 5.2 (A finite predictor list). *Fix the two low atoms on J , with complexity at most H_{low} . For every $B \geq 1$, their product in the form can be replaced, with absolute length-weighted scale-sum error at most*

$$e^{-B} \text{poly}(\log(2D_h), 1 + L) \|\mathbf{1}_J F_a\|_2 \|\mathbf{1}_J F_b\|_2,$$

by a finite list of terms

$$\psi_a(y_a) \psi_b(y_b) w_I(x, t) e(\lambda_* t^2), \quad |\lambda_*| |I|^2 \leq D_{\text{pred}}. \quad (6)$$

The scalar coefficients may depend on the scale. The list of triples $(\psi_a, \psi_b, \lambda_*)$ is fixed on J . Its size, the absolute coefficient bounds, and all atom complexities satisfy

$$\log D_{\text{pred}} \leq \text{poly}(1 + B, \log(2H_{\text{low}})).$$

The multipliers ψ_a, ψ_b may be normalized to have supremum norm at most one by incorporating their sizes into the coefficients.

Proof. Expand the low pair by Lemma 4.5. Its dyadic shell coefficients have the stated rapidly decreasing absolute sums. Apply Lemma 5.1 to each shell and sum its Fourier-supremum bands. First discard sufficiently high separation shells; then, for each retained term, discard sufficiently small A_r bands. The shell and band thresholds have logarithms polynomial in $1 + B + \log(2H_{\text{low}})$. The remaining scales admit the finite chart expansion from Lemma 4.4. Truncate its pooled tails in the absolute scale-sum norm exactly as in the proof of Lemma 5.1. Distribute the fixed modulation $e(\gamma t)$ between positions a, b . The surviving rational-height choices of quadratic parts and linear shifts form a fixed finite list. All of its bounds have the asserted logarithmic size. The discarded errors have the required sum by (5). $\square$

5.2. Predictor moments on a finer partition. A predictor for the multiplier ψ_a is the augmented column

$$\Phi_p = (\overline{\psi_a}, \sqrt{\epsilon} \mathbf{e}_p), \quad 0 < \epsilon \leq 1.$$

The conjugation is chosen so that the physical residual moment is bilinear against ψ_a .

Lemma 5.3 (Delayed predictor replacement). *Consider one fixed term of (6), born on J . Suppose the nested predictor family in position a contains Φ_p on every use interval and on all its descendants. Other columns are allowed, subject to the projection budget D_h . Let the other position b contain an admissible sequence. For a use interval I of depth s , replace F_a by the projection of F_a on the partition at depth $s + K_0$, where $K_0 \geq 0$ is an integer. The absolute length-weighted error, summed over any collection of uses, is at most*

$$C(2^{-K_0} + (1 + K_0)\sqrt{\epsilon}) \text{poly}(D_{\text{pred}}, \log(2D_h), 1 + L) \|\mathbf{1}_J F_a\|_2 \|\mathbf{1}_J F_b\|_2. \quad (7)$$

The assertion also permits sequential replacement in positions a, b .

Proof. The predictor identity expresses the multiplier-weighted mean of the physical residual as ϵ times the predictor's column coefficient. Smooth-bump approximation below the delayed partition supplies the 2^{-K_0} term, while subtracting the coarse mean and summing private-coordinate increments gives the $(1 + K_0)\sqrt{\epsilon}$ term.

Let E_h denote conditional expectation on the depth- h partition of J , and write P_h for the direct sum of its local projections. On descendants where the predictor is present, put

$$D_h = (f_a - (P_h F_a)_{\text{phys}}) \psi_a, \quad \mu_h = E_h D_h.$$

If $a_{h,p}$ is the predictor coefficient on a depth- h cell, orthogonality to Φ_p gives

$$\mu_h = \epsilon a_{h,p}.$$

Consequently $\mu_{h+1} - \mu_h$ is $\sqrt{\epsilon}$ times the change of the p -th private coordinate of $P_h F_a$. This identity continues to hold after arbitrary further column insertions.

For a smooth normalized bump ω_I on an interval of depth s , set $H = s + K_0$. The residual moment has the exact telescoping representation

$$\begin{aligned} \int_I D_H \omega_I &= \int_I \mu_H \omega_I \\ &+ \sum_{h \geq H} \int_I (D_h - D_{h+1})(\omega_I - E_h \omega_I) \\ &+ \sum_{h \geq H} \int_I (\mu_{h+1} - \mu_h)(E_{h+1} \omega_I - E_h \omega_I). \end{aligned}$$

To verify it, telescope $\int_I D_h(\omega_I - E_h \omega_I)$ and use $E_{h+1} D_{h+1} = \mu_{h+1}$; the term involving μ_h and $E_{h+1} \omega_I - E_h \omega_I$ vanishes by conditional expectation. The final remainder tends to zero because D_h has uniformly bounded $L^2(I)$ norm and the smooth bump's conditional-expectation error tends to zero. Thus no convergence of $P_h F_a$ to F_a is being assumed.

The smoothness of the bump gives a factor $O(2^{s-h})$ in each of the two infinite sums. Its other factor is a localized orthogonal projection increment, with an additional $\sqrt{\epsilon}$ for the private-coordinate term. At a fixed depth offset $h-s$, sum squared row norms over intervals of depth s , then sum over s using orthogonality of the increments $P_{h+1} - P_h$. Minkowski's inequality over the offsets gives an aggregate row norm

$$C2^{-K_0} \|\mathbf{1}_J F_a\|_2.$$

This estimate uses only the bounded multiplier ψ_a , and is unaffected by the size of the enlarged column catalog.

If ω_I has mean zero, the first term can be written with $\mu_H - \mu_s$. Decomposing it into the K_0 successive private increments gives aggregate row norm at most

$$CK_0\sqrt{\epsilon} \|\mathbf{1}_J F_a\|_2.$$

Apply the mean-zero tensor decomposition from the preceding proof to w_I , including the fixed distributed modulation in the fixed multipliers. Pair the resulting error rows in position a with the Bessel rows in position b . Compression handles a current projection in that position; an energy sequence is handled by its per-interval Bessel estimate. This proves (7) for the part of the kernel with $e(\lambda_* t^2)$ replaced by one.

For the difference kernel $w_I(e(\lambda_* t^2) - 1)$, an ordinary smooth tensor expansion suffices. Its derivative costs contain the factor $|\lambda_*| r^2 \text{poly}(D_{\text{pred}})$. The same telescoping argument estimates its expectation-free part. For the initial μ_H term, which need not be tested against a mean-zero bump, use the per-depth bound $C\sqrt{\epsilon} \|\mathbf{1}_J F_a\|_2$ and sum $|\lambda_*| r^2$ geometrically over the scales satisfying $|\lambda_*| r^2 \leq D_{\text{pred}}$. This accounts for the additional $\sqrt{\epsilon}$ term in (7).

For sequential replacement, write a previously delayed projection as its current projection plus a rolling projection difference. The latter has a square-energy bound with $L \leq C(1 + K_0)$, by telescoping its orthogonal increments. It is therefore an admissible opponent in the second application of the argument. $\square$

5.3. Adaptive approximants at staggered accuracies. For a physical residual r in position j , let $\mathfrak{d}_{j,I}(r)$ be the supremum of $|H_I|$ when that position is occupied by r and the other positions contain one normalized $L^q(I)$ input and two normalized $L^2(I)$ inputs. The supremum also ranges over which opposing position receives the L^q input.

Lemma 5.4 (Inherited adaptive approximation). *Suppose $\|f_j\|_{q,I} \leq 1$ throughout a finite normalized subtree. For every sufficiently large k , one can construct, on segments whose roots have bounded total length, local projections $g_I = P_I F_j$ such that*

$$\mathfrak{d}_{j,I}(f_j - (g_I)_{\text{phys}}) \leq e^{-k}, \quad \|g_I\|_{2,I} \leq 1.$$

Within each segment the columns are inherited and the projections are nested in depth. Column path counts, atom complexities, and the physical supremum bounds of g_I are at most $\exp(k^{O(1)})$. Fresh columns may all have private weight one. The polynomial degrees are independent of q .

Proof. Start with no columns at a segment root J and proceed in depth order. At a cell whose residual fails the desired inequality, Lemma 4.12 supplies a bounded atom φ with physical residual correlation at least ρ , where $\log(1/\rho) \leq k^{O(1)}$ and the atom complexity obeys the same logarithmic bound. Adjoin $(\varphi, \mathbf{e}_v)$ with a fresh private coordinate and refit. The old residual has zero component in this fresh direction. The squared projection energy therefore increases by at least $c\rho^2|I|$. Repeated fits at one cell and refining fits at successive depths form one nested sequence of subspaces in $L^2(J)$, so their total energy gain is at most $\|\mathbf{1}_J F_j\|_2^2 \leq |J|$.

Stop at the first cells where the cumulative number of insertions along a path would exceed $C\rho^{-2}$. Integrating the insertion count over the disjoint first-excess cells bounds their total length by C^{-1} times a fixed multiple of $|J|$. Choose C so this is at most $|J|/4$, and restart the same

procedure on those roots. Iteration gives a geometric bound for the sum of segment-root lengths. On every retained cell the greedy fit has terminated with the required residual estimate.

With private weight one, the coefficient ℓ^2 norm is bounded by the normalized projection norm. A catalog of at most $\exp(k^{O(1)})$ bounded scalar columns consequently gives the claimed physical supremum bound by Cauchy–Schwarz. Complexity and count bounds follow from detection and the insertion cap. The uniform polynomial-degree assertion is inherited from Lemma 4.12. $\square$

Use levels $t = 1, 2, \dots$, updating positions in a fixed cyclic order. Choose an absolute $A > 1$ and large initial k_1 , and set $k_{t+1} = k_t^A$. Apply Lemma 5.4 at level t with accuracy e^{-k_t} ; denote its projection in the updated position by $g_{t,I}$. Put $g_{t-4,I} = 0$ when the position has not yet been updated. The columns for a fixed level are inherited except at the sparse resets just constructed. When finitely many levels are under discussion, intersect their segments. For a fixed bounded number of levels this common refinement still has bounded total root length: every new common root is a root of one of the constituent segment families, or the original root.

Lemma 5.5 (The two-earliest-increments expansion). *Take the two earliest increment levels $u < t$ in distinct positions i, j , and let a, b be the complementary positions. Define*

$$X_i = g_u - g_{u-4}, \quad X_j = g_t - g_{t-4}, \quad X_a = F_a - g_{a,\text{prev}}, \quad X_b = F_b - g_{b,\text{prev}}. \quad (8)$$

Here $g_{e,\text{prev}}$ is the last update in position e before level t , and is zero if no such update exists. For a finite collection of intervals, the original form is the limit of the sum of the corresponding tuple forms $H_I(X)$ as the update cutoff tends to infinity. If A is sufficiently large, there is an absolute $\theta_0 > 0$ such that, whenever $t \geq 12$ and $k = k_t$, every pair of full arguments in (8) has bilinear form norm at most $e^{-2k^{\theta_0}}$ against two normalized $L^2(I)$ inputs. All four arguments have bounded normalized $L^2(I)$ size.

Proof. For a finite terminal level, decompose each actual input into its increments up to that level and its terminal residual. In every term having at least two increments, record the earliest two levels $u < t$. Summing all later choices in either other position gives exactly the input minus its last update before t . This yields (8) for all such terms. The terms with at most one increment consist of either four terminal residuals, or one whole terminal projection and three terminal residuals.

We record the comparison that controls these remainders and the claimed pair estimates. If $h > \ell$ are updates in distinct positions, replacing their pair of projections by the corresponding actual inputs has form-norm error at most

$$C_q(e^{-k_h} \exp(C_q k_\ell^d) + e^{-k_\ell})$$

for an absolute d . Replace the more accurate projection first, using the physical supremum bound of the other as its opposing L^q norm, and then replace the less accurate projection against the actual L^q input. Choose $A > d$, with fixed reserve. The same reasoning applies when a projection is already replaced by an actual input. In particular, a pair of terminal residuals has form norm tending to zero. Their two remaining opponents have bounded L^2 size, proving the assertion about the terminal remainder on a finite interval collection.

For the tuple estimate, first suppose $u < t - 4$. Whenever the early argument in position i is paired with a later difference or residual, compare that later argument directly with its actual input against each of the two earlier projections. The actual terms cancel in the later difference or residual. Its smallest accuracy index is at least $t - 4$, and exceeds the relevant earlier indices; the stagger therefore defeats their supremum bounds. Pairs not involving this early position use only indices at least $t - 4$, apart from the preceding projection in position j , whose index is exactly $t - 4$.

If instead $u \geq t - 4$, distinct positions force $u \geq t - 3$; all nonzero projections in the tuple then have indices at least $t - 7$. Expand a selected pair into projection/actual pairs and use the preceding comparison. At least one selected full argument has coefficient sum zero after replacing projections by their actual inputs, so the two-actual terms cancel. The same argument covers the

pairs in the first case that do not contain the early position. Each error is bounded by a fixed multiple of $\exp(-c_q k_{t-7})$ for large starting log. Since $k_{t-7} = k_t^{A^{-7}}$, decreasing a fixed exponent below A^{-7} gives the stated $e^{-2k^{\theta_0}}$ bound. The L^2 size assertion follows from contraction and the triangle inequality. $\square$

5.4. Delayed high fits and their coarse stopping cells. Fix one tuple and a common segment for its older lists. The two positions i, j in (8) are called low only to specify which arguments will be expanded into their original columns. The later construction does not expand all four arguments coefficientwise.

Proposition 5.6 (Truncating the two high positions). *For the tuple above, put $k = k_t$. There are an integer $K_0 \leq k^{O(1)}$, further bounded-packing segments, and nested projection families $P_a^{\text{new}}, P_b^{\text{new}}$ on those segments, for which*

$$Y_i = X_i, \quad Y_j = X_j, \quad Y_e = P_{e,s+K_0}^{\text{new}} F_e - g_{e,\text{prev}} \quad (e = a, b), \quad s = \text{depth}(I). \quad (9)$$

The absolute length-weighted replacement error in H_I , summed over the original normalized root, is at most Ce^{-k} times its length. Path counts, inverse private weights, atom complexities, and the physical supremum bounds on coarse evaluation cells are all at most $\exp(k^{O(1)})$. For $t \geq 12$, after decreasing an absolute $\theta > 0$ if necessary, every pair of full Y arguments has form norm at most $\eta = e^{-k^\theta}$ against normalized L^2 opponents. All Y arguments have bounded normalized L^2 size.

Proof. We give the choices in their dependency order. On a common older segment, the low columns, their coefficients, and their path counts have a bound $D_{\text{old}} = \exp(k^{O(1)})$. Apply Corollary 5.2 to every available pair of low labels, using a precision log B that is a sufficiently large fixed polynomial in k . Insert the conjugate predictor columns in positions a, b at the first joint availability of that pair, and retain them thereafter. Pairs inherited at the segment root have their predictors present from that root. Use private weight $\sqrt{\epsilon}$ for these predictors.

Choose K_0 and $\log(1/\epsilon)$ to be sufficiently large fixed polynomials in k , after D_{pred} and the original low coefficient/count bounds have been fixed. The choices make both the error of Corollary 5.2 and the error of Lemma 5.3 smaller than e^{-2k} after all original-label sums. Only a polynomial in the logarithm of a subsequently enlarged high catalog occurs in these errors. Thus the choices remain valid under any later fixed-polynomial growth of that log catalog bound, by increasing the starting k .

Here is the summation over low labels in detail. Expand the two low projections into their original labels only in this application. A fixed pair is available on the subtree rooted at its deeper birth cell, or on the segment root if it was inherited. Its coefficients on a use cell are constant and bounded by a fixed power of D_{old} . The number of such pair-birth roots on a path is also bounded by a fixed power of D_{old} . Consequently, on a normalized segment root R ,

$$\sum_{J \text{ pair-birth root}} \|\mathbf{1}_J F_a\|_2 \|\mathbf{1}_J F_b\|_2 \leq D_{\text{old}}^C |R|.$$

Indeed Cauchy–Schwarz bounds this sum by the product of the square roots of the corresponding two sums of squared masses, and the path multiplicity bounds each by $D_{\text{old}}^C \|\mathbf{1}_R F_e\|_2^2$. Apply Corollary 5.2 on both sides of each replacement, and then Lemma 5.3 to each retained term. A previously delayed opposing projection is its current projection plus a rolling square-energy sequence with $L = O(1 + K_0)$. This proves the absolute-sum error assertion, even after the extra fits now described.

The predictors alone need not give the pair estimates in the statement. We therefore add tuple-specific accurate fits. Proceed chronologically in projection depth h . At depth h , first admit any predetermined predictor columns becoming available there. A coarse decision cell at that step

has depth $s = h - K_0$. Test the physical residual

$$f_e - (P_{e,h}^{\text{new}} F_e)_{\text{phys}}$$

on that coarse cell in the norm $\mathfrak{d}_{e,I}$. The normalized $L^2(I)$ size of this residual is bounded, because $P_{e,h}^{\text{new}}$ is a contraction on the partition of I into its depth- h descendants. Require a precision whose log is a sufficiently large fixed power of k . On failure, detection gives a bounded scalar column with a coarse residual moment at least ρ' , where $\log(1/\rho') \leq k^{O(1)}$. Insert one new common label with private weight one on all the depth- h subcells of I , and retain that label on their descendants.

The joint gain from this insertion is at least $c(\rho')^2|I|$. To see this, let m_Q be the normalized residual moment on a depth- h subcell $Q \subset I$. The fresh private direction is absent from every old tested residual, so the gain on Q is at least $c|Q||m_Q|^2$. Cauchy–Schwarz and the coarse moment bound give

$$\sum_{Q \subset I, \text{depth}(Q)=h} c|Q||m_Q|^2 \geq c|I| \left| \int_I \text{residual } \bar{\varphi} \right|^2 \geq c(\rho')^2|I|.$$

All ancestor extra requests affecting a coarse cell are common to its fine partition. Cap their cumulative number along a coarse path at a sufficiently large multiple of $(\rho')^{-2}$. First excess coarse cells have small total relative length, by the joint gains and the orthogonal energy bound for the entire chronological iteration. Send their coarse summands and descendants to fresh attempts. Predictor admissions have their separate path cap from the older lists and the finite predictor catalogs.

There is no conflict between a coarse cut and a delayed projection already used above it. If a coarse cell is cut when projection depth is h , all its strict ancestors have already had their delayed evaluations computed, since their projection depths are smaller than h . Continue the stopped estimator on the cut subtree by inheritance and refitting, with its predictor moments retained; its later residual tests are no longer needed for the summands sent to the new attempt. This extension supplies all finer projections needed in the telescoping proof of Lemma 5.3. A fresh attempt for the descendant coarse summands may use a new extra-label catalog. Intersect the resulting coarse segment families between the two high positions and the older lists. Their roots still have bounded total length, and each resulting piece uses restrictions of genuine nested projection sequences, including their extensions beyond its boundaries.

The accuracy order is as follows. Choose the first new high precision to dominate all previous projection and predictor costs. Detection fixes the first new catalog bound. Choose the second new high precision to dominate that first catalog's physical supremum bound as well. For a fine subcell $Q \subset I$ at depth $s + K_0$,

$$\|F_e\|_{2,Q} \leq 2^{K_0/2} \|F_e\|_{2,I}.$$

The private-coordinate coefficient estimate thus bounds the first new projection's physical supremum on I by 2^{K_0} times a fixed power of its own count and conditioning bound. It does not involve the next high precision. The two choices are therefore successive fixed-polynomial log choices.

Apply the comparison argument of Lemma 5.5 to these two new approximants and the older projections. The new fits are more accurate than every older projection, and the second is more accurate than the first with sufficient reserve for its supremum bound. In each pair of full arguments in (9), the actual-input contributions cancel exactly as before. If the earliest low position has a much earlier index, the other selected full argument is a difference of later approximants and is tested directly against that earlier position. Otherwise all older indices involved are at least $t - 7$. Decreasing the absolute exponent from Lemma 5.5 yields the claimed η bound. Contraction on the fine partition and on the older coarse cells gives the local L^2 bounds. Finally, the new high fits are specific to this tuple; their larger precisions do not enter the global update recurrence $k_{t+1} = k_t^A$. $\square$

5.5. Base sides and the active collection. Work on one normalized segment root R from Proposition 5.6. Set

$$Y_l = B_l + U_l, \quad U_i = U_j = 0, \quad U_e = (P_{e,s+K_0}^{\text{new}} - P_{e,s}^{\text{new}})F_e \quad (e = a, b).$$

Thus each B_l is a difference of at most two current basic projections. These basic columns are inherited within the segment, and labels in different basic catalogs have distinct private directions. Choose D_{base} , with

$$e^k \leq D_{\text{base}} \leq \exp(k^C),$$

to dominate the base path counts, inverse private weights, complexities, and current coefficient sums. The local normalized sizes of B, U, Y are bounded. Rolling orthogonal projection differences give, also on any normalized subroot,

$$\sum_{I \subseteq R} |I| \|U_{e,I}\|_{2,I}^2 \leq C(1 + K_0)^2 |R|.$$

For example, expand the rolling difference into its K_0 orthogonal increments and sum at fixed depth offset; the triangle inequality in the resulting row space gives the displayed bound.

The term containing both possible U arguments is already summable. Its complementary two arguments are the full Y_i, Y_j , whose pair norm is at most η . Cauchy–Schwarz over intervals therefore bounds its absolute length-weighted sum by $C\eta(1 + K_0)^2 |R|$. For the finitely many leading levels the same assertion holds with the factor η omitted.

Lemma 5.7 (Few active scales for a fixed triple). *Fix three bounded basic scalar columns, in three distinct positions, on their common birth subtree. For a dyadic interval I let $\mathcal{A}_I$ be the norm of their form against a normalized $L^2(I)$ input in the fourth position. Along every spatial path, the number of scales for which $\mathcal{A}_I > \delta$ is polynomial in $\log D_{\text{base}} + \log(1/\delta)$. Moreover, for $0 < \delta \leq 1$,*

$$\sum_{I \text{ on the path: } \mathcal{A}_I \leq \delta} \mathcal{A}_I \leq \delta \text{poly}(\log D_{\text{base}}, \log(1/\delta)).$$

The bound is uniform in the locations of the path and in all unrestricted oscillation parameters.

Proof. Fix a path point y , and let the fourth position be l . Writing the transpose at $y + h$, its squared L^2 norm is an integral in the scaled variables $(h, t, t')/r$, $r = |I|$, containing the fixed triple and its conjugate copy and the two smooth kernels at the shared anchor. The six atoms combine into the multivariable chart setting of Lemma 4.3. Recenter at the fixed y . All unscaled horizontal, linear, and quadratic parameters are now fixed along the path, before multiplication by r or r^2 .

If $\mathcal{A}_I > \delta$, apply Lemma 4.3 to this integral at Fourier frequency zero, and then Lemma 4.4 with an error small compared with δ^2 . Every effective quadratic entry and every linear t -frequency in the truncated expansion is of the form $r^2 a \cdot \chi_*$ or $ra \cdot \chi_*$, respectively, for a fixed finite parameter vector χ_* . The coefficient vectors have rational height quasipolynomial in D_{base} and δ^{-1} ; the dimension is polynomial in the original chart dimension. This description also covers the choices of rational splitting.

Choose tiny/large gap thresholds after the finite truncation, with reserve for its absolute coefficient norm. Outside polynomially many intermediate scales, the height-bin observation makes every effective quadratic entry that is bounded in the Weyl expansion tiny, and every retained t -frequency either tiny or large. Remove the effective quadratic by uniform approximation. Large t -frequencies integrate to a negligible amount by smoothness of the kernel. For a tiny t -frequency, replace its oscillation by a constant; the remaining t -integral is zero by the exact marginal

$$\int w_I(y + h - lt, t) dt = 0$$

at fixed h, t' . The other kernel and the remaining linear oscillations are independent of this t variable. All errors, after coefficient summation, are smaller than δ^2 . This contradicts $\mathcal{A}_I > \delta$ and proves the count.

Apply that count to the successive bands $2^{-n-1}\delta < \mathcal{A}_I \leq 2^{-n}\delta$, using threshold $2^{-n-1}\delta$. Their total contribution is bounded by

$$\sum_{n \geq 0} 2^{-n}\delta \text{ poly}(\log D_{\text{base}} + \log(1/\delta) + n),$$

which is the asserted small-norm sum. The kernel may vary with I within its fixed class: only its uniform smoothness and exact marginal were used. $\square$

Proposition 5.8 (Reduction to the active cell minimum). *The remaining patterns, with at least three B arguments and optionally one U argument, can be restricted, with total absolute error at most $e^{-k}|R|$, to a common active collection whose path multiplicity satisfies*

$$M \leq D_{\text{base}}^C. \quad (10)$$

The finitely many leading update levels have a bound independent of the number of depths. For $t \geq 12$, it suffices on each segment and for each retained pattern z to estimate

$$\sum_{I \text{ active}} |I| \min(\eta, |H_I(z)|). \quad (11)$$

The two nonside arguments complementary to a pair of later side arguments are the whole B arguments of this tuple before further filters. Restoring them to their full Y arguments recovers the pair-smallness above. Further attempts may restart on normalized subroots with these basic arguments and their inherited lists unchanged.

Proof. Declare I active if some triple of available pure basic scalar columns in three positions has transpose norm greater than $D_{\text{base}}^{-C'}$, with C' a sufficiently large fixed constant. Along a path there are at most D_{base}^3 triples, and Lemma 5.7 gives only a logarithmic-polynomial number of active scales for each one. Enlarging the absolute power proves (10).

On an inactive interval, expand three of the B arguments into their basic columns. Their coefficient sums and the local L^2 size of the remaining argument cost only a fixed power of D_{base} . For a fixed triple, the small-norm assertion of Lemma 5.7 bounds its path sum by $D_{\text{base}}^{-C'} \text{ poly}(\log D_{\text{base}}, C')$. Sum over the pathwise triple count and integrate over path points. Choosing C' larger than all these fixed power costs, and using $D_{\text{base}} \geq e^k$, gives the stated $e^{-k}|R|$ error.

For each fixed leading level, all budgets and the multiplicity in (10) are fixed constants. Local absolute L^2 estimates and bounded sizes thus give a bound independent of the number of depths. For later levels, the pair estimate in Proposition 5.6 and the bounded sizes of the other two arguments give $|H_I(Y)| \leq C\eta$. Decrease θ once more, and increase the starting log, to absorb this fixed constant into the definition of η . Multilinearity expands $H_I(Y)$ into the retained patterns and the already estimated term with both U arguments. For nonnegative quantities, $\min(\eta, \sum_j a_j) \leq \sum_j \min(\eta, a_j)$. The absolute errors and the two- U term may therefore be taken outside the minimum, leaving (11) separately for the remaining patterns.

This reduction retains the whole coarse arguments B_l . In particular, later filter removal in two nonside positions returns exactly these arguments; substituting $Y_l = B_l + U_l$ changes the term only by terms with an additional basic side. The full complementary pair then has the pair-smallness already proved. All relevant current families remain inherited under restriction to a normalized subroot, and their rolling square estimates follow anew from orthogonal energy there. Hence a later stopping attempt can use the same base data. $\square$

A bound of the form $\eta^c(1 + C_N)|R|$, for some fixed $c > 0$, for (11) is sufficient for the normalized absorption argument. The original-label replacements have absolutely summable errors, segment

roots pack, and the number of earlier indices $u < t$ is at most t . The completion of the proof will sum the resulting positive powers of $e^{-k_t^\theta}$.

6. LOCALIZATION BY QUADRATIC CURVATURE

We work with one of the active patterns in (11), on a normalized segment with root R . Its original column lists, including their private coordinates, are fixed throughout this section. Write

$$P = \lceil \eta^{-\alpha} \rceil,$$

where the positive absolute constant α will be chosen in Section 10. All estimates below are uniform in the number of summation depths. In particular, $\log(2P) = O(k)$, whereas a factor

$$G_0 = \exp(\log^{O(1)}(2k))$$

is $P^{O(1)}$. We distinguish such factors from fixed powers of P . Further attempts can begin at normalized subroots, keeping the original basic arguments and inheriting their column lists.

The aim is the localization into current groups in Proposition 6.8, up to its stated summable error. A graph first records scale-dependent compatibility of quadratic curvatures. Decreasing colors, decreasing graph distances, and increasing vertex lists will give nested cumulative masks. The group masks also retain earlier central uses: the later square estimates need separation between distinct groups of the same color on overlapping whole cells that can still lead to an actual use. These are the usable endpoints defined below. The final transfer identity prepares the two-side estimates of Section 7; the linear component of those side states is constructed in Section 8.

6.1. Tags, an ambient graph, and short paths. Put $c = (-1, 3, -3, 1)$. A vertex is an occurrence of an original basic column, in any slot. Its *normalized curvature tag* $p_v = (\theta_v, C_v, \lambda_v)$ is defined by writing its quadratic chart phase in slot i as

$$c_i(\lambda_v y^2 + (\theta_v y)^T C_v (\theta_v y - n)).$$

Thus the curvature data of an original column in slot i are divided by c_i . A linear modulation is not part of its tag. The vertex lists $\mathcal{V}(I)$ increase on downward paths and have cardinality at most $D = D_{\text{base}}$, enlarging the original base bound if necessary. They are ordered by admission, with fixed tie breaking. Repeated tags can have different vertices and private labels.

Every additional main or helper column constructed below is attached to one of these original vertices. In slot j it has precisely that vertex's normalized tag, with phase multiplier c_j , a bounded smooth chart amplitude, and an unrestricted linear modulation. The single-column complexities remain at most $\exp(k^{O(1)})$, independently of the number of insertions. In particular, these insertions do not add vertices to the graph. The later linearized modes belong to a separate catalog and do not add graph vertices either.

At depth s , let r be the interval length. Use the ambient graph of *all* hypothetical curvature tags satisfying the fixed basic bounds, not just the currently present vertices. Identify equal tags at distance zero. Join two tags v, w when their difference data

$$V = r \begin{pmatrix} \theta_v \\ \theta_w \end{pmatrix}, \quad C = \text{diag}(C_v, -C_w), \quad \Lambda = r^2(\lambda_v - \lambda_w)$$

admit the splitting in (2), with heights and size bounds at most H_{ed} . The graph distance is denoted $d_s(v, w)$ and can be infinite. The threshold $H_{\text{ed}} = \exp(k^{O(1)})$ will be fixed after an a priori count of the main and helper catalogs. Refining one depth halves V and quarters Λ_* , without changing the rational fast space. Consequently edges persist and d_s is nonincreasing in s . These distances depend only on the tags and absolute depth, not on the location, birth cell, or availability of the vertices.

Lemma 6.1 (Separation of different tags). *Fix an upper bound $\exp(k^{O(1)})$ for the complexities of the single columns under consideration. Given $\tau_2 > 0$ with $\log(1/\tau_2) \leq k^{O(1)}$, one can choose $H_{\text{ed}} = \exp(k^{O(1)})$ so that the following statements hold. If $d_s(v, w) > 1$, the physical inner product of two same-slot pure columns of these tags, normalized on any interval of depth s , has absolute value at most τ_2 . If the two columns are in distinct slots, their four-linear form has norm at most τ_2 against two arbitrary normalized $L^2(I)$ arguments in the other slots.*

Proof. For the inner product, smooth the interval indicator with L^1 error much smaller than τ_2 and derivative bounds polynomial in τ_2^{-1} . The physical columns have bounded supremum norm. A larger inner product would therefore give a Fourier coefficient of their conjugate product above a fixed multiple of τ_2 after recentering and rescaling. Lemma 4.3 gives (2) for the difference data. The fixed factor c_i in a same-slot product only changes the allowable height by a fixed power. Choosing H_{ed} above this bound proves the first assertion.

For the second assertion, call the source slots i, j and the spectator slots a, b . In centered, rescaled spectator coordinates (z, w) , the linear parts of the source coordinates are $r(\xi_i z + \zeta_i w)$ and $r(\xi_j z + \zeta_j w)$, where

$$\xi_l = \frac{b-l}{b-a}, \quad \zeta_l = \frac{l-a}{b-a}.$$

All four coefficients for $l = i, j$ are nonzero fixed rationals. The identities $\sum_l c_l l^u = 0$ for $u = 0, 1, 2$ give

$$c_i \xi_i \zeta_i = -c_j \xi_j \zeta_j =: \kappa \neq 0.$$

First clip each spectator at a sufficiently large inverse polynomial in τ_2 . The discarded part has L^1 norm at most the inverse clipping height, since its L^2 norm is bounded. The absolute form estimate with exponents $1, 2, \infty, \infty$ bounds this error. We can therefore suppose that the spectators have both bounded L^2 norm and supremum norm at most an inverse polynomial in τ_2 .

Use the return calculation from the proof of Lemma 4.3 for the displayed difference data. Choose an integer period clearing the chart and interpolation denominators. For a small return $Vh = m + d_h$, with m divisible by this period, translate z by h and translate the source lift indices by $\xi_i m_v, \xi_j m_w$. The amplitude and kernel errors are bounded by $|h| + |d_h|$ times a quasipolynomial in the available complexity. The resulting modulation in w is

$$\kappa \Omega(h, m), \quad \Omega(h, m) = (2\Lambda + V^T C V)h - V^T (C - C^T)m.$$

Indeed the mixed quadratic contribution in each source contains $c_l \xi_l \zeta_l$; factoring out κ leaves exactly the difference of the two curvature data. The modulation in z can be included in the first spectator and has no effect on its L^2 norm.

Suppose the original correlation is larger than τ_2 . Applying the calculation to $\ell(h, m)$, for a sufficiently large quasipolynomial number of integers ℓ , produces correlations of size at least a fixed multiple of τ_2 between one fixed bounded kernel $K_1(z, w)$ and tests $f_\ell(z)g_\ell(w)$. Here the f_ℓ have bounded L^2 norms, and

$$g_\ell(w) = g(w)e(\ell \kappa \Omega(h, m)w)$$

up to a common bounded multiplier. The error remains small by choosing the return threshold sufficiently small before choosing these multiples. If $|\Omega(h, m)|$ exceeds a fixed constant, the separated exponentials, multiplied by the clipped spectator, are Bessel on the fixed box, with bound controlled by that spectator's squared supremum norm. Cauchy-Schwarz in z gives

$$\sum_\ell \left| \int K_1(z, w) f_\ell(z) g_\ell(w) dz dw \right|^2 \lesssim \int \sum_\ell \left| \int K_1(z, w) g_\ell(w) dw \right|^2 dz.$$

The right side is bounded by the same Bessel constant times $\|K_1\|_2^2$, while the left side grows with the number of returns. This is a contradiction. We have proved a uniform bound for $\Omega(h, m)$ on every return below a quasipolynomially small threshold.

The final, algebraic part of the proof of Lemma 4.3 now applies to these difference data: the alternating return identity gives an isotropic lattice space, successive minima give a rational fast projection and a bounded slow component, and one return in each coordinate direction bounds the effective quadratic form. This part uses only the just-established return bound. It yields (2) at quasipolynomial height, contradicting $d_s(v, w) > 1$ when H_{ed} is chosen sufficiently large. $\square$

Lemma 6.2 (Endpoint bounds along a short path). *Suppose two tags can be joined, at one scale, by a path of length $L \leq \text{poly}(P, k)$. Their difference data satisfy (2) in their own endpoint coordinates, with $\log H' \leq \text{poly}(P, k)$.*

Proof. List the path tags as $v_0, \dots, v_L$, and put all their horizontal coordinates in one direct sum. For each edge choose its fast space U_e from the definition of an edge. Let U be the subspace of the joint coordinate space whose restriction to each consecutive pair belongs to U_e . These are rational linear equations. Clearing denominators and applying determinant bounds to one maximal independent system shows that the orthogonal projection onto U has logarithmic height polynomial in L , the dimensions, and $\log H_{\text{ed}}$. The same determinant estimate bounds the norm of a right inverse on the range of this system.

Let V be the concatenation of the normalized horizontal vectors. Each defining equation has bounded error on V , because the edge slow component is bounded. The right-inverse bound therefore shows that V^F , the orthogonal projection of V onto U , satisfies $\|V - V^F\| \leq H_1$, where $\log H_1 \leq \text{poly}(P, k)$. On an edge, both its chosen fast component and the restriction of V^F lie in U_e , and their difference Δ is bounded by this new budget. Since U_e is isotropic for the edge skew form, replacing its fast component by this restriction changes the effective correction by

$$\text{sym}(\Delta^T C_{\text{sk},e} V_e) = \text{sym}(\Delta^T C_{\text{sk},e} V_{s,e}),$$

which is bounded at the new budget. In particular, no unbounded fast coordinate enters this error.

Sum the edge effective quadratic forms using these common joint fast and slow components. The diagonal symmetric terms, the skew mixed terms, and the original λ terms telescope, leaving only the two endpoints. Let U' be the image of U under endpoint restriction. It is rational with the same type of height bound. For two vectors of U , summing their edge skew pairings cancels all intermediate coordinates; each edge pairing is zero. Hence U' is isotropic for the endpoint difference skew form. The restricted joint fast vector lies in U' and is within the new bounded error of the endpoint V . Replacing it by the orthogonal projection of endpoint V onto U' changes it by a bounded vector of U' . The preceding isotropy calculation again bounds the change in effective curvature. These are all the assertions of (2) in endpoint coordinates. $\square$

6.2. Colors with controlled changes. The following construction is on original vertices only. The colors are ordered integers in a fixed range, and each vertex's color is nonincreasing along a downward path. A group will consist of vertices of one current color together with additional fixed winning data.

Set

$$d_0 = \lceil 1000 \log(2D) \rceil, \quad \mathcal{L} = d_0^{30}, \quad H_{\max} = 100P(\mathcal{L} + 1),$$

and choose the two distance scales

$$b_2 = 100(H_{\max} + 1), \quad b_1 = d_0^{10} b_2 (H_{\max} + 1).$$

These definitions fix the available color range before the coloring is performed.

In each of d_0 independent primary rounds, give every original vertex z a geometric delay D_z , truncated to $[0, d_0]$, with untruncated tail $\mathbb{P}(D_z \geq n) = 2^{-n}$. Arrays for different births and rounds are independent. At a present vertex v , the top score is

$$g = \max_{z \in \mathcal{V}(I)} \{D_z - \lfloor d_s(z, v)/b_1 \rfloor\}.$$

A round is *marked* when it has a unique winner z whose score exceeds every other score by at least 5. Use the first marked primary round.

Let i be the admission index of its winner and let $j \geq i$ be the current list size. If $j = i$, use an equality cohort and the prefix through i . Otherwise there is a unique smallest dyadic block of positive indices whose lower half contains i and whose upper half contains j . Throughout this upper-half cohort freeze W to be the prefix through that lower half. The winner z belongs to W . A cohort is specified by this block, and its length level has only $O(\log(2D))$ possibilities on a path.

For this primary round and frozen context define, at every absolute depth s' and for every hypothetical tag a ,

$$f_a(s') = \max_{w \in W \setminus \{z\}} (b_1 D_w - d_{s'}(w, a)).$$

An empty maximum is $-\infty$. These profiles are defined before and after the births of their arguments; availability is not used in this definition. They are nondecreasing in depth. Put

$$T_l = (g - 3)b_1 + \frac{lb_1}{2d_0}, \quad U = (g - 2)b_1, \quad 1 \leq l \leq d_0,$$

and write $C_l(x) = \min(U, \max(T_l, x))$, including the value at $-\infty$. The frozen pseudometrics are

$$d^{(l)}(a, b) = \sup_{s' \in \mathbb{Z}} |C_l(f_a(s')) - C_l(f_b(s'))|.$$

The supremum over *all absolute depths* is part of the definition. Thus these metrics are static within the frozen context, and $d^{(l+1)} \leq d^{(l)}$.

Let $R_l = 10l(d_0 + 1)b_2$. At v choose the least $l < d_0$ for which the balls with respective metrics and radii

$$B_{d^{(l)}}(v, R_l), \quad B_{d^{(l+1)}}(v, R_{l+1}),$$

counted on currently present vertices, have the same floor logarithm to base $3/2$. Call this common integer n . Such an l exists: the balls are nested as l increases, are nonempty, and their cardinalities are at most D , while d_0 exceeds the possible range of their floor logarithms.

Finally use d_0 independent high rounds of geometric delays, independent of the primary arrays. The high arrays can be shared among all frozen contexts. In the high round use the score defined by $d^{(l+1)}/b_2$ in place of d_s/b_1 and the same gap condition. Use the first marked high round, with winner a and score g_2 .

The color metadata are the primary round and score g , the equality case or separating length level, l, n , and the high round and score g_2 . At failed markings use fixed placeholders. Initialize each vertex's change counter to d_0^8 at admission. Whenever any of these data, or either winning identity, changes, decrement this counter. Order counter and metadata lexicographically, with the counter first, and embed the resulting finite ordered set into $\{1, \dots, \mathcal{L}\}$. A successful *group* additionally records z, a, W and the exact cohort block. It does not record the changing list size within an upper half or the identity of the marked central vertex.

Lemma 6.3 (Color budget and exceptional roots). *For sufficiently large starting k , the preceding construction uses at most $\mathcal{L}$ colors, with at most d_0^8 changes per vertex on a path. Delays can be chosen so that the first cells where a required primary or high marking fails have arbitrarily small fixed total length relative to R . They can be sent to fresh attempts. All statements below are on the remaining paths, where every needed marking exists.*

Proof. At a present vertex a round's top score lies in $\{0, \dots, d_0\}$: the vertex itself is a competitor of distance zero. Individual primary competitor scores only increase with depth. At a fixed top score a marked winner can occur in at most one episode, with a fixed identity. Once a rival has destroyed its gap, that rival's score cannot decrease; a different marked winner at the same top value would have the old winner as a rival of that value. Thus the first marked primary round, its score, and its winner have at most $5d_0^2$ episodes.

Within a primary episode there are at most d_0 cohorts. Within one cohort all profile metrics are fixed. The vector of floor log-sizes of all the balls has at most $2d_0^2$ epochs, including its initial value, since its coordinates increase and each has at most $O(\log D)$ possible log-sizes. Within each such epoch the selected l, n are fixed. In that epoch the high metric is fixed, so high scores increase only by admission of competitors. The same marked-episode argument gives at most $5d_0^2$ episodes of selected high data. Altogether there are at most $50d_0^7 \leq d_0^8$ changes when k is large. Seven metadata coordinates, each with at most $d_0 + 2$ states, and the change counter require fewer than $256d_0^{15} < d_0^{30}$ ordered states. Decrementing the counter at each change makes the assigned color nonincreasing even if the other metadata increase.

For completeness, a fixed untruncated score situation is marked with probability at least 2^{-5} . Choose a canonical winner in every delay outcome, resolving ties deterministically, and increase only its delay by 5. The resulting outcome has a unique winner with gap at least 5. This map is injective, because its image identifies the increased coordinate; geometric probabilities decrease by exactly 2^{-5} . Truncation changes this probability by at most $D2^{-d_0}$. In particular it remains at least $1/64$ for large k .

There are only polynomially many situations to check on a fixed spatial path. For the primary stage cap $\lfloor d_s(z, v)/b_1 \rfloor$ at $d_0 + 5$; larger values cannot affect any top or gap decision. The capped pair distances change only $O(D^2(d_0 + 1))$ times, by monotonicity. Counting the vertices to test gives $O(D^3(d_0 + 1))$ fixed primary situations. Conditional on all primary arrays, the primary, cohort, and ball epochs number at most $O(Dd_0^5)$ across vertices. Each has at most D present prefixes, so at most $O(D^2d_0^5)$ high situations require checking. These situations are fixed before revealing the independent high arrays. The probability that all d_0 rounds fail at any such situation is at most $(1 - 1/64)^{d_0}$. The union bound tends to zero with D , since $d_0 \geq 1000 \log(2D)$.

Integrate this pathwise failure probability over R . Some realization of the arrays has a small measure of failed paths. The first failure cells are disjoint, and their total length is at most that measure. This proves the claimed exceptional-root bound without a bound on the number of vertices across different spatial branches. $\square$

6.3. Historical ball masks and usable endpoints. Fix a successful group G . For an integer pad $1 \leq h \leq H_{\max}$, include a present vertex b if, for some historical central vertex v_0 assigned this same color and group at or above the current cell on its ancestor path,

$$\begin{aligned} b_1 D_z - d_s(z, b) &\geq (g - 1)b_1 - h, \\ d^{(l+1)}(b, v_0) &\leq h, \quad d^{(l)}(b, v_0) \leq 10R_l h. \end{aligned} \tag{12}$$

Denote this vertex mask by $\mathcal{M}_{G,h}$. It is empty before its first central assignment. Within the cohort it increases in depth: the score increases, the metrics are frozen, and historical witnesses are retained. After the cohort, where no further actual central use of this group occurs on that branch, retain the eligible vertex set by inheritance whenever a nested operator is needed. Births on other branches supply no historical data to the current branch.

An *actual use* of a group requires a current central vertex assigned to it. A *usable endpoint* for it is any cell with a nonempty pad mask and some actual use of that group at or below the cell. Thus a usable endpoint need not itself be central. Nonempty usable endpoints are still in their cohorts: an inherited mask after a cohort has no later actual use there.

Lemma 6.4 (Mask inclusion, closure, and separation). *The following assertions hold for every allowed pad.*

- (1) *At an actual use with current central vertex v , every present vertex b with $d_s(b, v) \leq h$ lies in $\mathcal{M}_{G,h}$.*

- (2) *At an actual use, every present graph neighbor of a vertex in $\mathcal{M}_{G,h}$ belongs to $\mathcal{M}_{G,h+1}$ if $h < H_{\max}$. Every vertex admitted to a mask has, already at admission, a graph path to z of length at most $\text{poly}(P, k)$.*
- (3) *For distinct groups of the same color, vertices belonging to their pad masks on overlapping usable endpoints are at graph distance greater than 1 at the finer endpoint.*

Proof. At a marked primary center v at depth s , each rival's true score is at most $(g - 5)b_1$, since its quantized score is at most $g - 5$. Consequently

$$f_v(s') \leq (g - 5)b_1 < T_l \quad (s' \leq s).$$

The winning true score is greater than $(g - 1)b_1$. If $d_s(b, v) \leq h$, the winner score at b loses at most h , and $f_b(s) \leq (g - 5)b_1 + h < T_l$. Both profiles are below the clipping range at every earlier depth. At all later depths the graph distance remains at most h , so their true profiles, and then their clipped profiles, differ by at most h . Thus both profile distances from b to v are at most h , which proves the first assertion with witness $v_0 = v$.

Next compare a historical central witness v_0 with the current center v of the same group. Their common high winner a has delay at most d_0 , and the marked top score g_2 is nonnegative. Hence

$$d^{(l+1)}(a, v_0), d^{(l+1)}(a, v) < (d_0 + 1)b_2.$$

The earlier low ball at v_0 , on the vertices present when it was chosen, is contained in the current high ball at v : use $d^{(l+1)} \leq d^{(l)}$ and $R_l + 2(d_0 + 1)b_2 < R_{l+1}$. This earlier low ball and the current low ball each have at least $(3/2)^n$ vertices, whereas the current high ball has fewer than $(3/2)^{n+1}$ vertices. The two low balls intersect. The triangle inequality in the low metric gives

$$d^{(l)}(v_0, v) \leq 2R_l.$$

For a pad- h vertex b it follows that $d^{(l)}(b, v) \leq A_h := 10R_l h + 2R_l$. Our constants give

$$A_h < T_{l+1} - T_l - 10 \quad (h \leq H_{\max}).$$

For example the gap on the right before subtracting 10 is $\frac{1}{2}d_0^9 b_2(H_{\max} + 1)$, whereas $A_{H_{\max}} \leq 10d_0(d_0 + 1)b_2(10H_{\max} + 2)$. Since the clipped profile of v at the use is T_l , this proves the *true* rival bound

$$f_b(s) \leq T_l + A_h < T_{l+1} - 10.$$

Let b' be a present graph neighbor of b at this use. Its winner score loses at most one. Both f_b and $f_{b'}$ are below T_{l+1} at all depths up to s , and their difference is at most one at all depths after s . Thus $d^{(l+1)}(b, b') \leq 1$. For the low metric, after s the distance from b' to v_0 costs at most $10R_l h + 1$. Before s , its clipped profile is between T_l and $T_l + A_h + 1$, while the clipped profile of v_0 is between T_l and $T_l + 2R_l$. Their difference is at most $A_h + 1 \leq 10R_l(h + 1)$. All three predicates of (12) therefore hold at pad $h + 1$. Finally, at admission the first predicate gives

$$d_s(z, b) \leq (d_0 + 1)b_1 + H_{\max},$$

which is polynomial in P, k and proves the path bound.

It remains to prove the usable-endpoint assertion. Let I, J be overlapping endpoints for distinct same-color groups G, G' , with J the finer endpoint, and choose an actual-use certificate u for G' at or below J . Let b, b' be the respective mask vertices. Suppose, for a contradiction, that $d_{\text{depth}(J)}(b, b') \leq 1$. This distance bound persists to u . The vertex b' remains in its mask at u . No future use of G on this same branch is required.

First suppose their primary winners z, z' differ. Their colors give the same primary round, score g , and equality case or separating length level. At a fixed length level, upper-half index cohorts on an increasing path either use the same dyadic block or increasing disjoint blocks. In either case the

earlier winner belongs to the later frozen prefix W' . This is immediate also for equality cohorts. The earlier winner score at b persists, so adjacency at u gives

$$f'_{b'}(u) \geq (g-1)b_1 - h - 1.$$

This contradicts the true rival bound for G' at u , which is less than $T_{l+1} - 10 \leq (g-5/2)b_1 - 10$, since b_1 is much larger than $H_{\max}$.

Now suppose $z = z'$. Its fixed admission index and the common separating level determine the same cohort block; its prefix W is inherited on the overlapping path. The metrics coincide, and distinct groups must have distinct high winners a, a' . Historical witnesses v_0, v'_0 for the two masks lie on the common ancestor path of J and can be ordered along it. At the later historical witness the earlier high winner remains available in the same high round and metric. The earlier winner's true score is greater than $(g_2-1)b_2$ at its own witness, whereas at the later witness it is at most $(g_2-5)b_2$. Therefore

$$d^{(l+1)}(v_0, v'_0) > 4b_2 > 2H_{\max} + 1.$$

On the other hand, at the certificate u the true rival bound puts $f_{b'}(u)$ below $T_{l+1} - 10$, and adjacency puts $f_b(u)$ below $T_{l+1} - 9$. Both profiles were below the high clipping range at all earlier depths, and they differ by at most one at all later depths. It follows that $d^{(l+1)}(b, b') \leq 1$. The two mask predicates then imply

$$d^{(l+1)}(v_0, v'_0) \leq h + 1 + h' \leq 2H_{\max} + 1,$$

a contradiction. This proves separation, including the case where the two endpoints have later certificates on different branches. $\square$

Lemma 6.5 (Bessel separation of usable endpoint spaces). *Fix a slot and a color. For each of finitely many groups G let Z_G be a finite sum of available main pure columns, with constant coefficients on entire usable endpoint cells whose pad masks contain their tags. Different terms may use arbitrary endpoint times, and a label may be used repeatedly. Suppose the number of groups and available column labels on a spatial path, and the inverse private weights, are bounded by $\exp(k^{O(1)})$. Choosing τ_2 sufficiently small after these bounds gives*

$$\left\| \sum_G Z_G \right\|^2 \lesssim \sum_G \|Z_G\|^2.$$

The constant has no dependence on the number of endpoint times or on the depth of a future use certificate.

Proof. For a fixed label v in G , combine all its occurrences into the scalar coefficient function

$$a_{v,G}(y) = \sum_I a_{v,G,I} \mathbf{1}_I(y).$$

The private coordinate of label v in $Z_G(y)$ is precisely $\epsilon_v^{1/2} a_{v,G}(y)$. Thus, if W_0 bounds inverse square-root private weights,

$$\sum_v |a_{v,G}(y)|^2 \leq W_0^2 \|Z_G(y)\|^2.$$

This uses the combined coefficients, so any cancellation between repeated uses of a label is retained.

Fix labels v, w in distinct groups. Their tags have a deterministic last separating depth: by monotonicity, the depths where $d_s(v, w) > 1$ form an initial interval. Cap this interval at the last depth in the finite calculation if needed, and call its endpoint s_* . If two supporting endpoint cells I, J overlap, their intersection is the finer cell. Lemma 6.4 says that its depth is at most s_* . Therefore every nonzero term in

$$a_{v,G} \overline{a_{w,G'}} = \sum_{I,J} a_{v,G,I} \overline{a_{w,G',J}} \mathbf{1}_{I \cap J}$$

is measurable on the grid at depth s_* . The whole product is measurable on that grid. Endpoint terms finer than s_* have no overlap with any term of the other label and contribute zero. If no separating depth occurs within the root, there are no overlaps at all.

Apply Lemma 6.1 on every grid cell. The two private labels cannot coincide on an overlap, since equal tags have distance zero. Hence

$$|\langle Z_G, Z_{G'} \rangle| \leq \tau_2 \sum_{v,w} \int |a_{v,G}(y) a_{w,G'}(y)| dy.$$

Pointwise Cauchy–Schwarz in groups and labels bounds the sum of all these cross terms by τ_2 times a fixed power of the pathwise count and private-weight bounds, multiplied by $\sum_G \|Z_G\|^2$. Taking τ_2 smaller than the reciprocal of this factor proves the assertion. There is no factor counting repetitions or endpoint times. $\square$

The whole-cell hypothesis of Lemma 6.5 will be used also for adjoint ranges. If an output is restricted to finer use cells, the adjoint puts that restriction on the input side of its endpoint projection. The projection then again produces constant column coefficients on the entire endpoint cell. Thus earlier endpoint cells are retained in full whenever they have a downstream certificate; spatial restrictions are not treated as arbitrary restrictions of the range spaces in the lemma.

6.4. Same-tag tests and finite projection catalogs.

Lemma 6.6 (A uniform same-tag test). *For each original vertex v there is a fixed template family in every slot, consisting of bounded pure columns of normalized tag v and unrestricted linear modulations, with the following property. If a pure source column of tag v in slot i has form norm $t_0 > 0$ against a physical residual in slot j and two normalized $L^2(I)$ spectators, then the residual has normalized moment at least t_0/H_0 against a template column of the same tag in slot j . Here $H_0 \leq \exp(k^{O(1)})$ is independent of all insertion ranks and of the linear modulations. The source can be either an original column or any column in these template families.*

Proof. Fix a nonnegative compact smooth partition function ρ with $\sum_{n \in \mathbb{Z}^d} \rho(u - n) = 1$. Its support and Gevrey bounds can be chosen within the basic budget. Choose a sufficiently large integer period P_0 , fixed for this vertex, accommodating the supports of both its original amplitude and these partition functions. The templates have amplitude

$$\rho(d) e(\ell^T d / P_0), \quad d = \theta_v y - n,$$

where ℓ ranges over residue vectors modulo P_0 , in addition to an arbitrary external linear modulation. Their physical supremum norms are bounded by one because the partition weights are nonnegative.

Insert this partition in the other three slots and use lifts $d_l = \theta_v y_l - n_l$ in all four slots. Extract the matching phase $c_l(\lambda_v y_l^2 + (\theta_v y_l)^T C_v d_l)$ from each input, putting its conjugate into the corresponding input multiplier. On the progression,

$$d_l = (1 - l)d_0 + ld_1 + k_l,$$

with bounded integer carries k_l . The common quadratic terms cancel because $\sum_l c_l l^u = 0$ for $u = 0, 1, 2$. The phase left by the carries is

$$\sum_l c_l (\theta_v y_l)^T C_v k_l,$$

which is a sum of linear modulations. There are quasipolynomially many possible carries within the fixed support bounds.

To separate the lifts, impose each specified carry equality by a smooth function of $d_l - (1 - l)d_0 - ld_1$ that is one at that integer and zero at all other possible integers. Such a function is exact here, since its argument on the progression is integer valued. Expand these coupled smooth functions in Fourier series on a box of period P_0 containing the bounded lifts. After multiplication by the

partition functions, every nonsource factor has the required template amplitude, possibly with an unrestricted character index. Write an index as $\ell = P_0 q + r$; then

$$e(\ell^T d / P_0) = e(q^T \theta_v y) e(r^T d / P_0),$$

since $q^T n$ is integral. Thus only the residue r enters the template, and the other part is an allowed linear modulation. The source factor, after cancellation of its phase, periodizes a smooth horizontal function; expand that function in ordinary torus characters. Its remaining dependence on y_i is only a linear modulation. Derivatives of sufficiently high order, polynomial in the dimension, bound all these absolute Fourier coefficient sums by a quasipolynomial in the fixed basic budget. For a template source, reducing its own character to a residue gives the same bound. There is no iterative increase of nonlinear complexity.

It remains to prove the assertion for one separated term. Express y_i, y_j in the two spectator coordinates y_a, y_b . Separate the smooth rescaled kernel by an absolutely summable Fourier expansion in these coordinates; source linear modulations can be absorbed into the two spectator tests. The target coordinate has the form

$$y_j = \alpha y_a + \beta y_b,$$

with both α, β nonzero fixed rationals. Zero-extend the demodulated target, consisting of the physical residual multiplied by the conjugate of its bounded template, from I to a fixed larger interval and Fourier-expand it there. Its coefficients are normalized moments of the residual with same-tag templates and arbitrary linear modulations. If their supremum is M , the resulting integral is bounded by

$$M \sum_m |A_m B_m| \leq M \left(\sum_m |A_m|^2 \right)^{1/2} \left(\sum_m |B_m|^2 \right)^{1/2}.$$

Here A_m, B_m are the spectator coefficients at frequencies affine in m , with nonzero spacings proportional to α, β . These families are Bessel on the fixed integration intervals. Multiplication of a spectator by its bounded demodulating template preserves its L^2 bound. The last display is therefore at most a fixed constant times M . Fourier expansions can first be finite; L^2 convergence and the same Bessel estimate pass to the limit. Summing the absolutely convergent separated terms gives the form bound $H_0 M$. A form norm larger than t_0 consequently forces a target moment at least t_0/H_0 , after enlarging H_0 by a fixed factor if necessary to avoid an unattained supremum. $\square$

Besides the ball masks, define

$$\mathcal{U}_{m,h}(I) = \{v \in \mathcal{V}(I) : d_s(v, \{w : \text{color}(w) \leq m\}) \leq h\}.$$

They increase in depth, because distances decrease, colors do not increase, and vertices are inherited. They also have the immediate neighbor-closure property with pad $h + 1$.

Give the original basic labels rank zero. In a slot, a projection of rank r and a vertex mask means the ordinary cell-local orthogonal projection onto all available columns of ranks at most r whose tags belong to that mask. These are projections of vector functions with constant column coefficients on the cell. Newly inserted columns have private weight one and a new private coordinate. A positive smeared filter averages P such projections at the same rank, with pads $h + p$, $1 \leq p \leq P$. At a fixed cell these projections are nested in p ; at a fixed pad they are nested in depth. Their averages and complements are contractions, although products of them need not be projections.

For the cumulative filter at a split m , use rank $\mathcal{L} - m + 1$ and base pad $10P(\mathcal{L} - m + 1)$. At a final group, use four ball filters of ranks $\mathcal{L} + j$, $1 \leq j \leq 4$, in their insertion order, with base pads

$$20P(\mathcal{L} + 1) + 10(j - 1)P.$$

All these pads, including one-neighbor enlargements used below, are within $H_{\max}$.

We require the following finite closure of the catalogs. At a target rank r and one of its individual masks, apply its residual projection to a required history of length $O(\log \mathcal{L})$ on a fixed basic B or U

argument. Test that physical residual against every pure available lower-rank source whose tag lies in the target mask and two arbitrary normalized L^2 spectators. The required form norm is at most

$$\frac{\tau_3}{D\sqrt{n_{r-1}}} \|\text{base of target history}\|_{2,I}, \quad \tau_3 = D^{-C_\tau}. \quad (13)$$

Here n_r is an a priori bound for all labels through rank r on a path, and C_τ is a sufficiently large fixed exponent. The required histories include all individual pad choices of the products and complements used in the localization and transfer below. Thus the same tests hold for averaged histories. Columns, once inserted, are inherited in descendants.

There is also a helper rank $\mathcal{L} + 5$, constructed after the side states in Section 7 are fixed. Its sources are main columns only. Its additional bases are the whole side states, with at most $\exp(k^{O(1)})$ bases per active cell over slots, groups, and patterns. These bases are fixed independently of helper insertions; they are not individual frequency expansions. They may have extra private coordinates, all chosen distinct from fresh helper coordinates. The same history length and tolerance (13) are required.

Lemma 6.7 (Catalog closure and its budget). *The main closure and then the helper closure can be performed at every active cell with path counts satisfying $\log n_{\max} \leq k^{O(1)}$. The exponent is fixed independently of H_{ed} and of subsequent linearization precision. All fresh columns have the template complexities of Lemma 6.6.*

Proof. Hold the original graph vertices, masks, and named histories fixed. The numbers of masks and histories at one cell are $\exp(k^{O(1)})$: there are polynomially many original-vertex contexts, polynomially many ranks and colors, P pad choices per factor, and $O(\log \mathcal{L})$ factors. This count uses names of contexts and masks, not their numeric distance values or the number of inserted columns. The helper bases have the stipulated same count.

Upon a failed test, Lemma 6.6 supplies a bounded physical column with moment at least the threshold in (13) divided by H_0 . Insert a fresh copy at the target rank. Its new private coordinate is zero in the old tested state and in the old projection. Thus that physical moment equals the vector inner product with the old orthogonal residual. If $P_{\text{old}}, P_{\text{new}}$ are the target projections before and after insertion and X is the old tested state, then

$$\|(P_{\text{new}} - P_{\text{old}})X\|_{2,I} \gtrsim \frac{\tau_3}{D\sqrt{n_{r-1}}H_0} \|\text{base}\|_{2,I}.$$

Indeed the fresh vector column has uniformly bounded norm, and projection onto its component orthogonal to the old range gives this inequality. If the base has zero norm no test fails.

Although X can change when other projections change, its squared increments can still be charged. Fix a named history, a target mask, and a trial sequence of at most T insertions. Every individual positive projection in that history is nested in the insertion index. Expand complements and use the binary-prefix argument of Lemma 3.4. The target increments are orthogonal rows; the right-hand history is evaluated at the old insertion index. The sum of their squared actions on this changing history is at most

$$(C \log(2T))^{O(\log \mathcal{L})} \|\text{base}\|_{2,I}^2.$$

This is precisely the row version of that binary argument, and it does not require treating a changing projected state as a fixed vector.

It follows that rank- r insertions at a cell are bounded by

$$n_{r-1} \exp(k^{C_1}) (C \log(2T))^{O(\log \mathcal{L})}.$$

The exponential factor includes the configuration count, the squared inverse tolerance, and H_0^2 . It does not include the number of possible triggering source columns. For a fixed target history, any triggering source produces a fresh moment in that history's old residual; the resulting target increment is charged to the same orthogonal row sequence regardless of that source's identity.

There are at most M active cells on a path. Absorb this factor, and fixed factors for ranks and slots, in $\exp(k^{C_1})$. Starting with a basic bound n_0 , choose

$$n_r/n_{r-1} = \exp(k^{C_2})$$

with C_2 sufficiently large, and set the global trial cap to the resulting $n_{\max}$, with spare fixed factors. Since there are only $\mathcal{L} + 5 = k^{O(1)}$ ranks, $\log n_{\max} = k^{O(1)}$. With $T = n_{\max}$ the binary-prefix factor is much smaller than the chosen ratio for large k . Thus no trial cap is attained. This argument also accommodates simultaneous changes of histories caused by insertions at other ranks: all the underlying positive projections remain nested along the common insertion sequence. It proves termination of the main closure, and the identical argument proves termination of the later helper closure on its fixed bases.

All constants used in this count were determined by the basic template bounds and the number of named graph contexts. They do not depend on H_{ed} . We may therefore first fix these bounds, then choose τ_2 sufficiently small for Lemma 6.5 and all pair-mismatch errors, and finally choose H_{ed} by Lemma 6.1. This order is not circular. $\square$

We may summarize the resulting main count, coefficient, inverse private weight, and active-multiplicity bounds by D_m , where $\log D_m \leq k^{O(1)}$. The dimensions of individual tags and unions of two tags obey the same polynomial bound, independently of P . Whenever an outer primal range of rank at most r is expanded, its coefficient ℓ^1 norm is at most

$$D\sqrt{n_r} \|\text{its input}\|_{2,I}.$$

This follows from the private-coordinate lower bound and contraction of the projection. It is the reason for the normalization in (13); the full column coefficient sum will not be charged in the main estimates.

6.5. Binary localization.

Proposition 6.8 (Localization into current groups). *For an active pattern in (11), there are $\text{poly}(k)$ fixed search leaves such that, up to total absolute length-weighted error at most $Ce^{-k}|R|$, each leaf contributes*

$$\sum_{G \text{ current}} H_I(Z_{0,G,s}, Z_{1,G,s}, Z_{2,G,s}, Z_{3,G,s}), \quad Z_{l,G,s} = \bar{L}_{l,G,s} \mathcal{B}_{l,s} z_l. \quad (14)$$

Here $\bar{L}_{l,G,s}$ is one positive smeared ball filter, and $\mathcal{B}_{l,s}$ is a product of cumulative filters or complements of length $O(\log \mathcal{L})$, independent of G . The product and slot choices are fixed by the search leaf, uniformly over cells. A group is summed only at an actual current central use. The minimum with η in (11) may be applied separately per leaf.

Proof. Maintain a range of colors $[\ell, h]$, an upper primal guard slot, and when $\ell > 1$ a lower residual guard slot. Initially the range is $[1, \mathcal{L}]$; use any of the at least three basic coarse arguments as the upper guard and take no lower guard. At a midpoint $m = \lfloor (\ell + h)/2 \rfloor$, choose a third slot distinct from the guards and split it by the smeared cumulative filter at m and its complement. In the positive term that slot becomes the upper guard, the former upper guard is released, and the range becomes $[\ell, m]$. In the residual term it becomes the lower guard, the former lower guard is released, and the range becomes $[m + 1, h]$. A guard is not postcomposed before it is released. Use a deterministic slot-choice rule for each branch of this binary search, even if some colors are empty at a particular cell. There are $O(\log \mathcal{L})$ splits per path and at most $\mathcal{L}$ leaves.

Consider a leaf with color k' . Its upper guard is either the initial basic coarse argument or has outer cumulative rank $\mathcal{L} - k' + 1$ and the corresponding pad. Expand this one outer primal support, keeping its average when present. For the initial basic guard, set the upper pad to zero. Call a source tag *old* if it lies within this upper pad of a vertex of color at most $k' - 1$. If $k' > 1$, the

lower residual guard has rank $\mathcal{L} - k' + 2$ and a base pad larger by at least $10P$. Every old source therefore lies in all its residual masks and has lower rank. Equation (13) kills it, with arbitrary normalized L^2 spectators. For $k' = 1$ there are no old tags.

Every remaining upper source is near a current vertex of color k' . In the initial basic case this is the source's own original vertex; in the other case it follows from membership in the cumulative mask and exclusion of old tags. Let G be the current group of such a central vertex. The upper pad is smaller than the first ball pad, so Lemma 6.4 puts the source in every needed mask of that ball filter. The source cannot match two distinct groups: it would then violate same-color mask separation at the current cell. Moreover it is separated from the primal sources of every other current group's ball filter.

Choose a third slot, distinct from the upper and lower guards, and insert its first-rank ball projection, summed over current groups. For the matching group, the difference from the identity is bounded by (13), using the upper source. For any other group, expand the new primal ball source and apply Lemma 6.1. Old-source terms on either side are still bounded using the untouched lower residual guard. This yields an identity, with small error, in which the third slot has pure support in one group's first ball mask.

Keep that third slot as the source, and postcompose the other three slots with their own larger-rank, larger-pad ball filters in order. The first ball source has lower rank than each new target, and its mask is contained in all the new target masks. Thus (13) bounds each residual insertion. The third slot is left unchanged. The result is (14); each slot's inner cumulative history depends only on the search path, and only its outer ball filter depends on the group.

We specify why the numerical errors are summable without spending a full coefficient bound in a main term. Each test using a rank- r residual expands only an outer lower-rank source. Its coefficient bound is at most $D\sqrt{n_{r-1}}$ times the local input norm, canceled by the denominator in (13). Basic local sizes and all same-time contractions are bounded. The remaining counts of active cells on a path, leaves, current groups, and filter-insertion steps are fixed powers of D ; choosing C_τ sufficiently large makes their total length-weighted contribution at most $e^{-k}|R|$. Pair-mismatch errors can additionally contain fixed powers of the full main count, but τ_2 was chosen after that count and makes the same bound hold. No expansion of a full column coefficient sum is retained in (14). Finally, $\min(\eta, \sum_j a_j) \leq \sum_j \min(\eta, a_j)$ for $a_j \geq 0$, so the minimum can be distributed over the fixed leaves as claimed. $\square$

For a basic coarse input $z_l = B_l$, the histories before its ball filter are fixed signed sums of products of averaged nested column projections on F_l , shared across groups. Their product lengths are $O(\log \mathcal{L})$, so Lemma 3.4 gives losses at most G_0 . Before an attempt root, ball outputs can be set to zero. Between actual uses the positive masks and histories are still evaluated on their inherited endpoint cells whenever needed for nesting.

6.6. Transferring complementary filters to a side.

Lemma 6.9 (Outgoing filters and near-band defects). *At a use of (14), suppose slots a, b are fixed side states S_a, S_b , chosen independently of the helper catalog. Suppose the other two slots have whole arguments $Z_{c,G,s}, Z_{d,G,s}$ with basic inputs B_c, B_d . Expand every factor $1 - \bar{Q}$ in slots c, d as the difference of the identity and a positive average. In each resulting term, all positive filters on these two slots can be removed, one at a time, by transferring corresponding helper averages to slot a . The main term at the end is*

$$H_I(\mathcal{D}_{a,s}S_a, S_b, B_c, B_d),$$

where $\mathcal{D}_{a,s}$ is a product of helper averages of length $O(\log \mathcal{L})$.

For a step transferring a filter from c to a , let $Q_{c,p}$ be its individual outgoing projections, $1 \leq p \leq P$, and let $Q_{a,p}$ be the helper-rank projections with the same masks and pads. For $l = a, c$,

put

$$Q_{l,0} = 0, \quad Q_{l,P+1} = 1, \quad A_l^p = Q_{l,p} - Q_{l,p-1}, \quad \overline{Q}_l = P^{-1} \sum_{p=1}^P Q_{l,p}.$$

Let x_c be the inner history on B_c after deleting the outgoing filter. In the other slots, let $x_a = \mathcal{D}_{a,s} S_a$ use the helper averages already transferred, let $x_b = S_b$, and let x_d be the whole remaining history on B_d . The exact difference

$$H_I(x_a, x_b, \overline{Q}_c x_c, x_d) - H_I(\overline{Q}_a x_a, x_b, x_c, x_d)$$

is

$$\sum_{p,q=1}^{P+1} \frac{p-q}{P} H_I(A_a^p x_a, x_b, A_c^q x_c, x_d). \quad (15)$$

Only $|p - q| \leq 2$ need be retained. The discarded far terms at each cell and group have total absolute size at most $D^{-C_3} \|S_a\|_{2,I} \|S_b\|_{2,I}$, for any prescribed fixed C_3 , after the indicated choices of numerical accuracies.

Proof. Expand each complementary factor $1 - \overline{Q}$ as a difference of identity and a positive average. The number of resulting products is at most $2^{O(\log \mathcal{L})} = \text{poly}(k)$. In any such product, remove the larger rank among the current outermost filters of c, d , regarding an unfiltered basic input as rank zero. Positive ranks are distinct: each cumulative rank was inserted in exactly one slot at one search split, and the four final ball ranks are distinct. Thus the other complementary source always has strictly lower rank than the filter being removed.

Suppose the next filter is on c , and use the projections and arguments defined in the statement. The A_l^p are orthogonal band projections at the current cell and sum to the identity.

On band p , the positive average $P^{-1} \sum_{u=1}^P Q_{l,u}$ has weight $(P - p + 1)/P$, including weight zero at $p = P + 1$. Therefore the difference in the statement is exactly Equation (15), with the arguments written in role order. This proves the algebraic identity, including the terminal bands.

If $p > q + 2$, the lower c band is a difference of primals, so expand its pure sources. Their rank is a main rank below the helper rank, and their tags belong to both helper masks needed to write the higher a band as a difference of residuals. The helper version of (13), applied to the history on the fixed base S_a , bounds the resulting terms. The two other arguments are arbitrary L^2 spectators.

If $q > p + 2$, instead expand the outer primal source of x_d , using the basic expansion if it is already unfiltered. This is a main source of rank strictly below the outgoing rank on c . The lower a band is primal. Let $\mathcal{M}_p$ be the outgoing vertex mask at its pad $h + p$. Classify a pure source of x_d as near if its tag is within one edge of some vertex in $\mathcal{M}_p$, and as far otherwise. For a near source, neighbor closure gives membership in $\mathcal{M}_{p+1}$, hence in both masks needed for the higher c residual band. Apply the main test with the entire helper band $A_a^p x_a$ as one L^2 spectator. For a far source, expand the helper primal: every pair of source tags is separated, so Lemma 6.1 applies. Neighbor closure is the triangle inequality for a cumulative mask and Lemma 6.4 at the current actual use for a ball mask. The gap $q - p > 2$ leaves at least this one-pad margin. The main version of (13), already imposed on the inner history x_c , then bounds it. Thus helper insertions need not alter any main catalog.

In the first direction the coefficient bound of the main source is canceled by the helper test's denominator. In the second direction the lower-rank main source coefficient is canceled by the main test's denominator; the helper primal serves only as an L^2 spectator. For pair mismatch, expanding the two primal supports can cost fixed powers of the helper and main counts. Those counts were anticipated before τ_2 was selected. There are only $O(P^2)$ pairs of bands and polynomially many transfer positions. Since $\log P = O(k)$, these factors are absorbed by the prior accuracy choices. All filters are contractions, so the remaining norm factors are $\|S_a\|_{2,I} \|S_b\|_{2,I}$ and bounded

complementary basic sizes. This proves the stated far-error estimate. Averages in $\mathcal{D}$ are bounded directly by their tested averaged histories and do not cost the number of pad choices.

Retain the near bands and forward only the transferred main term to the next step. Iterating deletes all complementary filters and gives the asserted final main term. The number of helper factors remains $O(\log \mathcal{L})$. Restoring B_c, B_d to Y_c, Y_d in that main term uses the full pair estimate; every resulting difference contains an additional basic side U . These are the three-side terms estimated in Section 7. $\square$

We record the range convention for the retained defects. Keep their band sums together at each fixed value of $q - p$. If the outgoing filter is a ball filter, its final complementary band $A_c^{P+1} = 1 - Q_{c,P}$ is split into the group-independent identity and the positive ball term. All other complementary band endpoints have usable ball ranges. If the outgoing filter is cumulative, its outer ball has already been removed, and the whole complementary band stack is shared across groups. Likewise slot d either has one positive ball filter outermost or is shared. These conventions concern the estimation of defects, not the transfer of its main term, and will permit the joint square bounds of Section 7.

7. SHORT DEPTH BLOCKS AND SIDE TERMS

Three side factors can be summed at a power $p \in (2/3, 1)$, using their square budgets and the retained normalized L^2 bound for the fourth input. For exactly two sides, transferring the complementary filters recovers the full-pair estimate; the remaining paired near-band terms require an absolute estimate. We first prove a short-block comparison with $C_N(q)$, and then reduce the two varying sides to fixed stacks. After summing the products of their sizes, the transfer coefficient $1/P$ leaves a gain $P^{-1/2}$, up to small overhead and the factor $(K+2)^{C(q-2)}$ from the short-block comparison. Throughout the side argument, Equation (17) is an explicit hypothesis for the blockwise constant linear sides; Section 8 verifies it uniformly on fresh normalized roots.

7.1. A mixed-norm maximal estimate. We record the precise maximal estimate needed below. All the maximal operators in this subsection act in the real variable y alone.

Lemma 7.1. *Let (X, μ) and (W, ν) be finite measure spaces. If $1 < r, s, t < \infty$ and $v \in A_r(\mathbb{R})$, then*

$$\|M_y g\|_{L_y^r(v; L_\xi^s(X; L_w^t(W)))} \leq C_{r,s,t}([v]_{A_r}) \|g\|_{L_y^r(v; L_\xi^s(X; L_w^t(W)))}.$$

The constant is independent of the numbers of atoms in the measure spaces. In particular, for finite exponents $p, b, h > q \geq 1$,

$$\|M_q f\|_{L_y^p L_\xi^b L_w^h} \leq C_{p,b,h,q} \|f\|_{L_y^p L_\xi^b L_w^h}.$$

Proof. Here $v \in A_a$ means that $v > 0$ almost everywhere and

$$[v]_{A_a} = \sup_J \left(\int_J v \right) \left(\int_J v^{-1/(a-1)} \right)^{a-1} < \infty.$$

We use the scalar weighted maximal theorem and the following family-of-pairs form of Rubio de Francia extrapolation: if a family $\mathcal{F}$ of nonnegative pairs (U, V) satisfies, for some $a_0 \in (1, \infty)$ and every $v \in A_{a_0}$,

$$\int U^{a_0} v \leq C_{a_0}([v]_{A_{a_0}}) \int V^{a_0} v \quad ((U, V) \in \mathcal{F}),$$

uniformly in the pair, then the corresponding estimate holds at every $a \in (1, \infty)$ and every $v \in A_a$. No operator relation between U and V is assumed; see [1, Theorem 3.9].

Initially let $g \geq 0$ be bounded and supported in a bounded interval in y . All the weighted left-hand sides used in the following applications are finite: the inner norms are bounded by a constant times the scalar maximal function of the indicator of that interval. Thus the version

of extrapolation with finite left-hand sides suffices. The scalar weighted L^t maximal inequality, integrated in w , gives the weighted L^t hypothesis for the family of pairs

$$U_\xi(y) = \|M_y g(y, \xi, \cdot)\|_{L_w^t}, \quad V_\xi(y) = \|g(y, \xi, \cdot)\|_{L_w^t}.$$

Extrapolation gives their weighted L^a inequality for every $a > 1$. Choose $a = s$ and integrate this inequality in ξ . The resulting pairs

$$U(y) = \|U_\xi(y)\|_{L_\xi^s}, \quad V(y) = \|V_\xi(y)\|_{L_\xi^s}$$

satisfy the weighted L^s hypothesis for every A_s weight. A second application of extrapolation, now with target exponent r , proves the first assertion. This argument never interchanges the three norms and requires no ordering among r, s, t .

For general nonnegative g , apply the result to $\min(g, n)\mathbf{1}_{[-n, n]}(y)$ and use monotone convergence, including $M_y g_n \uparrow M_y g$. Absolute values treat general g . Finally apply the first assertion to $g = |f|^q$, with

$$r = p/q, \quad s = b/q, \quad t = h/q,$$

and take the q th root. Each exponent is strictly greater than one, and the power identity for the iterated norms is exact. $\square$

7.2. Comparison on a short block. A finite dyadic tree piece is completed by its terminal intervals, called leaves; they partition its root. Only internal intervals are used in its form sum. Consequently, at a contributing point in an internal interval I , the leaf containing that point is a strict descendant of I .

Lemma 7.2 (Short-block comparison). *Let $2 < q < 3$. Let $\mathcal{T}$ be a finite dyadic tree piece with root R , whose internal summation intervals belong to at most K consecutive depths and to the original range of at most N depths. Suppose its leaves occur in the same depth block or one depth below it. In slot $j \in \{0, 1, 2, 3\}$ let $\mathbf{z}_j$ be a deterministic Hilbert-space-valued function supported on R , with*

$$\|\mathbf{z}_j\|_{2, L} \leq s_j \quad \text{for every leaf } L.$$

Let z_j^σ be random scalar inputs such that, for every fixed finite $a \geq 1$ used below,

$$(\mathbb{E}_\sigma |z_j^\sigma(y)|^a)^{1/a} \leq C_a \|\mathbf{z}_j(y)\|.$$

The constants C_a are independent of the tree and K ; products of at most two independent Rademacher randomizations of a vector stack have this property. For arbitrary measurable kernel signs $|\epsilon_I(\sigma)| \leq 1$, and any subcollection of the internal intervals,

$$\mathbb{E}_\sigma \left| \sum_I |I| \epsilon_I(\sigma) H_I(z_0^\sigma, z_1^\sigma, z_2^\sigma, z_3^\sigma) \right| \leq (1 + C_N(q)) C_q (1 + \log(K + 2))^C (K + 2)^{C(q-2)} |R| \prod_{j=0}^3 s_j. \quad (16)$$

Here C is absolute. The constant C_q may depend on the fixed moment bounds, as well as the fixed kernel bounds.

Proof. By homogeneity it suffices to take $s_j = 1$. A zero size gives a zero input almost everywhere and hence a zero form. To estimate the expected absolute value, insert a common σ -dependent scalar of modulus one into the kernel signs. It is therefore enough to estimate the absolute value of the expected signed sum.

Rectangles of leaf depths. For each slot write $k_j(y_j)$ for the depth of its leaf. There are at most $K + 1$ possible depths. Pad this list to an interval of consecutive integers of dyadic length at most $4(K + 1)$ and use its dyadic subdivision. Partition all four-tuples of leaf depths as follows.

If the minimum t occurs in at least two slots, specify its set of attaining slots. Those slots have depth t , and all the others have depth greater than t . Designate two attaining slots a, b in a fixed order. If the minimum is unique, designate its slot a . The minimum and the second minimum have a unique smallest dyadic index interval whose lower and upper halves, denoted L, U , contain them respectively. Specify the nonempty set of slots other than a whose depth lies in U ; all remaining slots have depth greater than every index in U . Choose a designated slot b in that nonempty set and put $t = \min U$. This gives a disjoint partition into rectangles of leaf masks.

There are $O(1 + \log(K + 2))$ length levels and a bounded number of slot assignments at each level. At a fixed length level and assignment, write $E_{j,\iota}$ for the union of the leaves selected in slot j at position ι . The sets $E_{a,\iota}$ are disjoint as ι varies, and so are $E_{b,\iota}$. There are $O(K + 1)$ positions. The nonunique-minimum cases are treated as an additional level with singleton positions.

Fix one rectangle and suppress ι . Every slot has leaf depth at least t , except possibly slot a in the unique-minimum case. Call the former slots regular and the latter slot heavy. We may restrict the summation depths to $s < t$. Indeed, if $s \geq t$, one of the selected leaves has depth at most s and cannot meet the interior of the internal summation interval at depth s . This observation also excludes such leaves after the small coordinate enlargements below, by the fixed kernel support margin.

A slow lattice and a fast torus. Write r_s for the interval length at depth s , and choose $\delta = \beta r_t$, where $\beta > 0$ is a sufficiently small fixed dyadic number depending only on the support margin. Write

$$y_0 = \delta(n_0 + u), \quad y_1 = \delta(n_1 + v), \quad n_0, n_1 \in \mathbb{Z}, \quad u, v \in [0, 1).$$

On a progression this gives

$$n_j = (1 - j)n_0 + jn_1, \quad w_j = (1 - j)u + jv, \quad y_j = \delta(n_j + w_j).$$

Take a fixed torus $\mathbb{T}_0 = \mathbb{R}/(T_0\mathbb{Z})$ of sufficiently large length, with normalized Haar measure. In slots 0, 1 define $F_j^\sigma(n, w) = z_j^\sigma(\delta(n + w))\mathbf{1}_{E_j}(\delta(n + w))$ on $0 \leq w < 1$ and set it to zero elsewhere on this torus. In slots 2, 3 use the same formula on a fixed bounded lift interval containing all the required w_j , and set it to zero outside that interval. For example, $T_0 = 16$ and lift intervals contained in $[-4, 4]$ suffice. With fixed normalization factors, the fast Haar progression integral then reproduces the original integral: the first two factors restrict u, v to $[0, 1)$, and the other lifts do not wrap around the torus.

Replace the kernel at $\delta(n_j + w_j)$ by its value at the slow coordinates

$$(x, t') = (\delta n_0, \delta(n_1 - n_0)).$$

We use t' here for the real progression parameter, to distinguish it from the depth index t . The difference of the kernels is $O(\delta/r_s)$, with the same fixed support enlargement. The corresponding normalized summand has the factor $(\delta/r_s)^2$ times a lattice sum and a fast progression integral.

Here and below the lattice form bounds follow from the same pair-coordinate argument as Lemma 2.1. For two different progression slots the integer pair map is injective onto its image, and the pair map on the connected torus is Haar-surjective. Thus the two- L^1 , two- L^∞ estimates hold on the product of the normalized local counting measure and the fast torus. Interpolation gives the local absolute estimate whenever the four reciprocal exponents have sum at most two.

For clarity, all kernel-sampling errors can be summed as follows. Let $M_{j,I}$ be the squared mass of the deterministic stack on the sampling windows relevant to I , including their bounded overlap multiplicity. The support margin and the choice of β place these windows inside I . Hence

$$M_{j,I} \leq C|I|, \quad \sum_{I: \text{depth}(I)=s} M_{j,I} \leq C|E_j|.$$

The absolute L^2 progression estimate, followed by Hölder in σ , bounds the sum of the errors at depth s by

$$C \frac{\delta}{r_s} \left(\sum_{\text{depth}(I)=s} M_{a,I} \right)^{1/2} \left(\sum_{\text{depth}(I)=s} M_{b,I} \right)^{1/2} \leq C \frac{\delta}{r_s} |E_a|^{1/2} |E_b|^{1/2}.$$

We used the normalized mass bounds in the other two slots. Since $\sum_{s < t} \delta/r_s \leq C$, these errors cost at most $C|E_a|^{1/2}|E_b|^{1/2}$ for the rectangle. Summing positions by the two designated disjointness properties costs $C|R|$ at each length level.

Quadratic randomization on the fast torus. Put $c = (-1, 3, -3, 1)$. On Fourier coefficients in the fast variable define the unitary maps

$$\widehat{U_{j,\gamma} F}(n, m) = e(\gamma m^2 / c_j) \widehat{F}(n, m), \quad \gamma \in \mathbb{R}/(3\mathbb{Z}).$$

The fast progression integral is unchanged when all four factors are replaced by $U_{j,\gamma} F_j$. Indeed its Fourier expansion is supported on

$$\sum_j m_j = \sum_j j m_j = 0.$$

The nullspace of these two equations is spanned by $(c_j)_j$ and $(j c_j)_j$, so $m_j = c_j(h + j h')$ for real h, h' . Since $\sum_j j^\ell c_j = 0$ for $\ell = 0, 1, 2$, we have

$$\sum_j m_j^2 / c_j = 0.$$

Moreover, for each single slot and each fixed n, σ ,

$$\mathbb{E}_\gamma \int_{\mathbb{T}_0} |U_{j,\gamma} F_j^\sigma(n, w)|^4 dw \leq 2 \|F_j^\sigma(n, \cdot)\|_2^4.$$

To see this, integration in w equates the sums of two Fourier indices, and integration over the common period 3 equates their sums of squares. The two pairs are consequently equal as multisets. The two possible pairings give the displayed bound. The identities are first proved for Fourier polynomials. The pairwise L^2 progression estimate and the fourth-moment bound, also applied to differences, justify passage to L^2 limits.

Let $e_j(n)$ be the fast L^2 norm of the deterministic stack on the corresponding sampling window. Minkowski and the pointwise sign moment hypothesis give, for every needed finite a ,

$$(\mathbb{E}_\sigma \|F_j^\sigma(n, \cdot)\|_2^a)^{1/a} \leq C_a e_j(n).$$

The windows overlap a bounded number of times, so

$$\delta \sum_n e_j(n)^2 \leq C |E_j|.$$

If slot j is regular, then $e_j(n) \leq C$. In fact, all selected leaves have length at most r_t . Disjoint leaves meeting a window of length $O(\delta)$ have total length at most $O(\delta) + 2r_t$: all but at most two are contained in that window. Summing their complete leaf masses and dividing by δ proves the bound, with a constant depending only on the fixed β . On the lattice indices relevant to any summation interval I , the local version of the preceding mass estimate is

$$\frac{\delta}{|I|} \sum_{n \text{ relevant to } I} e_j(n)^2 \leq C$$

in all four slots, including the heavy slot. All the required windows are inside I , whose leaves have normalized squared mass at most one.

Clipping the possible heavy slot. Write $G_j^{\gamma,\sigma} = U_{j,\gamma} F_j^\sigma$. If a heavy slot is present, clip G_a radially at $A = (K+2)^{10}$; write the clipped function as G_a^A . For the normalized local counting measure in an interval I and Haar measure in w , unitarity gives, at every γ ,

$$\|G_a - G_a^A\|_1 \leq A^{-1} \|G_a\|_2^2 = A^{-1} \|F_a^\sigma\|_2^2.$$

Its L^4 norm in the random variables is $O(A^{-1})$, by the local mass estimate and the eighth sign moment. For each regular slot, the local L^3 norm of G_j has bounded fourth moment in (γ, σ) . Indeed the local measure is bounded, and the fourth-moment identity bounds its fourth power by a constant times

$$\frac{\delta}{|I|} \sum_{n \text{ relevant to } I} e_j(n)^4 \leq C.$$

The absolute progression estimate with exponents $(1, 3, 3, 3)$ and Hölder of exponent four in the shared random variables therefore bound the expected tail error by C/A per normalized cell. Even the crude sum over $O(K+1)$ positions and $O(K+1)$ depths costs only $C(K+2)^2/A$ times $|R|$ per length level. This is bounded uniformly in K . Henceforth G_a denotes the clipped function when there is a heavy slot.

Return to the continuous form by bursts. For fixed γ, σ, u, v , with u, v now ranging over the whole fast torus, set

$$g_j(y) = \sum_n G_j^{\gamma,\sigma}(n, (1-j)u + jv) \mathbf{1}_{\{|y-\delta n| < c\delta\}}(y),$$

where $c > 0$ is a sufficiently small fixed constant. The torus argument in this formula is taken modulo T_0 . If a real progression meets four such bursts, then the integer centers satisfy exactly $n_j = (1-j)n_0 + jn_1$. For example, the discrepancies for $j = 2, 3$ are integers of absolute value less than one when $c < 1/20$. For a fixed slow pair, the admissible offset polygon

$$\{(a_0, a_1) : |(1-j)a_0 + ja_1| < c, 0 \leq j \leq 3\}$$

has a fixed positive area. Thus, after replacing the continuous kernel by its value at the center, its integral on these bursts is that fixed area times δ^2 times the desired sampled product.

This second replacement has the same summed error bound as the first one. On the relevant indices, all the original fast sampling windows still lie inside I . Unitarity and clipping bound the transformed local L^2 sizes by those of F_j^σ . The lattice-times-torus pairwise estimate, then Hölder in the random variables, therefore uses the same masses $M_{j,I}$ and the same geometric factor δ/r_s . Although Fourier evolution spreads the fast support on the torus, it does not create new slow indices, so this support argument remains valid.

We may now apply the definition of $C_N(q)$ to the burst inputs with the original admissible kernels and original summation intervals. Only the comparison error used sampled kernels; no marginal cancellation hypothesis has been changed. It remains to bound

$$C_N(q) \int_{\mathbb{R}} \mathbb{E}_{\gamma,\sigma} \int_{\mathbb{T}_0^2} \prod_{j=0}^3 M_q g_j(y) du dv dy.$$

The exponents and the dependence on K . Put

$$p_a = p_b = p_* = 2 + 2(q-2), \quad p_j = \frac{2p_*}{p_* - 2} \quad (j \notin \{a, b\}).$$

These four exponents are finite, strictly greater than q , and have reciprocals summing to one. Use fast exponents $h_a = p_*$ for a heavy slot, and $h_j = 4$ for every regular slot. The sum of their reciprocals is at most two. The fast progression Hölder estimate, followed by Hölder of exponent

four in (γ, σ) and spatial Hölder with the exponents p_j , bounds the last display by

$$C_q C_N(q) \prod_{j=0}^3 \|M_q g_j\|_{L_y^{p_j} L_{\gamma, \sigma}^4 L_w^{h_j}}.$$

All three exponents in each slot exceed q . Apply Lemma 7.1 in exactly this norm order.

For a regular slot the fourth-moment estimate gives

$$\|g_j\|_{L_y^{p_j} L_{\gamma, \sigma}^4 L_w^4} \leq C \left(\delta \sum_n e_j(n)^{p_j} \right)^{1/p_j} \leq C |E_j|^{1/p_j}.$$

For the heavy slot, clipping and unitarity give pointwise in (γ, σ)

$$\|G_a(n, \cdot)\|_{p_*} \leq A^{1-2/p_*} \|F_a^\sigma(n, \cdot)\|_2^{2/p_*}.$$

Taking the fourth random moment and then the spatial p_* norm yields

$$\|g_a\|_{L_y^{p_*} L_{\gamma, \sigma}^4 L_w^{p_*}} \leq C_q A^{1-2/p_*} |E_a|^{1/p_*}.$$

No dimension or number of signs occurs in these bounds.

At a fixed length level and assignment there are $J \leq C(K+1)$ positions. Designated disjointness and Hölder give

$$\sum_{\iota=1}^J |E_{a,\iota}|^{1/p_*} |E_{b,\iota}|^{1/p_*} \leq J^{1-2/p_*} |R|^{2/p_*}.$$

The other masks have length at most $|R|$, so their factors contribute $|R|^{1-2/p_*}$. The resulting power of $K+2$ is at most

$$A^{1-2/p_*} (K+2)^{1-2/p_*} = (K+2)^{11(1-2/p_*)} = (K+2)^{11(q-2)/(q-1)}.$$

The number of length levels is logarithmic; the two sampling errors and the clipping errors have already been bounded at that cost. This proves Equation (16), with an absolute exponent C . All Fourier and measurability approximations are harmless: there are finitely many slow indices, the fourth-moment estimate controls differences, clipping is continuous in L^2 , and the mixed-norm estimate gives convergence of the final bounds. $\square$

7.3. The side hypotheses and variation of coarse outputs. Return to one attempt on a normalized root R , one fixed active pattern in Equation (11), and one search leaf in Equation (14). Write D_m for the main catalog budget. Thus $\log D_m \leq k^{O(1)}$, the number $\mathcal{L}$ of colors is $k^{O(1)}$, and each relevant history has length $m = O(\log \mathcal{L})$. Write $G_0 = \exp(\log^{O(1)}(2k))$; increasing its fixed exponent or taking a fixed power is allowed. Since $K \leq \text{poly}(P, k)$, all factors $(C \log(K+2))^{O(m)}$, as well as the compression factors with budget D_m , are of this form.

Partition the depths into contiguous blocks $[b, b+K)$ and choose a common pre-depth $u_b \leq b$ with $b - u_b \leq CK$. Outputs before the attempt are set to zero. In a basic- B slot put

$$S_{l,G,s}^{\text{roll}} = Z_{l,G,s} - Z_{l,G,u_b}, \quad b \leq s < b+K.$$

We also allow a function $V_{l,G,b}$ which is fixed throughout that block, may be piecewise in space, and is supported on branches with needed outputs or downstream use certificates. The hypothesis on this linear side is

$$\sum_{G,b} \|V_{l,G,b}\|_2^2 \leq G_0 |R|, \quad G_0 = \exp(\log^{O(1)}(2k)). \quad (17)$$

This is assumed uniformly whenever the construction is started on a normalized root. On a fresh attempt the linear sides may be reconstructed from the inherited basic data; no proportional local energy bound is being inferred by merely restricting an arbitrary family satisfying the displayed global inequality. In a basic- B slot, a side state $S_{l,G,s}$ is a sum of its rolling side and $V_{l,G,b}$. In a

basic- U slot the entire current output $Z_{l,G,s}$ is a mandatory side. The basic- U square bound from Section 5 is available on every normalized subroot. We shall prove the estimates uniformly over these finitely many choices of side component.

The helper projections in Lemma 6.9 are constructed after these entire side states have been fixed. Their individual mask projections are nested in depth, including their inherited values between actual uses. Their histories have length $O(m)$. Only whole states and those histories are used here; no choice of a mode, height, or amplitude inside a side state is a helper base.

Lemma 7.3 (Variation on certified endpoint cells). *Let $Z_{G,s}$ be a basic coarse output in Equation (14). Consider a common collection of disjoint, or boundedly overlapping, depth intervals $[r, t]$. At each right endpoint, restrict to a union of t -cells having a downstream actual use certificate for G . Then*

$$\sum_{G,[r,t]} \|\mathbf{1}_{E_{G,r,t}}(Z_{G,t} - Z_{G,r})\|_2^2 \leq G_0 \|\mathbf{1}_R F\|_2^2.$$

The same conclusion holds for a shared coarse history with no outer ball, omitting the group sum. The constant only acquires the given overlap multiplicity.

Proof. For this estimate expand complements by linearity and couple each pad choice across all times in an average. Jensen permits us to treat a fixed choice. We have

$$Z_{G,s} = L_{G,s} B'_s F,$$

where $L_{G,s}$ is an individual nested ball projection and B'_s is a shared product of $O(m)$ nested main projection families. Basic B inputs are fixed signed sums of at most two such current histories, which changes only a fixed factor. Split

$$Z_{G,t} - Z_{G,r} = (L_{G,t} - L_{G,r}) B'_r F + L_{G,t} (B'_t - B'_r) F.$$

For a fixed group, the rows $\mathbf{1}_{E_{G,r,t}}(L_{G,t} - L_{G,r})$ have a square Bessel bound, since increments of a nested orthogonal projection family on disjoint depth intervals are orthogonal. Their adjoints have ranges in sums of whole ball ranges at the two endpoints. A right-endpoint certificate also certifies the required earlier input cell when its mask is nonempty. Splitting at the r -cells if necessary therefore places these adjoint ranges in the usable endpoint spaces of Lemma 6.5. That lemma makes the rows jointly Bessel across groups. They are local to their r -cells, so Lemma 3.4 inserts the shared product B'_r at a cost G_0 .

For the second term, the usable-range Bessel estimate at each common right endpoint first sums the outer ball projections over groups. Telescope $B'_t - B'_r$. In each telescoping term, discard the left contractions. The changed middle factor is an orthogonal increment on $[r, t]$, and the factors to its right are evaluated at r . These middle increments form Bessel rows, and Lemma 3.4 inserts their right factors. Summing the $O(m)$ telescoping terms is again absorbed in G_0 .

An interval coloring treats bounded overlap. Expansions of the $O(m)$ complements cost $2^{O(m)}$, and averages are restored by Jensen, without a factor counting pad choices. The proof without an outer ball is the same middle-increment argument. Intervals crossing the start of the attempt occur with bounded overlap; the zero convention and the one-time projection bound treat them. $\square$

The integrated local squared sizes of all the side components obey

$$\sum_{s,G} \sum_{I \text{ used at } s} |I| \|S_{l,G,s}\|_{2,I}^2 \leq P^C |R|.$$

For a basic- U side, current group Bessel and the basic square bound give this even with G_0 on the right. For a rolling side apply Lemma 7.3 to $[u_b, s]$; these intervals have $O(K)$ overlap. A fixed $V_{l,G,b}$ is counted at most K times, so Equation (17) gives the assertion for it. First-crossing stops

for the partial path sums consequently give, outside subroots of arbitrarily small fixed total relative length,

$$\sum_{s,G} \|S_{l,G,s}\|_{2,I_s(x)}^2 \leq P^C$$

on retained paths. The fixed exponent can be increased to impose all polynomially many power-cost caps used below. In this notation only actual uses are counted. The same convention applies to shared side data, with no group index.

7.4. Power estimates and frozen complements. We explain which terms can be estimated before using the stronger stack argument. Suppose that a term has three strong size factors and a fourth bounded factor. For two mandatory group-indexed sides write, along a path,

$$a_s = \left(\sum_G a_{s,G}^2 \right)^{1/2}, \quad b_s = \left(\sum_G b_{s,G}^2 \right)^{1/2}.$$

The third size is either shared or is bounded by its analogous group-aggregated size c_s . Cauchy–Schwarz in the group index bounds the absolute group sum by $P^C a_s b_s c_s$; the factor P^C also covers a retained supremum bound for the fourth input. For $2/3 < p < 1$,

$$\sum_s (a_s b_s c_s)^p \leq \left(\sum_s a_s^{3p} \right)^{1/3} \left(\sum_s b_s^{3p} \right)^{1/3} \left(\sum_s c_s^{3p} \right)^{1/3} \leq \left(\sum_s a_s^2 \right)^{p/2} \left(\sum_s b_s^2 \right)^{p/2} \left(\sum_s c_s^2 \right)^{p/2}.$$

The path caps and integration in the path point give a total $P^C |R|$ at power p . A fourth strong factor may use its retained supremum. In particular, this estimate takes the power after the group sum; it does not apply the cell minimum separately to each group.

Consider a two-side term with whole current complements as in Lemma 6.9. On removing the complementary filters, the main term has inputs

$$(\mathcal{D}_{a,s} S_{a,G,s}, S_{b,G,s}, B_c, B_d).$$

Replacing B_c, B_d by the full Y_c, Y_d makes errors with a third basic side. The full complementary pair then has norm at most η , by the base construction. Local contractions on the first side and Cauchy–Schwarz in the two side sizes give total main cost $\eta P^C |R|$. The numerical errors of the transfer identity have an arbitrarily small inverse base-budget factor times those same two sizes and are negligible. The only remaining terms are the coupled near bands of Equation (15).

In a block, denote the outgoing complementary band stack by $(C_{G,s}^q)_q$ and the other complementary input by $D_{G,s}$. Freeze both at b . Current pad differences are mutually orthogonal projections, so the current local stack norm of $(C_{G,s}^q)_q$ is bounded; the current norm of $D_{G,s}$ is bounded as well. Their differences from their values at b have squared budgets $P^C |R|$, by Lemma 7.3, paying the polynomial number of band indices and the $O(K)$ overlap. When the last complementary band is $1 - Q_P$, split it into an identity term and a positive ball term, as in Lemma 6.9. The positive outer-ball terms have joint group budgets, and the identity or cumulative histories have shared budgets. Thus this argument does not count the same shared history once for every group.

Expanding a freeze error gives a third strong factor. If its other frozen complement is unbounded, write that complement as its bounded current value plus its strong difference. The preceding power estimate still applies. The same observation allows us to discard uses at which a frozen complement is large, either at that use or at an ancestor in its block. More explicitly, if a current stack has size at most C_0 and its frozen stack has size greater than $2C_0$, their difference has size greater than C_0 ; hence the indicator of this event is bounded by C_0^{-2} times the squared difference size. Lemma 7.3 applies at all potential ancestor cells with downstream certificates. Propagating such an event down its block costs at most K in the integrated square budget. Its square-root indicator is therefore a third strong factor. Include these budgets in the power-cost path caps. On all remaining uses, both

frozen complements have bounded local L^2 norms on their entire ancestor hull inside the block, using the stack norm for the banded complement.

7.5. Fixed stacks for the two sides.

Lemma 7.4 (Side stacks). *Fix a near offset $q - p$ in Equation (15), a transfer step, and a search path. The two time-dependent side inputs can be written, at their uses, as a sum of at most small-overhead many configurations of fixed vector stacks for each (G, b) . One position is selected at a use from each stack. The banded side stack also includes the band index. After absorbing the number of configurations in G_0 , their energies satisfy*

$$\sum_{G,b} \|\text{stack}_{a,G,b}\|_2^2 \leq PG_0|R|, \quad \sum_{G,b} \|\text{stack}_{\text{untouched},G,b}\|_2^2 \leq G_0|R|. \quad (18)$$

The functions in these stacks are fixed before subsequent cuts of the summation set.

Proof. For a sequence W_s , $b \leq s < b + K$, write

$$W_s = W_b + \sum_{[r,t] \in \mathcal{P}(b,s)} (W_t - W_r),$$

where $\mathcal{P}(b, s)$ is the usual disjoint binary decomposition of the prefix $[b, s]$. There is at most one interval of each length in this decomposition. For a fixed length, its intervals are disjoint as depth intervals, and the selected endpoint is at most s . Restrict each resulting fixed function to the union of the use cells selecting it. This preserves its value at every required use and only decreases its norm. The base position and the choices of lengths account for logarithmic factors.

For a basic- U input, use each depth s directly as a position. The current side square bound controls the sum of its energies; subsequent helper products are local contractions. On the banded side, paying at most $P + 1$ indices is permitted in the first inequality of Equation (18).

For a fixed linear side $V_{G,b}$, let T_s be the product of its subsequent averaged projections. At a fixed band index, the outer band difference is a difference of two such products. The binary-prefix product estimate in the proof of Lemma 3.4, now on a sequence of at most $K + 1$ times, gives

$$\|T_b V_{G,b}\|_2^2 + \sum_{[r,t] \text{ at a fixed length}} \|(T_t - T_r) V_{G,b}\|_2^2 \leq (C \log(K + 2))^{O(m)} \|V_{G,b}\|_2^2.$$

Individual projections are nested; averages are coupled across times and restored by Jensen. Sum the length levels and use Equation (17). Pay at most $P + 1$ on the banded side.

For a rolling input write $W_s = T_s(Z_{G,s} - Z_{G,u_b})$. The base term is controlled by Lemma 7.3 on the intervals $[u_b, b]$, whose overlap across blocks is bounded. Its increments split exactly as

$$W_t - W_r = T_t(Z_{G,t} - Z_{G,r}) + (T_t - T_r)(Z_{G,r} - Z_{G,u_b}).$$

The first term is bounded directly by Lemma 7.3. When estimating norms, keep the entire endpoint cells containing the selected use cells: these cells still have downstream certificates, and locality of T_t makes this enlargement legitimate.

For the second term, also expand $[u_b, r]$ by binary prefix intervals in a common window of $O(K)$ depths. Fix an outer length, an inner length, and an inner interval $[a, v]$. For all outer positions $[r, t]$ using this same inner interval, use the fixed input

$$X = \mathbf{1}_\Omega(Z_{G,v} - Z_{G,a}),$$

where Ω is the union of the needed r -cells. This input is independent of the outer position. Since $v \leq r$, locality on r -cells shows that it gives the required outputs of $T_t - T_r$. Its norm is bounded by keeping the union of the v -cells containing those r -cells; all of these have downstream use certificates. For this fixed input, the product-jump estimate bounds the sum over outer positions by $(C \log(K + 2))^{O(m)} \|X\|_2^2$. Now sum the squared inputs over groups and inner intervals at the fixed inner length, using Lemma 7.3. The windows for different blocks overlap a bounded number

of times. Finally sum the two length choices. This proves the rolling estimate with only small overhead, and with the allowed factor P for a banded side.

The number of search leaves, transfer steps, choices of side component, and length configurations is small overhead. Individual pad choices are not additional configurations: they remain inside the actual averaged operators throughout. Thus summing all required configurations preserves the asserted bounds after increasing G_0 . $\square$

The reduction to fixed stacks does not require synchronizing their two position decompositions. For each fixed configuration choose independent Rademacher signs (ε_i) and (η_j) for the two position indices, and a third independent family (ζ_p) for the paired band indices. Randomize the banded fixed inputs by $\varepsilon_i \zeta_p$, the untouched side inputs by η_j , and the frozen complementary band stack by ζ_p . At a use, multiply the kernel by the selected position signs $\varepsilon_{i(s)} \eta_{j(s)}$. Expectation selects exactly the two required positions and the matched bands. The remaining coefficient is $(p-q)/P = O(1/P)$. Iterated Khintchine gives every fixed pointwise moment bound required in Lemma 7.2, with the deterministic stack norms. Initial kernel signs may be arbitrary.

7.6. Hull maxima and common cuts. We give the support convention needed to use Equation (18) without an additional power of P . For each index $\nu = (G, b, \text{fixed configuration})$, let Ω_ν be the union of its use cells. Its maximal use cells are disjoint roots. The hull consists of the cells between those roots and their uses. All fixed stack functions are kept unchanged when uses are subsequently removed. A maximal function on a hull is defined on the entirety of each of its intervals, including missing branches.

The sum of the lengths of Ω_ν is at most $M_0|R|$, where $M_0 \leq \exp(k^{O(1)})$. This follows from the active path multiplicity, the main group count, and the small-overhead number of configurations. Suppose one family of stacks has total energy at most $E|R|$. Its squared hull maxima

$$m_\nu(x) = \sup_{\substack{I \text{ in its hull} \\ x \in I}} \int_I \|\text{stack}_\nu\|^2$$

satisfy, by the weak L^1 inequality for the dyadic maximal operator,

$$\sum_\nu |\{m_\nu > T\}| \leq \min(M_0|R|, CE|R|/T).$$

First stop at individual crossings of $C_1 E$, losing at most C/C_1 of the root length. Integrating the last distributional bound up to $C_1 E$ gives

$$\sum_\nu \int_R \min(m_\nu, C_1 E) \leq CE(1 + \log(2 + C_1 M_0))|R|.$$

Markov's inequality then gives a joint square cap at a further small fixed loss of root length. The logarithm, and the constants needed for the small-overhead many configurations, are absorbed in G_0 . Applying this to the two energies in Equation (18) gives caps PG_0 and G_0 respectively.

These stops are made on partial ancestor suprema, not merely on a set specified at terminal points. More precisely, at depth s sum the suprema over the hull ancestors of depth at most s and stop at the first threshold crossing. Remove that cell and its descendants from every original summation set. Each triggering cell is removed before its value is included in a retained cap. Consequently the maxima of the retained hulls satisfy the joint caps everywhere, even inside a stopped region: only its earlier retained ancestors still contribute there. This construction is dominated by the preceding full-tree distributional bounds and has the same measure cost. Combine these cuts with the first-crossing cuts for the partial power-cost budgets, including the budgets of the large-frozen-complement flags, by taking their union and using the original summands at each first cut. A use flagged for a large frozen complement is removed only from the critical near-band sum and is charged to the powered error estimate; the flag itself does not create a restart root.

7.7. The near-band estimate and the conditional side bound. Fix one of the configurations in Lemma 7.4. On each retained hull, make a sparse stopping decomposition for the two fixed side stacks. From a piece root Q , stop at the first descendants where either squared stack average exceeds sixteen times its average on Q . Their total length is at most $|Q|/8$. If a root average is zero, the corresponding stack is zero on Q and there is no contribution there. The sets E_Q left after removing the next roots have measure at least $7|Q|/8$ and are disjoint within each fixed hull construction.

Complete every piece by children immediately below its included cells. This includes branches without a use and boundaries made by the common cuts. A child's squared average is at most twice its parent's. Thus both side stacks have bounded normalized sizes, relative to their root sizes, on every leaf of the piece. The frozen complementary stack and the other frozen complement have uniformly bounded sizes on retained hull cells, and hence on these boundary leaves as well. The piece uses at most K depths, with leaves at most one step below. Apply Lemma 7.2 to the randomized fixed inputs.

Let a_Q, b_Q be the two side stack sizes at the piece root. On every $x \in E_Q$, including a missing or stopped branch, the retained hull maxima satisfy

$$a_Q b_Q \leq \sqrt{m_{a,\nu}(x) m_{b,\nu}(x)}.$$

Indeed Q itself is one of the intervals in each supremum. Hence

$$\sum_{\nu, Q} |Q| a_Q b_Q \leq C \int_R \sum_{\nu} \sqrt{m_{a,\nu} m_{b,\nu}} \leq C \int_R \left(\sum_{\nu} m_{a,\nu} \right)^{1/2} \left(\sum_{\nu} m_{b,\nu} \right)^{1/2} \leq P^{1/2} G_0 |R|.$$

The logarithms in Equation (16) are small overhead. After the coefficient $O(1/P)$ and the small-overhead number of configurations, the total near-band contribution is at most

$$(1 + C_N(q)) C_q P^{-1/2} G_0 (K + 2)^{C(q-2)} |R|. \quad (19)$$

This bound holds with arbitrary initial signs at the original cells. It therefore controls the absolute sum of the cell contributions, provided the prescribed paired band sums at a fixed offset are retained inside each contribution. No absolute summation over their individual band indices was made.

Proposition 7.5 (Side estimate). *Assume the base and localization hypotheses above, the helper conclusions of Lemma 6.9, and the linear-side square bound (17). Fix $p \in (2/3, 1)$. After common first-crossing cuts of arbitrarily small fixed total relative length, terms having at least three side factors have total length-weighted p th-power cost at most $P^C |R|$, provided the fourth factor has retained normalized L^2 size at most P^C . This includes a fourth side, a bounded current argument, or a bounded current argument minus side factors.*

Terms having exactly two side factors and whole current coarse complements admit the following estimates. Their freeze errors, large-complement terms, and restoration errors have total length-weighted p th-power cost at most $P^C |R|$. Their transferred main terms cost at most $\eta P^C |R|$ absolutely. Their numerical errors are negligible after the stated choice of the main precision. Their remaining coupled near-band terms obey Equation (19) without taking a power less than one.

All statements are uniform on normalized subroots with the same basic data. Subsequent common cuts may remove original summands without changing the fixed functions used in these estimates.

Proof. The three-side power bound was proved above using the path square caps and $3p > 2$. Lemma 6.9 gives the decomposition into transferred main terms, numerical errors, and coupled near bands. Restoring the full complementary pair in a main term introduces a third basic side and leaves the pair-small estimate η on its two arbitrary side inputs. The frozen-complement argument accounts for every freeze error and every discarded large complement as a third strong factor. Lemma 7.4, the common maximal cuts, and the sparse comparison then give Equation (19) for the remaining terms.

All cuts were imposed on nonnegative partial ancestor quantities and removed the original summands at and below their first violation. The normalized base hypotheses, the basic square estimate, and the root-uniform hypothesis (17) are available anew when the construction is restarted on such subroots. The fixed functions in the parent attempt remain unchanged. The thresholds may be chosen to make the union of the bad roots an arbitrarily small fixed fraction of $|R|$. This proves the asserted uniformity and support convention.

Finally, the estimates have the required relation to the original cell minimum. The subadditivity of $x \mapsto \min(\eta, x)$ removes absolute numerical errors and the critical unpowered terms. For the remaining cell sum use

$$\min(\eta, x) \leq \eta^{1-p} x^p.$$

The group sum and the prescribed coupled bands are formed before this inequality is used. Thus every power cost gains η^{1-p} , whereas Equation (19) is used exactly as an absolute estimate. This completes the conditional side argument. $\square$

8. LINEARIZATION, PERSISTENT COMPONENTS, AND TOPS

We work on one normalized attempt with root R , for one fixed leaf of the localization identity in Proposition 6.8. The estimates will be uniform over its leaves. All trees, lists, and collections in this section are finite. Interval norms are normalized, whereas norms without an interval subscript use Lebesgue measure.

This section constructs the linear sides required in Section 7 and the core data used in Section 9. We expand each matched column once and inherit its mode labels, while freezing its coefficients over blocks of dyadic depths. The PRE and CURRENT frequency graphs separate the linear side from the core and give the square bound in Equation (17). Hereditary representative labels and their first actual-use cells reaching a fixed amplitude threshold, called tops, then give the pathwise count in Equation (28).

We write D_m for a main-catalog budget and enlarge it by fixed powers when necessary:

$$\log D_m \leq k^{O(1)}.$$

This budget covers main-column counts, their coefficient bounds and inverse private weights, and the multiplicity of relevant groups and actual uses along a path. The dimension of the horizontal parameters of a vertex or of two vertices together is at most $d_{\max} \leq k^{O(1)}$; we take $d_{\max} \geq 2$. As before, a factor

$$G_0 = \exp(\log^{O(1)}(2k))$$

is a small overhead. Its value can increase from one estimate to the next. It must be distinguished from a fixed power of P .

8.1. The data retained from localization. We specify the properties of the localization construction used here. In a fixed basic- B role l , its output at depth s has the form

$$Z_{l,G,s} = \bar{L}_{l,G,s} \mathcal{B}_{l,s} B_{l,s},$$

as in Equation (14). The outer operator is an average of main ball projections of one fixed rank. Each individual projection family is nested in depth. The product $\mathcal{B}_{l,s}$, including the constituent projections in $B_{l,s}$, is a fixed signed sum of averaged products of $O(\log \mathcal{L})$ nested main projection families, independent of G . Every coefficient choice in these averages is fixed across time.

We use the following conclusions of the preceding constructions. First, columns and ball-mask matches are inherited along a path, and a matched main column has a short path to the primary center of its group. The endpoint version of Lemma 4.3 and Lemma 4.4 therefore apply at its first match. Second, an endpoint cell with a nonempty ball mask can be retained whenever it has a downstream actual-use certificate. The whole usable-range spaces of distinct groups of the fixed color satisfy Lemma 6.5. This statement concerns the spans of main columns on the entire

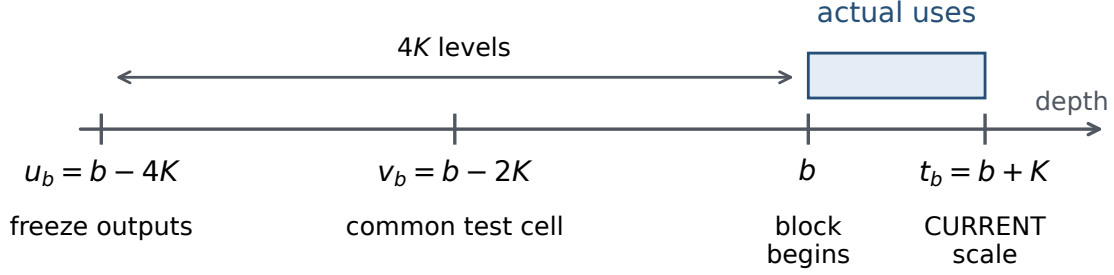


FIGURE 1. A block $[b, b + K)$ of dyadic depths. Depth increases to the right and spatial scale decreases. Coefficients are frozen at $u_b = b - 4K$. The PRE and CURRENT graphs defined below are both evaluated on the same $v_b = b - 2K$ cell, using comparison scales r_{v_b} and r_{t_b} , respectively, where $t_b = b + K$. Each primary's first-match expansion is inherited unchanged across blocks. The counting blocks in Figure 2 use a different partition.

endpoint cells, including all repetitions of such cells; it is stronger than a separation statement only at actual uses. Third, all the endpoint projections and shared histories are among the column families covered by Lemma 3.4. Their column budgets are D_m , rather than the larger mode budget introduced below.

These properties also hold on the nested extensions past stopping boundaries used to define the original summands. We keep entire input endpoint cells when applying a projection; only its output need be restricted to branches with a downstream certificate.

8.2. A single expansion at the first match. Choose a dyadic integer K , whose size will be specified below. Partition the depth schedule of the attempt into blocks $[b, b + K)$, and set

$$u_b = b - 4K, \quad v_b = b - 2K, \quad t_b = b + K.$$

Let r_s be the interval length at depth s . In a basic- B role freeze Z_{l,G,u_b} , and set it to zero if u_b precedes the attempt. Positive ball masks and their histories are evaluated at these pre-depths even when the group has no actual use there. We retain a v_b -cell J for this purpose only when a descendant actual use of G occurs in the block. Basic- U roles are not linearized. Figure 1 records these depths and the common test cell.

Fix a group G , with primary center z , and a basic- B role l . A *primary* in this discussion is an individual main column Φ available to the maximal pad of this role's outer ball filter. Distinct linear shifts of one vertex tag are distinct primaries. At its first match, on a cell I_0 , make one expansion choice and retain that choice on every relevant descendant of I_0 . The lists of primaries and their modes are therefore nested. Separate first-match branches may have separate expansions and private labels. In particular, nothing in an already chosen mode list is resampled at a later freezing time.

Choose an integer Q_0 , common to the roles of this group and within the main budget, clearing the denominators of C_z , and put $\vartheta = \theta_z/Q_0$. For a uniformly random branch offset ρ on the unit cube, set

$$d^\rho(y) = \rho + \{\theta_z y - \rho\}, \quad p^\rho(y) = \lambda_z y^2 + (\theta_z y)^T C_z d^\rho(y).$$

The braces are coordinatewise fractional parts. Choose a product Gevrey cutoff ζ^ρ , periodic in $\theta_z y - \rho$, with $0 \leq \zeta^\rho \leq 1$, vanishing within distance δ_c of a coordinate cut and equal to one outside distance $2\delta_c$, where

$$\delta_c = \exp(-k^{C_c}).$$

The exponent C_c is fixed sufficiently large after the main budgets. Translation invariance of the offset gives

$$\mathbb{E}_\rho \zeta^\rho(y) = c_\zeta \geq \frac{1}{2}, \quad \mathbb{E}_\rho |1 - (\zeta^\rho(y))^2|^2 \leq C d_{\max} \delta_c$$

at every y . A zero-dimensional horizontal parameter has the constant convention $\zeta^\rho = 1$. For each role and group the offset will be common to every branch and time, and independent of the already completed main selections.

Lemma 8.1 (First-match expansion and bounded lift). *At each first match I_0 , there is a finite, permanently labeled expansion*

$$\zeta^\rho(y) \varphi_\Phi(y) e(-c_l p^\rho(y)) = \sum_m a_{\Phi,m} e(\lambda_m y^2 + \gamma_m y) + O(\delta_c), \quad y \in I_0,$$

where all the modes belonging to this primary have the same λ_m . We may use augmented mode atoms

$$\psi_m(y) = \left(\zeta^\rho(y) e(c_l p^\rho(y)) e(\lambda_m y^2 + \gamma_m y), \epsilon_m^{1/2} \mathbf{e}_m \right). \quad (20)$$

Their private indices distinguish primaries and first-match expansions. For an admissible budget D_1 , with $\log D_1 \leq \text{poly}(P, k)$, we have

$$|\lambda_m| |I_0|^2 \leq D_1, \quad \sum_m |a_{\Phi,m}| \leq D_1,$$

and the number of modes is at most D_1 . For two modes of this primary,

$$\gamma_m - \gamma_{m'} = a \cdot (\theta_v, \vartheta), \quad a \in \mathbb{Q}^{\dim(\theta_v, \vartheta)} \text{ has height at most } D_1. \quad (21)$$

There is a block-independent pointwise linear operator $\mathcal{R}_{l,G}$, of uniformly bounded norm, mapping each frozen outer ball output to exactly the corresponding synthesis of Equation (20).

Proof. The branch cutoff makes the conjugating factor a smooth quadratic chart supported in a bounded branch cube. At I_0 , the joint difference data of v and z obey the endpoint version of Equation (2), with the harmless fixed factor c_l . Recenter at I_0 , choose one rational splitting there, and apply Lemma 4.4. Its tail estimate gives the stated uniform error and finite coefficient and count bounds. A single splitting gives a single effective quadratic coefficient for this primary. Recentring adds a common unrestricted linear shift; the differences of the remaining linear shifts are rational combinations of the two endpoint horizontal vectors. Clearing Q_0 gives Equation (21). All these choices are made once.

Here is an explicit lift, which will also explain its operator properties. Write the main atom as $\Phi = (\varphi_\Phi, \epsilon_\Phi^{1/2} \mathbf{e}_\Phi)$, and set

$$q_\Phi(y) = \sum_m a_{\Phi,m} (\psi_m)_{\text{phys}}(y) - (\zeta^\rho(y))^2 \varphi_\Phi(y).$$

It satisfies $|q_\Phi| \leq \delta_c$ on the matched subtree. For a main augmented input F , define the physical part of $\mathcal{R}_{l,G} F$ by

$$(\zeta^\rho)^2 F_{\text{phys}} + \sum_{\Phi \text{ matched at } y} \frac{F_\Phi(y)}{\epsilon_\Phi^{1/2}} q_\Phi(y).$$

Its mode coordinate m , belonging to Φ , is

$$\epsilon_m^{1/2} a_{\Phi,m} \frac{F_\Phi(y)}{\epsilon_\Phi^{1/2}},$$

and is zero before the appropriate match. The main count and inverse weight bounds, followed by Cauchy–Schwarz, bound the physical correction’s pointwise operator norm by $\delta_c D_m^C$. Choose $\epsilon_m > 0$ so small that the squared norm of the mode-coordinate map is bounded by one. The

preceding count and coefficient estimates allow $\log(1/\epsilon_m) \leq \text{poly}(P, k)$. Thus the whole lift has a uniform pointwise norm.

An outer ball output on a cell has a representation $\sum_{\Phi} \beta_{\Phi} \Phi$ with constant coefficients, including when the projection is averaged. Its main private coordinate is exactly $\epsilon_{\Phi}^{1/2} \beta_{\Phi}$. Substitution in the displayed definition gives $\sum_{\Phi, m} \beta_{\Phi} a_{\Phi, m} \psi_m$, as required. The lift is pointwise and hence cell-local; it has not been commuted through any projection. $\square$

Put

$$\tilde{Z}_{l, G, b} = \mathcal{R}_{l, G} Z_{l, G, u_b}$$

on the retained branches. It is a mode sum $\sum_m \alpha_m \psi_m$, with coefficients constant on the u_b -cell. Its available list may include every primary matched for the maximal pad and rank, whether or not its coefficient is zero. Fix a budget

$$D_* \geq e^k, \quad \log D_* \leq \text{poly}(P, k),$$

large enough for all the preceding mode counts, rational heights, coefficient sums, inverse private weights, and Q_0 . It also bounds the total number of available modes over all relevant groups along a path, all actual cell/group uses, and the sums $\sum_m |\alpha_m|$ in normalized frozen evaluations. These bounds follow from the main coefficient bounds and Lemma 8.1; groups with a certified nonempty endpoint already have historical data on that path. We use disjoint private coordinates to place all the required spaces in one Hilbert space.

8.3. The common progression multiplier and parameter order. The common gauge has an additional property needed in the core estimate. Allow four different offsets ρ_j , and let $y_j = y_0 + jt$. On the supports of their four cutoffs, $\prod_j e(c_j p^{\rho_j}(y_j))$ agrees with a smooth periodic multiplier in $(\vartheta y_{l'}, \vartheta t)$, for any anchor l' . This multiplier can be chosen bounded by one.

To verify this, write $d_j = d^{\rho_j}(y_j)$. Each $d_j - \theta_z y_j$ is an integer vector, so

$$k_j = d_j - (1 - j)d_0 - jd_1$$

is an integer vector. It is bounded, and locally constant away from the branch cuts. The identities $\sum_j c_j = \sum_j j c_j = \sum_j j^2 c_j = 0$ cancel the quadratic terms and the terms containing d_0, d_1 , leaving

$$\prod_j e(c_j p^{\rho_j}(y_j)) = e \left(\sum_j c_j (\theta_z y_j)^T C_z k_j \right).$$

For each fixed carry choice the right side is an ordinary character on the indicated torus, since Q_0 clears C_z . Multiply it by slightly enlarged smooth cutoffs equal to one on the required supports, and extend by zero at the cuts. The pieces then give the asserted smooth periodic function. The same construction works after changing the anchor, because the coordinate change has integer coefficients.

Gevrey bounds for these cutoffs and the finitely many carry pieces give admissible absolute Fourier coefficient sums, uniformly in the offsets. Enlarge D_* to cover those sums, including cutoff products such as $(\zeta^{\rho})^2$. Only after this enlargement, truncate their Fourier series with absolute tails at most D_*^{-300} . Denote a common frequency cutoff by B_* , where $\log B_* \leq \text{poly}(P, k)$. Thus there is no circular choice between a coefficient-sum budget and a Fourier truncation radius.

We now fix the rest of the parameters. Let $\Delta = 10$ and choose a sufficiently small absolute $\delta_* > 0$. Set

$$C_* = (10d_{\max} D_*)^{100d_{\max}^4}, \quad R_*^{\text{ht}} = \lceil \log_2 C_* \rceil + 2\Delta + 10.$$

Cramer's rule shows that expressing a height- D_* vector in a basis of such vectors, in dimension at most $d_{\max}$, uses coefficients whose absolute values and common denominator are at most C_* .

Choose an integer J_* sufficiently large compared with

$$\delta_*^{-2} d_{\max} R_*^{\text{ht}} \log(2D_m),$$

and put

$$L_h = 6^h, \quad E_h = A_0^{h+1} \quad (0 \leq h \leq J_*).$$

Here A_0 is an integer chosen as a sufficiently large fixed power of $C_* 6^{R_*^{\text{ht}}} B_* D_*$. The notation E_h denotes a shift bound, not a projection. Set $\Xi = D_*^{1000}$. Choose a rational height bound

$$\mathcal{H} > (d_{\max} C_* E_{J_*} B_* L_{J_*} Q_0 \Xi)^{10},$$

and then a gap factor

$$\mathcal{T} > (d_{\max} \mathcal{H})^{C d_{\max}^4 \Xi}$$

with a sufficiently large absolute C . All their logarithms are polynomial in P, k .

Finally choose the dyadic block length K , still polynomial in P, k , sufficiently large after $\text{poly}(d_{\max})(1 + \log \mathcal{H}) + \log \mathcal{T}$. In particular, if W is the width of a log packet in Lemma 3.5 at height $\mathcal{H}$, require $K \geq W + 2 \log_2 \mathcal{T}$, after converting W to base-two logarithms. Increase K further whenever needed to make $D_* 2^{-4K}$ smaller than any fixed inverse power of D_* . These choices have fixed polynomial degrees independent of q .

For a mode let $\omega_m(x) = 2\lambda_m x + \gamma_m$. Its first match is no later than u_b if it is in the list frozen at u_b . Lemma 8.1 therefore gives, on a v_b -cell,

$$r_{v_b} \text{osc}_J \omega_m \leq C D_* 2^{-4K}.$$

This estimate also applies to every earlier retained mode when it is viewed on a later pre-cell.

8.4. Two component graphs. On a retained v_b -cell J , use the common list frozen at u_b . At height h , join two modes in the PRE graph if the first bound below holds, and in the CURRENT graph if the second holds, for some integral n with $|n|_\infty \leq E_h$:

$$\sup_{x \in J} |\omega_m(x) - \omega_{m'}(x) + n \cdot \vartheta / L_h| \leq \Xi / r_{v_b} \quad \text{or} \quad \leq \Xi / r_{t_b}, \quad \text{respectively.} \quad (22)$$

Write $\mathcal{E}_h(m)$ and $\mathcal{F}_h(m)$ for their components. PRE refines CURRENT. Both graphs increase with h : an old witness is replaced by $6n$, which is allowed since $A_0 \geq 6$. They also increase along successive blocks on a path, on retained lists, since the pre-cells shrink and the thresholds increase. For a set of modes $\mathcal{D}$, let $\pi(\mathcal{D})$ be its set of primary labels. A *minimal component* means the component at height zero.

A primary with parameter vector $\Theta = (\theta_v, \vartheta)$ is *nonexceptional* at this block if every rational vector a of height at most $\mathcal{H}$ satisfies

$$|a \cdot \Theta| \leq \mathcal{T}^{-1} / r_{v_b} \quad \text{or} \quad |a \cdot \Theta| \geq \mathcal{T} / r_{t_b}. \quad (23)$$

Lemma 8.2 (Exceptional blocks and the tiny rational space). *Each primary is exceptional in at most $C d_{\max}$ blocks on a path. At a nonexceptional block, let K_{rat} be the rational span of all height- $\mathcal{H}$ vectors satisfying the tiny alternative in Equation (23). If a vector of height at most $\mathcal{H}$ belongs to K_{rat} , then it too satisfies that tiny alternative.*

Proof. Fix the primary, put $r_s = r_0 2^{-s}$, and write $\tau = \log_2 \mathcal{T}$. For a nonzero evaluation set $x(a) = \log_2(|a \cdot \Theta| r_0)$. Exception at block start b means

$$b - 2K - \tau < x(a) < b + K + \tau.$$

Lemma 3.5 places all these $x(a)$ in at most $d_{\max}$ packets of width W . A packet $[\alpha, \alpha + W]$ can cause an exception only when

$$\alpha - K - \tau < b < \alpha + W + 2K + \tau.$$

This interval has length at most $4K$ by our choice of K , so it contains at most five starts of the K -spaced block schedule. Zero evaluations are always tiny.

For the second assertion, choose a basis of tiny vectors from the defining set. Cramer's rule expresses any height- $\mathcal{H}$ vector in its span with coefficients of total absolute value at most $(d_{\max} \mathcal{H})^{C_{\max}^3}$. Its evaluation is consequently bounded by this factor times $\mathcal{T}^{-1}/r_{v_b}$. The choice of $\mathcal{T}$ makes this strictly smaller than $\mathcal{T}/r_{t_b}$. The gap in Equation (23) then forces the tiny alternative. $\square$

Put

$$K_B = \{w \in \mathbb{Q}^{\dim \theta_z} : (0, w) \in K_{\text{rat}}\}.$$

The space K_{rat} , and hence K_B , can depend on the primary and block.

Lemma 8.3 (Height stabilization). *For a nonexceptional primary, all but $Cd_{\max}R_*^{\text{ht}}$ heights have the following property: if two of its modes are in one CURRENT component at height h , then they are directly adjacent in the PRE graph at height $h - \Delta$. We call these heights height-good for the primary.*

Proof. Abbreviate $R' = R_*^{\text{ht}}$, and discard $h < R'$. If two modes of this primary are connected in CURRENT, sum the edge witnesses along a simple connecting path. There are at most D_* edges. Their quadratic parts cancel between the endpoints, because the primary has one effective quadratic coefficient. With a from Equation (21), this gives an integral m_h with $|m_h|_{\infty} \leq D_*E_h$ and

$$|a \cdot \Theta + m_h \cdot \vartheta / L_h| \leq D_*\Xi / r_{t_b}.$$

The vector $a + (0, m_h/L_h)$ has height at most $\mathcal{H}$. Since $D_*\Xi < \mathcal{T}$, the gap implies

$$a + (0, m_h/L_h) \in K_{\text{rat}}.$$

For each height i , let V_i be the rational span of the height- D_* vectors q' for which

$$q' + (0, m/L_i) \in K_{\text{rat}} \quad \text{for some } m \in \mathbb{Z}^{\dim \theta_z}, \quad |m|_{\infty} \leq D_*E_i.$$

Let U_i be the rational span of the integer vectors in K_B with size at most $C_*^3 D_*E_i$. Both families increase. Discard every h for which the rank of either family changes between $h - R'$ and h . There are at most $2d_{\max}R'$ such heights, since each rank can increase only $d_{\max}$ times.

At a remaining height, $a \in V_h = V_{h-R'}$. Express $a = \sum_j b_j a_j$ using a basis from the defining vectors at $h - R'$, with witnesses m_j . The b_j and their common denominator are bounded by C_* . Subtracting the corresponding relations in K_{rat} shows that

$$m_h - 6^{R'} \sum_j b_j m_j \in K_B.$$

After clearing the common denominator this is an integer vector whose size is at most $C_*^3 D_*E_h$; this follows directly from the witness bounds and the choice of A_0 . It belongs to $U_h = U_{h-R'}$, which we denote by U .

The lattice $U \cap \mathbb{Z}^{\dim \theta_z}$ is saturated: if a nonzero integer multiple of an integer vector belongs to U , the vector itself belongs to the rational space U . Thus the quotient $\mathbb{Z}^{\dim \theta_z} / (U \cap \mathbb{Z}^{\dim \theta_z})$ is free. In an integer basis of this quotient,

$$[m_h] = 6^{R'} \sum_j b_j [m_j].$$

The right side has integral coordinates. The common denominator bound and $R' > \log_2 C_* + 2\Delta$ show, coordinate by coordinate, that each is divisible by 6^Δ : the valuations at both 2 and 3 are at least Δ . Other denominator primes do not affect these divisibilities. Choose an integral lift m' of $[m_h]/6^\Delta$.

Then $m' - 6^{R'-\Delta} \sum_j b_j m_j \in U$. Choose linearly independent integer vectors of size at most $C_*^3 D_*E_{h-R'}$ spanning the rational space U , from the definition of $U_{h-R'}$. Subtract integer multiples

of these vectors from m' , rounding the coefficients of the preceding difference. The result remains an integer lift and satisfies

$$|m'|_\infty \leq 6^{R'-\Delta} \sum_j |b_j| |m_j|_\infty + d_{\max} C_*^3 D_* E_{h-R'} \leq E_{h-\Delta}.$$

The last inequality is one of the permitted fixed-power choices of A_0 . This rounding argument uses a small integer spanning set of the rational space, not a lattice basis, and therefore incurs no covolume factor.

Finally, $m_h - 6^\Delta m' \in U \subseteq K_B$, so

$$a + (0, m'/L_{h-\Delta}) \in K_{\text{rat}}.$$

Its height remains at most $\mathcal{H}$. Lemma 8.2 makes its evaluation at most $\mathcal{T}^{-1}/r_{v_b}$, which is below the PRE threshold. Equation (21) gives the required direct edge. $\square$

8.5. Good modes and the bad-density count. For $h \geq \Delta$, call a mode *good at h* when

$$\lfloor \log_{1.1} |\pi(\mathcal{E}_{h-\Delta}(m))| \rfloor = \lfloor \log_{1.1} |\pi(\mathcal{F}_h(m))| \rfloor$$

and at least nine tenths of $\pi(\mathcal{F}_h(m))$ are nonexceptional and height-good at h . Every $h < \Delta$ is bad. A mode is a *core mode* if at most $\delta_*(J_* + 1)$ of its heights are bad; all other modes are *linear-side modes*.

Lemma 8.4 (Coherence of good predicates). *Good modes in one CURRENT component at height h have the same PRE component at height $h - \Delta$. The core/side decision is constant on each minimal PRE component. Moreover, core modes in one minimal CURRENT component have identical good predicates throughout*

$$\mathcal{H}_{\text{up}} := \{h \in \mathbb{Z} : \lceil 4\delta_* J_* \rceil \leq h \leq J_*\},$$

and have the same PRE components at height $h - \Delta$ there. Every core mode has bad proportion at most $2\delta_$ in $\mathcal{H}_{\text{up}}$.*

Proof. Write $F = \pi(\mathcal{F}_h)$. Each predecessor of a good mode has more than $|F|/1.1$ primaries. Two such predecessor primary sets intersect in more than $(2/1.1 - 1)|F|$, and thus their intersection contains a nonexceptional height-good primary, since fewer than one tenth of the primaries are excluded. The witnessing modes of that primary belong to $\mathcal{F}_h$. Lemma 8.3 joins them directly in PRE at $h - \Delta$, so the two predecessor components agree.

Modes in the same minimal PRE component have the same components at every larger height and in CURRENT, so all their predicates agree. This proves the second assertion. For the last assertion, two core modes have at most $2\delta_*(J_* + 1)$ bad heights between them. Our lower bound on J_* ensures that the interval of integer heights below $4\delta_* J_*$ contains more heights than this. They therefore have a common good height h_0 in that interval. Their CURRENT components at h_0 agree, since their minimal CURRENT components agree. The first assertion identifies their PRE components at $h_0 - \Delta$. Nesting then identifies all later PRE and CURRENT components, which also identifies the corresponding predicates. Finally, $|\mathcal{H}_{\text{up}}| \geq (1 - 4\delta_*)J_*$, so the full-range bad count $\delta_*(J_* + 1)$ is at most $2\delta_* |\mathcal{H}_{\text{up}}|$ for our small δ_* and sufficiently large J_* . $\square$

We give the counting argument for side modes explicitly. The simple summation principle behind it will be used twice.

Lemma 8.5 (Entry-rank summation). *Let $S_1 \subseteq \dots \subseteq S_T$ be nonempty finite sets with $|S_T| \leq M$. Suppose $X_t \subseteq S_t$, and each element belongs to at most A of the sets X_t . Then*

$$\sum_{t=1}^T \frac{|X_t|}{|S_t|} \leq AH_M, \quad H_M = \sum_{r=1}^{\lfloor M \rfloor} \frac{1}{r} \leq 1 + \log M.$$

Proof. Enumerate elements by first entry into the increasing sets, breaking ties arbitrarily. A containing set for the element of entry rank r has at least r elements. Its total contribution is at most A/r . Sum this bound over the elements. $\square$

Lemma 8.6 (Bad-density chains). *Fix $0 < \delta, \beta < 1$, integers $J \geq \Delta \geq 1$, and $M \geq 1$. For $1 \leq i \leq N$, $0 \leq h \leq J$, let $P_{i,h}$ and $C_{i,h}$ be nonempty sets of primary labels, each increasing in h , with cardinality at most M , and satisfying*

$$P_{i,h} \subseteq C_{i,h} \subseteq P_{i+1,h} \quad (i < N).$$

Let X_i be the exceptional primaries at position i , each primary exceptional at at most B positions. For a fixed position, suppose each nonexceptional primary is height-bad at at most A heights; write $Y_{i,h}$ for these height-bad primaries. Declare $h < \Delta$ bad, and for other heights declare h bad if the floor logs of $|P_{i,h-\Delta}|$ and $|C_{i,h}|$, to base 1.1, differ, or if

$$\frac{|(X_i \cup Y_{i,h}) \cap C_{i,h}|}{|C_{i,h}|} > \beta.$$

Suppose each position has more than $\delta(J+1)$ bad heights. Set $L = \lfloor \log_{1.1} M \rfloor$ and

$$Q = \Delta(L+1) + \beta^{-1}AH_M.$$

If $J+1 \geq 2Q/\delta$, then

$$N < \frac{2}{\delta}(L + \beta^{-1}BH_M).$$

Proof. Write $p_{i,h}$ and $c_{i,h}$ for the two floor logs at height h . Their values are integers between 0 and L . For $h \geq \Delta$, the first failure indicator is bounded by

$$c_{i,h} - p_{i,h-\Delta} = (p_{i,h} - p_{i,h-\Delta}) + (c_{i,h} - p_{i,h}).$$

At fixed i , the sum of the first term over h is at most ΔL , because each increment of the increasing sequence $p_{i,h}$ is counted at most Δ times. At fixed h , the temporal nesting gives

$$\sum_i (c_{i,h} - p_{i,h}) \leq c_{N,h} - p_{1,h} \leq L.$$

Including the initially bad heights, this contributes at most $N\Delta(L+1) + (J+1)L$ bad incidences.

The second failure indicator is bounded by β^{-1} times the sum of the exceptional and height-bad fractions. At fixed i , Lemma 8.5, applied as h increases, bounds the height-bad fractions by AH_M . At fixed h , apply that lemma as i increases to bound the exceptional fractions by BH_M . The total bad count is therefore at most

$$NQ + (J+1)(L + \beta^{-1}BH_M).$$

On the other hand it is greater than $\delta N(J+1)$. The assumed lower bound for $J+1$ absorbs NQ into half of this last quantity, proving the result. $\square$

8.6. Separated projection rows. We record the analytic consequence of graph nonadjacency that is used both for side events and for tops.

Lemma 8.7 (Separated mode rows). *Fix a role and group. Consider a family of dyadic cells J_e , with a nonempty set $\mathcal{M}_e$ of available mode labels on each cell. Suppose that along every path:*

- (1) *the sets $\mathcal{M}_e$ for cells containing that path are pairwise disjoint, and their total cardinality is at most D_* ;*
- (2) *for distinct overlapping cells, every cross pair of mode atoms has normalized inner product on the smaller cell at most D_*^{-200} in absolute value.*

Assume the same cross-pair bound for distinct sets on the same cell. Let P_e be orthogonal projection onto the restrictions of $\{\psi_m : m \in \mathcal{M}_e\}$, with output supported on J_e . Then

$$\sum_e \|P_e F\|^2 \leq C \|F\|^2$$

with an absolute constant.

The cross-pair hypothesis holds if the sets have no shared or adjacent modes in the later PRE graph and J_e are the associated pre-cells. It also holds for actual-use cells if the sets have no shared or adjacent modes in the later CURRENT graph.

Proof. First verify the asserted oscillatory bound. The common factor $e(c_l p^\rho)$ cancels between two physical atoms. Fourier-expand $(\zeta^\rho)^2$ on the ϑ torus, retaining frequencies of size at most B_* , with absolute coefficient sum at most D_* and tail at most D_*^{-300} . At height zero these shifts are allowed because $E_0 = A_0$ exceeds B_* . At a larger height the corresponding shifts are multiplied by L_h , and $B_* L_h \leq E_h$, by the choice of A_0 .

For each retained shift, nonadjacency gives a point of the relevant pre-cell where the absolute phase derivative exceeds its graph threshold. The affine-slope oscillation bound makes the absolute derivative at every point of that cell at least half the threshold. The derivative is affine and of constant sign, so the elementary first-derivative oscillatory estimate on any interval applies. For PRE on a pre-cell it bounds the normalized integral by C/Ξ . For CURRENT on an actual cell I , it gives the same bound, since $|I| \geq r_{t_b}$. The cutoff coefficient sum and tail give

$$CD_*/\Xi + D_*^{-300} \leq D_*^{-200}$$

after increasing the starting budget. Distinct labels have orthogonal private coordinates, so they contribute no other inner product. Earlier modes satisfy the same slope-variation estimate on the later pre-cell by inheritance.

For the Bessel conclusion it suffices, by duality, to bound the synthesis of arbitrary f_e in these projection ranges. Write $f_e = \mathbf{1}_{J_e} \sum_{m \in \mathcal{M}_e} \beta_{e,m} \psi_m$, and put $b_e = (\sum_m |\beta_{e,m}|^2)^{1/2}$. Private coordinates give

$$|J_e| b_e^2 \leq D_*^2 \|f_e\|^2.$$

For an overlapping pair, constancy of these coefficients on their cells and the cross-pair estimate bound its inner product by

$$D_*^{-199} |J_e \cap J_f| b_e b_f.$$

There are at most D_* participating rows at a path point. Consequently

$$\begin{aligned} \sum_{e \neq f} |\langle f_e, f_f \rangle| &\leq D_*^{-199} \int_R \left(\sum_{e: x \in J_e} b_e \right)^2 dx \\ &\leq D_*^{-198} \sum_e |J_e| b_e^2 \leq D_*^{-196} \sum_e \|f_e\|^2. \end{aligned}$$

Adding the diagonal proves the synthesis bound and hence the row Bessel bound. This calculation uses a pathwise label count; it does not count the number of unused depths between rows. $\square$

Lemma 8.8 (Composition with the lift and main histories). *Fix a role. For each group, suppose a family of mode projection rows P_e , supported on cells J_e , has squared Bessel bound at most G_0 . Suppose $J_e \subseteq u_e$, where u_e is a usable main endpoint cell with a downstream actual-use certificate. The mode list of P_e is drawn from the list available at u_e . Then the frozen row outputs*

$$P_e \mathcal{R}_{l,G} \bar{L}_{l,G,u_e} B_{l,u_e} B_{l,u_e}$$

satisfy a joint squared bound $G_0 |R|$, over all groups and rows. Here B_{l,u_e} is the basic-B input generated by its specified constituent main projection families from the original normalized input.

Proof. Initially omit the shared histories. For one group, $P_e \mathcal{R}_{l,G}$ is a Bessel family with the same type of bound, since $\mathcal{R}_{l,G}$ is one fixed bounded operator. It depends only on J_e , and therefore also only on the containing decision cell u_e . Lemma 3.4 applies to the outer individual main ball projection at that cell. It permits arbitrarily many rows with the same decision cell. Its loss is a fixed power of $\log(2D_m)$. Averaging the pad choices uses the triangle inequality in the Hilbert sum of row outputs and costs no number-of-pads factor.

Let A_G be the resulting row operator for this group. For an individual outer projection its row adjoint has the order

$$L_{l,G,u_e} \mathcal{R}_{l,G}^* P_e.$$

Its range is thus in the main ball range on the entire cell u_e . With an averaged projection, it lies in the sum of these usable ranges. Lemma 6.5 gives

$$\left\| \sum_G A_G^* w_G \right\|^2 \leq C \sum_G \|A_G^* w_G\|^2 \leq G_0 \sum_G \|w_G\|^2.$$

Duality proves the joint Bessel bound before the shared histories. This is the step for which full endpoint cells and their downstream certificates are needed.

Finally apply Lemma 3.4 again to these joint rows, with decision cells u_e , to insert the shared histories. Expand complements and the fixed signed sum in B_l , and keep the positive averages as averages. Their product lengths are $O(\log \mathcal{L})$, their column log-budgets are polynomial in k , and the number of signed terms is a small overhead. The compression loss is therefore G_0 . The original input has squared norm at most $|R|$. All operators have remained in their indicated order. $\square$

Proposition 8.9 (Square energy of the linear side). *For each basic-B role, on every retained v_b -cell let $V_{l,G,b}$ be the exact subtotal of the linear-side modes in $\tilde{Z}_{l,G,b}$, including their private coordinates. Set it to zero on branches without a needed output or certificate. Then*

$$\sum_{G,b} \|V_{l,G,b}\|^2 \leq G_0 |R|.$$

In particular these functions satisfy Equation (17), uniformly in the branch offsets.

Proof. Fix l, G . An event is a triple $(b, J, \mathcal{E})$, where J is a retained v_b -cell and $\mathcal{E}$ is a minimal PRE component of side modes. Such a component consists entirely of side modes by Lemma 8.4. Split block indices into their ten congruence classes. Within one class, link an event to a later event on a descendant path when some of their modes agree or are adjacent in the later minimal PRE graph.

For two linked events, every earlier CURRENT edge persists in the later PRE graph. Indeed the later pre-cell is contained in the earlier pre-cell, the mode lists persist, and the separation of block starts gives $v_{\text{later}} \geq t_{\text{earlier}}$. The minimal PRE link also persists at every height. Consequently, choosing any mode in each event of a linked chain gives

$$\pi(\mathcal{E}_h^{(i)}) \subseteq \pi(\mathcal{F}_h^{(i)}) \subseteq \pi(\mathcal{E}_h^{(i+1)})$$

at every height.

Apply Lemma 8.6 with

$$M = D_m, \quad \beta = \frac{1}{10}, \quad A = C d_{\max} R_*^{\text{ht}}, \quad B = C d_{\max}, \quad \delta = \delta_*.$$

The height multiplicity comes from Lemma 8.3; the exceptional-block multiplicity comes from Lemma 8.2. Every event mode has more than $\delta_*(J_* + 1)$ bad heights. Our choice of J_* meets the lemma's lower bound. Thus every chain has length at most

$$C_{\delta_*} d_{\max} \log(2D_m).$$

For clarity, the inequality being absorbed is

$$\delta_* N(J_* + 1) \lesssim N d_{\max} R_*^{\text{ht}} \log(2D_m) + (J_* + 1) d_{\max} \log(2D_m).$$

Although R_*^{ht} depends on the mode budget, it disappears from the final chain-length bound.

Give each event its longest-chain level. Linked events have different levels. Distinct overlapping events in one level therefore have no shared or adjacent modes in the later minimal PRE graph. Events on one cell are distinct components and have the same separation. Their label sets are pairwise disjoint on a path, with total count at most D_* . Lemma 8.7 applies to the orthogonal projection $P_{\mathcal{E},J}$ onto the augmented columns of $\mathcal{E}$, restricted to J . Summing the ten congruence classes and the bounded number of levels gives squared Bessel bound G_0 for the event rows.

The decision cell for a row is its containing u_b -cell. Apply Lemma 8.8 to obtain

$$\sum_{G,b,J,\mathcal{E}} \|P_{\mathcal{E},J} \tilde{Z}_{l,G,b}\|^2 \leq G_0 |R|. \quad (24)$$

Zero outputs from pre-depths before the attempt were omitted.

It remains to pass from projections to exact coefficient subtotals. Write $T_{\mathcal{E},J}$ for that subtotal. Every mode outside $\mathcal{E}$ is PRE-nonadjacent to every mode inside it. The oscillatory estimate in Lemma 8.7, the private-coordinate lower bound, and $\sum_m |\alpha_m| \leq D_*$ give

$$\|P_{\mathcal{E},J} \tilde{Z}_{l,G,b} - T_{\mathcal{E},J}\|_{2,J} \leq D_*^{-190}.$$

For example, pairing the complementary subtotal against a unit vector in this range costs its coefficient ℓ^1 norm, at most $D_*^{3/2}$, times the outside coefficient sum and the pair bound D_*^{-200} . Within one event level the number of occurrences on a path is at most D_* . The number of levels is G_0 , and the budgets also include all groups present on these full-cell paths. Thus the squared additive errors summed over all events are negligible compared with $|R|$.

Finally, on each J the different minimal PRE subtotals are almost orthogonal by the same coefficient calculation. Their sum is $V_{l,G,b}$. Summing their squared norms in Equation (24) therefore proves the assertion. $\square$

8.7. Fixing offsets and separating the side terms. Define $A_{l,G,b}$ as the exact core subtotal of $\tilde{Z}_{l,G,b}$. In a basic- B role put

$$S_{l,G,s} = Z_{l,G,s} - Z_{l,G,u_b} + V_{l,G,b}, \quad b \leq s < b + K.$$

In a basic- U role set $S_{l,G,s} = Z_{l,G,s}$, with no core term. Taking physical components gives

$$Z_{l,G,s} = A_{l,G,b} + S_{l,G,s} + \mathcal{E}_{l,G,b}, \quad \mathcal{E}_{l,G,b} = (Z_{l,G,u_b})_{\text{phys}} - (\tilde{Z}_{l,G,b})_{\text{phys}}.$$

Only this displayed identity uses the scalar error $\mathcal{E}_{l,G,b}$; the mode subtotals themselves retain their augmented coordinates.

We can choose the offsets so that the errors have negligible integrated square energy on all actual uses. To see this without any independence assumption about the eventual core choices, let

$$Z_{l,G,u_b} = \sum_{\Phi} \beta_{\Phi} \Phi.$$

The explicit lift gives

$$\mathcal{E}_{l,G,b} = (1 - (\zeta^{\rho})^2)(Z_{l,G,u_b})_{\text{phys}} - \sum_{\Phi} \beta_{\Phi} q_{\Phi}.$$

The coefficients β_{Φ} , their normalized main bounds, and the full collection of actual main uses were fixed before drawing the offsets. On its own u_b -cell, an outer main projection has coefficient sum and physical supremum at most D_m^C . This bound follows from its private weights and the normalized input on that cell and does not lose a factor from the lag $4K$. Therefore

$$\mathbb{E}_{\rho} |\mathcal{E}_{l,G,b}(y)|^2 \leq D_m^C d_{\max} \delta_c$$

on every needed branch. Summing over all original actual cell/group uses, before any offset-dependent mode selection or pruning, gives

$$\mathbb{E}_\rho \sum_{l,G,I \text{ used}} |I| \|\mathcal{E}_{l,G,b(I)}\|_{2,I}^2 \leq D_m^C d_{\max} \delta_c |R|.$$

By enlarging C_c , this is smaller than any required fixed inverse main-budget power times $|R|$. Choose offsets satisfying the resulting bound and keep them fixed. Subsequent core decisions, restrictions, and stopping only remove nonnegative terms from this error sum. They need not be independent of the offsets. Proposition 8.9 was uniform in the offsets, so its estimate remains valid for this choice.

The rolling variation bounds and the path pruning in Section 7, now supplied with Proposition 8.9, give

$$\sum_{s,G} \|S_{l,G,s}\|_{2,I_s}^2 \leq P^C$$

along a retained path. Individual core sizes are also at most P^C : use the bounded lift, $Z_{u_b} = Z_s - (Z_s - Z_{u_b})$, and the corresponding bounds for $V_{l,G,b}$. The local absolute L^2 form bound and Cauchy–Schwarz over the original actual uses now make every term containing a physical error negligible; its other three sizes are at most P^C , and the original path multiplicity is within the main budget.

Expand Equation (14) with these arguments. Keep the terms with at least three core roles for the next section. For a term with exactly two side roles, replace its two core complements by the whole current $Z_{G,s}$ arguments. The corrections have at least three sides, apart from the physical errors just treated. Proposition 7.5 applies to the resulting two-side terms with whole complements, through Lemma 6.9; it also bounds the terms with three or four sides. Its powered costs use the minimum in Equation (11) and obtain the factor η^ε for any fixed sufficiently small $\varepsilon > 0$. An original basic- U role has remained a mandatory side throughout. In particular, the restored complementary arguments are the specified basic- B inputs required by the pair-smallness estimate.

8.8. Hereditary representatives. Fix a height $h \in \mathcal{H}_{\text{up}}$ and a basic- B role. Call a CURRENT component $\mathcal{F}_h$ *successful* if it contains a core mode good at this height. By Lemma 8.4, all its good modes have one predecessor $\mathcal{E}_{h-\Delta}$. Its *type* is the pair

$$(\lfloor \log_{1.1} |\pi(\mathcal{E}_{h-\Delta})| \rfloor, \lfloor \log_{1.1} |\mathcal{E}_{h-\Delta}| \rfloor).$$

The first coordinate counts primaries and the second counts modes. Both are needed in the following construction.

Lemma 8.10 (Hereditary representatives). *Fix the height, role, group, and type. We can choose a representative mode $a \in \mathcal{E}_{h-\Delta}$ for every successful component so that it is inherited whenever an earlier successful predecessor on the same path is contained in the current predecessor. Distinct representatives on comparable actual-use cells have no adjacent pair of component modes in the later CURRENT graph. In particular, if I, I' are such cells and $x_* \in I \cap I'$, then their distinct representatives satisfy*

$$|\omega_a(x_*) - \omega_{a'}(x_*) + n \cdot \vartheta/L_h| > \frac{\Xi}{2} \max(|I|^{-1}, |I'|^{-1}), \quad n \in \mathbb{Z}^{\dim \theta_z}, \quad |n|_\infty \leq E_h. \quad (25)$$

Proof. Say an earlier successful component links to a later one if a mode of the earlier component is equal or CURRENT-adjacent, at the later height- h graph, to a mode of the later component. Persistence of the earlier CURRENT edges then gives $\mathcal{F}_{\text{early}} \subseteq \mathcal{F}_{\text{late}}$. Write $F = \pi(\mathcal{F}_{\text{late}})$. Later goodness and the common first type coordinate imply

$$|\pi(\mathcal{E}_{\text{late}})| > |F|/1.1, \quad |\pi(\mathcal{E}_{\text{early}})| > |\pi(\mathcal{E}_{\text{late}})|/1.1 > |F|/1.1^2.$$

Both predecessor primary sets lie in F . Their intersection therefore has more than $(1/1.1 + 1/1.1^2 - 1)|F|$ elements, a proportion greater than one tenth. It contains a later nonexceptional height-good primary. Two witnessing modes of that primary lie in $\mathcal{F}_{\text{late}}$, one from each predecessor. Lemma 8.3 joins them in the later PRE graph at $h - \Delta$. Earlier PRE edges persist, so

$$\mathcal{E}_{\text{early}} \subseteq \mathcal{E}_{\text{late}}.$$

The same argument at one block reduces to uniqueness of the good predecessor.

Now process the considered pre-cells in block order. If a current predecessor contains an earlier successful predecessor of this type on its path, inherit its representative. Otherwise choose any mode of the current predecessor by a fixed ordering of labels. To prove consistency, take any two earlier predecessors contained in the current one. Equality of the second type coordinate gives each more than $1/1.1$, and in particular more than one half, of the current mode count. They intersect. They lie on one path, so their shared mode is a link from the earlier to the later of them. The inclusion just proved makes them nested. By induction all their inherited representatives agree. Comparable pre-cell data occur on ancestors, and the list and graph are constant throughout each pre-cell, so this defines representatives consistently for all its actual descendants.

If two components on comparable uses with this type had equal or adjacent modes in the later CURRENT graph, the same inclusion argument would force inheritance of the earlier representative. Thus distinct representatives imply the asserted nonadjacency of every cross pair of modes. Suppose I' is the later, smaller actual cell, in a block with pre-cell J' and comparison depth $t_{b'}$. For each allowed n , nonadjacency gives

$$\sup_{x \in J'} |\omega_a(x) - \omega_{a'}(x) + n \cdot \vartheta / L_h| > \Xi / r_{t_{b'}}.$$

The slope oscillation on J' is negligible compared with this threshold, so its value at x_* is greater than $\Xi / (2r_{t_{b'}})$. Since $|I'| \geq r_{t_{b'}}$, this implies Equation (25). $\square$

The representatives are mode labels, not numerical frequencies; different labels may initially have identical scalar atoms. Equation (25) is the separation obtained for the distinct labels that actually occur with one fixed type on a path. All lattice shifts in this equation, and in the later frequency boxes, are integral.

8.9. Sizes of components. At an actual cell I in the block, for each successful $\mathcal{F} = \mathcal{F}_h$ define

$$\sigma_{\mathcal{F}}^2 = \sum_{\mathcal{D} \subseteq \mathcal{F}} \left(\|A_{\mathcal{D}}\|_{2,I} + D_*^{-30} \sum_{\substack{m \text{ core} \\ m \in \mathcal{D}}} |\alpha_m| \right)^2. \quad (26)$$

Here $\mathcal{D}$ ranges over the minimal CURRENT components, and $A_{\mathcal{D}}$ is its full core subtotal, in the augmented mode space. The small coefficient term will absorb later uniform linearization and Fourier-truncation errors without replacing the component's L^2 amplitude by its coefficient sum.

Let $P_{\mathcal{F},I}$ denote orthogonal projection on the restrictions to I of all mode labels in $\mathcal{F}$, and let $V_{\mathcal{F}}$ be the exact linear-side subtotal on those labels.

Lemma 8.11 (Component-size comparison). *At every actual cell,*

$$\begin{aligned} \sigma_{\mathcal{F}}^2 &\lesssim \|P_{\mathcal{F},I} \tilde{Z}_{l,G,b}\|_{2,I}^2 + \|V_{\mathcal{F}}\|_{2,I}^2 + D_*^{-50}, \\ \sum_{\mathcal{F}} \|V_{\mathcal{F}}\|_{2,I}^2 &\lesssim \|V_{l,G,b}\|_{2,I}^2, \end{aligned} \quad (27)$$

where the second sum can be over the successful components. The same almost orthogonality controls arbitrary coefficient subtotals separated into distinct minimal CURRENT components.

Proof. Modes in different minimal CURRENT components are nonadjacent in that graph. The oscillatory proof of Lemma 8.7 applies on I , because $|I| \geq r_{t_b}$. Its synthesis calculation on this single cell shows that for arbitrary subtotals $f_{\mathcal{D}}$ in distinct components,

$$\left\| \sum_{\mathcal{D}} f_{\mathcal{D}} \right\|_{2,I}^2 \asymp \sum_{\mathcal{D}} \|f_{\mathcal{D}}\|_{2,I}^2.$$

The constants are uniform, since the off-diagonal norm is at most a fixed small inverse power of D_* after the private-coordinate bound. In particular this is a two-sided estimate and applies to any subset of labels inside each component.

Let $T_{\mathcal{F}}$ be the full coefficient subtotal on $\mathcal{F}$. Its core subtotal is $A_{\mathcal{F}} = T_{\mathcal{F}} - V_{\mathcal{F}}$. The preceding estimate bounds the sum of the squared core component norms in Equation (26) by $C\|A_{\mathcal{F}}\|_{2,I}^2$. The contribution of the coefficient floors is at most

$$CD_*^{-60} \left(\sum_m |\alpha_m| \right)^2 \leq CD_*^{-58}.$$

Every mode outside $\mathcal{F}$ is CURRENT-nonadjacent to the modes inside. Pairing against a unit vector of its augmented span, as in the proof of Proposition 8.9, gives

$$\|P_{\mathcal{F},I} \tilde{Z}_{l,G,b} - T_{\mathcal{F}}\|_{2,I} \leq D_*^{-190}.$$

Combine these observations with $\|T_{\mathcal{F}} - V_{\mathcal{F}}\|^2 \leq 2\|T_{\mathcal{F}}\|^2 + 2\|V_{\mathcal{F}}\|^2$ to obtain the first bound, with its stated generous error.

Finally decompose $V_{l,G,b}$ by all CURRENT components at height h , whether successful or not. They too are separated, so the two-sided almost orthogonality gives the second bound after discarding the unsuccessful terms. $\square$

8.10. First tops and the pathwise label bound.

Proposition 8.12 (Top count). *After removing summands on first stopping cells of arbitrarily small fixed total relative length, the following holds on every retained path. Fix a core role j , a height $h \in \mathcal{H}_{\text{up}}$, a type, and a dyadic amplitude bin $\sigma_j \leq \sigma_{\mathcal{F}} < 2\sigma_j$. Let $\mathcal{A}_j^G$ be the distinct representatives of successful components of this type actually used in that bin along the path. Then*

$$\sum_G |\mathcal{A}_j^G| \leq \min(D_*, P^C \sigma_j^{-2}). \quad (28)$$

The exponent C is fixed. Enlarging D_ by a fixed power is harmless in this statement. On retained uses, all component sizes and full local side sizes are at most P^C , and for a side role*

$$\sum_{s,G} \|S_{l,G,s}\|_{2,I_s}^2 \leq P^C.$$

Proof. First fix one height, type, and dyadic threshold $D_*^{-1} \leq \sigma \leq D_*^2$. For each group and representative, take the first actual-use cells on each spatial branch for which $\sigma_{\mathcal{F}} \geq \sigma$. Call them its first tops. First tops of the same representative do not overlap. For two overlapping first tops of different representatives, Lemma 8.10 gives CURRENT nonadjacency of all their component modes at the later block. The label sets are therefore disjoint on a path, with total mode count at most D_* . Lemma 8.7 proves a uniform Bessel bound for the rows $P_{\mathcal{F},I}$ at these tops.

The row is supported on its actual cell I , which is contained in the u_b -cell of its frozen list. This remains a valid decision cell for Lemma 8.8, even if many rows share it. All these endpoint cells have the top itself as a downstream actual-use certificate. That lemma gives

$$\sum_{\text{tops}} \|P_{\mathcal{F},I} \tilde{Z}_{l,G,b}\|^2 \leq G_0 |R|.$$

No restriction of the input ball projection to the smaller top cell is made in this step.

For the side contributions in Equation (27), sum first at one depth of one block. The actual cells there are disjoint; at each cell the component sum is bounded by $\|V_{l,G,b}\|_{2,I}^2$. Summing the at most K depths and then applying Proposition 8.9 gives

$$\sum_{\text{tops}} |I| \|V_{\mathcal{F}}\|_{2,I}^2 \leq CK \sum_{G,b} \|V_{l,G,b}\|^2 \leq KG_0 |R|.$$

Since $\sigma_{\mathcal{F}} \geq \sigma$, the size comparison now yields

$$\sigma^2 \sum_{\text{tops}} |I| \leq C(1+K)G_0 |R| + CD_*^{-50} \sum_{\text{tops}} |I|.$$

The final term can be absorbed, because $\sigma^2 \geq D_*^{-2}$. Thus

$$\sigma^2 \sum_{\text{tops}} |I| \leq C(1+K)G_0 |R|.$$

There are only polynomially many choices in P, k of height, type, and threshold: the counts are controlled by J_* and powers of $\log(2D_*)$. For each choice let $N_{\sigma}(x) = \sum_{\text{tops}} \mathbf{1}_I(x)$, summing also over groups. The displayed integral bound and Markov's inequality give a pathwise bound $N_{\sigma}(x) \leq P^C \sigma^{-2}$ outside a set of as small a fixed measure as desired, after enlarging C and the constant in the stopping threshold. Summing these exceptional measures over all the choices still costs a small fraction of $|R|$. Implement the cut by stopping at the first actual cells where the corresponding partial top count exceeds its threshold. All the estimates are valid again on normalized subroots, so these summands are assigned to fresh attempts.

If a representative is used in the bin $[\sigma_j, 2\sigma_j)$ on a retained path, its first σ_j -top is an ancestor of that use. It too is retained; otherwise the descendant summand would already have been sent to a fresh attempt. Different representatives have different such tops on the path, so their count is at most $N_{\sigma_j}(x)$. The full mode-label budget gives the additional bound D_* . For $\sigma_j < D_*^{-1}$, that full bound alone implies the minimum in Equation (28). Above the tested threshold range the bins are empty by the coefficient and atom bounds, after the allowed budget enlargement.

The side square bound is the pruning already made following Proposition 8.9. For amplitude control, $Z_{u_b} = Z_s - (Z_s - Z_{u_b})$, the bounded pointwise lift, and the side bounds give $\|\tilde{Z}_{l,G,b}\|_{2,I} + \|V_{l,G,b}\|_{2,I} \leq P^C$. Projection is a contraction on I , so Lemma 8.11 bounds $\sigma_{\mathcal{F}}$ by P^C as well. $\square$

The statements of Lemmas 8.4 and 8.10, the size definition Equation (26), and Proposition 8.12 are the inputs to the core estimate. In particular, upper-range good-height masks select unions of full core subtotals in minimal CURRENT components, and the representatives used by a fixed role, height, type, and amplitude bin satisfy both the separation in Equation (25) and the joint group count in Equation (28).

9. SUCCESSFUL HEIGHTS AND CONCENTRATION ACROSS SCALES

We estimate the terms containing at least three core arguments. Throughout this section we work on one retained attempt with root R , and on one of the search leaves in the localization. All the conclusions are uniform in that leaf. We use the mode construction, the retained path bounds, and the notation of Section 8. In particular, $\log D_* \leq \text{poly}(P, k)$, whereas every loss described below as P^C has an absolute exponent. The exponent can be increased a finite number of times without changing this convention.

We first sum the common-height histories and reduce each fixed configuration to frequency tests with torus amplitude bounds. A finite sparse-count lemma then controls the tested tuples across scales. Finally, interpolation sums the group, amplitude, and frequency parameters. The original cell minimum is applied before subadditivity, keeping each tuple count inside its power; numerical errors are estimated separately in absolute value.

9.1. A common successful height. Fix a term with $m_0 \in \{3, 4\}$ core roles. If $m_0 = 3$, denote the remaining, side role by l . Fix an actual use $I = I_s$, a group G , and a point $x_* \in I$. Write $r = |I|$ and $R_s = r^{-1}$. The choice of x_* is arbitrary; eventually we integrate the resulting estimates over such points. All mode lists here are finite.

Sample heights independently and uniformly from the upper height range. For a fixed tuple of core modes, retain the first sampled height at which all its roles are good. A core mode has bad density at most $2\delta_*$ in this range, so the probability of success in one trial is at least $1 - 2m_0\delta_* > 0$. Consequently the first-success rule gives an exact identity in expectation for each pure tuple, and hence for the finite mode expansion of the form.

Suppose that success occurs at trial $n + 1$, with final height h . Assign each of the previous n failed trials to its first failing role, in the natural order of the roles. There are at most m_0^n assignments. For a fixed assignment the conditions in each role form a product mask on its modes: that role must fail at the trials assigned to it, and must succeed at the earlier roles' required positions. Group the selected modes at the final CURRENT components $\mathcal{F}_j$. The coherence of the upper-height predicates on a minimal CURRENT component implies that the selected modes are unions of entire core subtotals $A_{\mathcal{D}}$ from (26). Final success puts all of them in the unique good predecessor of $\mathcal{F}_j$. Let a_j denote the inherited representative of that predecessor.

Put

$$\xi = \frac{\vartheta}{L_{h-\Delta}}, \quad \gamma_j^* = \omega_{a_j}(x_*).$$

A PRE path from a selected mode to its representative gives an integral vector n_j with $|n_j|_\infty \leq D_* E_{h-\Delta}$ such that its instantaneous slope at x_* differs from $\gamma_j^* + n_j \cdot \xi$ by at most $D_* \Xi / r_{v_b}$. The pre-lag makes the product of this error with r arbitrarily small at a prescribed inverse power of D_* . The quadratic Taylor error is equally small: the mode was admitted by u_b , and its curvature at its admission scale was bounded by D_* . The phase constant at x_* can be absorbed into its coefficient. Thus the physical selected argument, up to a uniform error bounded by D_*^{-150} times its absolute coefficient sum, has the form

$$e(c_j p^{\rho_j}(y_j)) e(\gamma_j^*(y_j - x_*)) g_j(\xi(y_j - x_*)). \quad (29)$$

Here g_j is the true periodic cutoff times a finite trigonometric polynomial. Indeed the cutoff is periodic on the ϑ torus, and dilation by the integer $L_{h-\Delta}$ makes it periodic on the ξ torus as well. We first use this true cutoff when identifying the progression multiplier, and then use its Fourier truncation.

The cutoff frequencies on this torus have size at most $B_* L_{h-\Delta}$. The selected mode frequencies, the cutoff factors, and the progression multiplier therefore have, in progression coordinates anchored at any role, their relevant Fourier supports in

$$|n|_\infty, |m|_\infty \leq B'_h, \quad B'_h = D_*^5 E_{h-\Delta}.$$

The absolute Fourier costs, apart from the selected absolute mode coefficient sums themselves, are bounded by fixed powers of D_* . Dilating the torus frequency indices does not change those absolute coefficient sums. All precision exponents used here and below are fixed before the parameter choices; increasing the pre-lag and the Fourier cutoff by the required fixed powers preserves their polynomial logarithmic budgets.

If there is a side role, insert in it a fresh cutoff ζ^{ρ_l} , with ρ_l uniform, divided by its constant expectation. That expectation is at least $1/2$. Set, on I ,

$$S' = \zeta^{\rho_l} e(-c_l p^{\rho_l}) S_{\text{phys}}.$$

Then $\|S'\|_{2,I} \leq \|S\|_{2,I}$. On the supports of the true cutoffs the product of all four branch phases is the smooth periodic progression multiplier constructed in Section 8. Write its shifted, truncated

version as $R_z(u, v)$, where $u = \xi(y_{l'} - x_*)$ and $v = \xi t$. Its supremum is bounded by an absolute constant, uniformly in every offset and in the anchor l' .

9.2. Torus norms and the frequency tests. Successful height h supplies a nonexceptional primary. We can therefore apply (23) to the pure ϑ shifts in the present Fourier boxes. Let K_B be its rational tiny-frequency subspace in the horizontal coordinates and set

$$\mathbb{H} = (K_B \cap \mathbb{Z}^{\dim \theta_z})^\perp \subseteq \mathbb{T}^{\dim \theta_z}.$$

The lattice $K_B \cap \mathbb{Z}^{\dim \theta_z}$ is saturated: if a nonzero integer multiple of an integer vector belongs to it, the vector itself belongs to the rational subspace K_B . Its annihilator $\mathbb{H}$ is consequently a connected torus. For every character difference in the Fourier ranges under consideration, $rn \cdot \xi$ is tiny when $n \in K_B$, and is huge otherwise.

Expand the squared modulus of g_j in characters. Averaging its restriction to $\xi(I - x_*)$ gives 1 up to a tiny error for a character in K_B , and a tiny error for every other character, by integration of an exponential over an interval. Haar integration on $\mathbb{H}$ gives exactly 1 and 0, respectively. The difference between these averages is bounded by D_*^{-100} times the square of the selected absolute coefficient sum. The approximation in (29), the almost orthogonality of distinct minimal CURRENT components on I , and the floors in (26) thus give

$$a'_j := \|g_j\|_{L^2(\mathbb{H})} \leq C \left[\sum_{\mathcal{D} \text{ selected}} \left(\|A_{\mathcal{D}}\|_{2,I} + D_*^{-30} \sum_{m \text{ core in } \mathcal{D}} |\alpha_m| \right)^2 \right]^{1/2}. \quad (30)$$

The augmented norm on the right also bounds the physical norm, so discarding the private coordinates causes no loss. The floors absorb the stated errors even after summing over all components.

We shall use the following Haar estimate. On $\mathbb{H}^2$, any pair of distinct progression coordinates $u + (j - l')v$ and $u + (j' - l')v$ has independent Haar distribution. The matrix defining this map has nonzero integer determinant $j' - j$ and is surjective on a connected torus; the pushforward of Haar probability is therefore Haar probability. Pairwise independence proves the absolute form bound with two L^1 factors and two L^∞ factors. Multilinear interpolation among these six bounds gives

$$\int_{\mathbb{H}^2} \prod_{j=0}^3 |h_j(u + (j - l')v)| \, du \, dv \leq \prod_{j=0}^3 \|h_j\|_{L^2(\mathbb{H})}.$$

A uniformly bounded multiplier may be inserted at a fixed absolute cost. This statement includes the zero-dimensional torus.

Let $U_0 = D_*^{250}$. First suppose that all four roles are core, and use anchor $l' = 0$. Put

$$Q = \sum_j \gamma_j^*, \quad T = \sum_j j \gamma_j^*.$$

Seek a pair n, m in the full shift box with

$$q' = Q + n \cdot \xi, \quad t' = T + m \cdot \xi, \quad \max(|q'|, |t'|) \leq U_0 R_s.$$

If such a pair exists, choose one by a fixed rule independent of all sampling histories and mode masks. Let $\mu \leq 1 \leq \nu$ be dyadic parameters, with fixed endpoint conventions, such that

$$\mu \asymp \min \left(1, \min_{0 \leq j \leq 3} \frac{|t' - j q'|}{R_s} \right), \quad \nu \asymp \max \left(1, \frac{|q'|}{R_s}, \frac{|t'|}{R_s} \right). \quad (31)$$

We discard a tuple if no reference exists or if the inner minimum in (31) is smaller than U_0^{-1} . These discards have the small absolute errors specified below.

Here is the Fourier estimate behind this assertion. In normalized anchor coordinates write the kernel as $W_s(h, v)$ and put

$$D(u, v) = R_z(u, v) \prod_j g_j(u + jv).$$

Expanding D samples the Fourier transform of W_s at the frequency pairs $r(Q + n \cdot \xi, T + m \cdot \xi)$. Two pairs that are both moderate must have the same pair of character cosets modulo K_B , by the tiny/huge dichotomy. Within those cosets their real frequencies differ from the reference by a tiny quantity, so the sampled Fourier values may be replaced by the reference value. Every other coset contributes only the rapidly decaying Fourier tail. The sum of the coefficients in the retained coset pair is

$$\int_{\mathbb{H}^2} D(u, v) e(-n \cdot u - m \cdot v) du dv,$$

whose absolute value is at most $C \prod_j a'_j$ by the preceding Haar estimate. The four vanishing marginals of the kernel say that its Fourier transform vanishes on all four lines $t' = jq'$. Smoothness, the mean value theorem near those lines, and rapid decay away from the origin show that its reference value is bounded by $C_L \mu \nu^{-L}$ for every fixed L . We obtain $C_L \mu \nu^{-L} \prod_j a'_j$. If no moderate reference exists, rapid decay instead gives a negligible contribution. If the minimum is less than U_0^{-1} , the vanishing-line bound does the same.

Now suppose that there are three core roles and use anchor $l' = l$. Set

$$Q = \sum_{j \neq l} \gamma_j^*, \quad T_l = \sum_{j \neq l} (j - l) \gamma_j^*.$$

Seek m in the shift box with $|T_l + m \cdot \xi| \leq U_0 R_s$ and choose a reference independently of masks as above. Write $\beta = r(T_l + m \cdot \xi)$, and discard when no reference exists or when $|\beta| < U_0^{-1}$. In this case set $\mu \asymp \min(1, |\beta|)$ and $\nu \asymp \max(1, |\beta|)$.

Fourier integration in the t variable gives a smooth function $W_\beta(h)$ supported in the normalized interior of the anchor interval. Every fixed number of its h derivatives is bounded by $C_L \mu \nu^{-L}$. Indeed the anchor marginal vanishes at $\beta = 0$, and integration by parts in t gives decay for large β . Only the reference m coset can contribute appreciably. With

$$D(u, v) = R_z(u, v) \prod_{j \neq l} g_j(u + (j - l)v), \quad D_m(u) = \int_{\mathbb{H}} D(u, v) e(-m \cdot v) dv,$$

duality and the four-factor Haar estimate give

$$\|D_m\|_{L^2(\mathbb{H})} \leq C \prod_{j \neq l} a'_j.$$

Collect its horizontal Fourier coefficients by character cosets and choose one reference n in the box for each coset. Parseval bounds the resulting coefficient sequence in ℓ^2 by the displayed norm. Different reference frequencies $rn \cdot \xi$ are separated by more than 1. Replacing the other frequencies of the coset by its reference again costs only a tiny absolute error.

Expand S' in normalized Fourier coefficients on I , writing $\widehat{S}'_I(k')$ for the coefficient of $e(k'(y - x_*)/r)$, $k' \in \mathbb{Z}$. The matrix of remaining pairings has entries bounded, for arbitrarily large fixed A , by

$$C_{L,A} \mu \nu^{-L} (1 + |k' + r(Q + n \cdot \xi)|)^{-A}.$$

In a shell of radius Λ , the number of nonzero entries in any row or column is $O(\Lambda)$, since the k' are integers and the reference real frequencies are 1-separated. Schur's inequality, with A increased to absorb this count, bounds that shell by $C_L \mu \nu^{-L} \Lambda^{-L} \prod_j a'_j$ times the ℓ^2 mass of $\widehat{S}'_I$ in

the corresponding windows. Averaging the fresh offset and applying Cauchy–Schwarz proves the bound

$$C_{L\mu\nu}^{-L} \prod_{j \neq l} a'_j \sum_{\substack{\Lambda=1,2,4,\dots \\ \Lambda \leq U_0}} \Lambda^{-L} b_\Lambda, \quad b_\Lambda^2 = \mathbb{E}_{\rho_r!} \sum_{\substack{k' \in \mathbb{Z}: |k' + r(Q+n \cdot \xi)| \leq 2\Lambda \\ \text{for some } n \in \mathbb{Z}^{\dim \theta_z}, |n|_\infty \leq B'_h}} |\widehat{S'}_I(k')|^2. \quad (32)$$

Shells larger than U_0 have negligible total contribution. Notice that the windows in this definition use the whole Fourier box and therefore do not depend on the mode masks. The same fresh offset can be used for all triples being tested at this cell.

For completeness, all the scalar errors and discarded terms just used are bounded by

$$CD_*^{-100} \prod_{j \text{ core } m \text{ selected}} \sum |\alpha_m|,$$

multiplied by the local side size when a side is present. This follows term by term from the Fourier tails, the pre-lag, the tiny-frequency widths, and the smooth kernel estimates. For the last high-shell error in (32), use ordinary L^2 pairing or the same Schur estimate. These errors do not pay for the number of failure masks: expand the absolute coefficient sums first, and each pure tuple has total first-success weight exactly one. The path count and coefficient bounds in D_* , together with the retained size caps, make their integrated sum $O(e^{-k}|R|)$. They will always be estimated absolutely, before taking a power less than one.

Fix the final height and its component tuple. If an assignment of n previous failures assigns n_j of them to role j , (30) and the predicate coherence give

$$a'_j \leq C\sigma_{\mathcal{F}_j}, \quad \mathbb{E}(a'_j)^2 \leq C(2\delta_*)^{n_j} \sigma_{\mathcal{F}_j}^2.$$

To see the second inequality, apply the probability bound to each selected minimal-component summand of (30); ignoring additional success conditions only increases the probability. Hölder with m_0 factors, using the deterministic bound to pass from second to m_0 th moments, yields

$$\mathbb{E} \prod_j a'_j \leq C^{m_0} (2\delta_*)^{n/m_0} \prod_j \sigma_{\mathcal{F}_j}.$$

The sum over m_0^n assignments and all n is finite, since the fixed δ_* was chosen sufficiently small. Summing over final heights, types, and dyadic amplitude bins, we have therefore proved an absolute cell bound, apart from the errors, by the sum of terms

$$C_{L\mu\nu}^{-L} \Lambda^{-L} \left(\prod_{j=0}^3 \sigma_j \right) N_s^G. \quad (33)$$

For a core role, σ_j is the dyadic bin of the size in (26); for a side it is the bin of b_Λ in (32). Zero-sized factors have zero contribution. For four cores we put $\Lambda = 1$. With all these choices fixed, N_s^G is the number of core label tuples passing the nonnegligible tests at $I_s \ni x_*$. A label tuple is counted at most once at a given scale. The useful power estimate will retain this count inside the power in (33).

9.3. A finite sparse-count lemma. The following finite statement isolates the combinatorial argument. The fourth role in its three-core case is represented by a square budget; that role is essential to the proof.

Lemma 9.1 (Sparse mass of separated progression tests). *There are absolute constants C and C_{res} with the following property. There are four roles $0, 1, 2, 3$, of which either four or exactly three are designated core roles; write J for their set. Let $M, L \geq 1$, $d \geq 0$, $B \geq 1$, and $0 < \mu \leq 1 \leq \nu$. In each core role i let $\mathcal{A}_i$ be a finite set of labels, identified injectively with real numbers, with $|\mathcal{A}_i| \leq M$. Fix $\zeta \in \mathbb{R}^d$.*

For finitely many distinct integer indices s , let $R_s = R_0 2^s$ and let $\mathcal{E}_s \subseteq \prod_{i \in J} \mathcal{A}_i$ be a set of tuples; put $N_s = |\mathcal{E}_s|$. Assume the following.

- (1) Two labels in distinct core roles determine at most one full core tuple among all the $\mathcal{E}_s$, and the pair has at most L scale occurrences.
- (2) If l is a side role, there are $w_s \geq 0$ with $\sum_s w_s \leq M$ such that, at scale s , the number of tuples incident to any fixed core label is at most w_s .
- (3) If distinct labels a, a' in the same core role occur at scales s, t , respectively, then

$$|a - a' + n \cdot \zeta| > \Sigma \max(R_s, R_t) \quad (n \in \mathbb{Z}^d, |n|_\infty \leq Q).$$

- (4) For every triple $T \subseteq J$ of core roles, let l' be the excluded role and set $\alpha_i = i - l'$ for $i \in T$. Every tuple at scale s has, for this triple, a witness $n \in \mathbb{Z}^d$, $|n|_\infty \leq B$, such that

$$\mu R_s \leq \left| \sum_{i \in T} \alpha_i a_i + 36n \cdot \zeta \right| \leq \nu R_s.$$

- (5) At every occupied scale s ,

$$|n \cdot \zeta| \notin [\mu R_s/1000, 1000\nu R_s] \quad (n \in \mathbb{Z}^d, |n|_\infty \leq Q).$$

Suppose that

$$Q \geq C_{\text{res}}(d+1)^2 B, \quad \Sigma \geq C_{\text{res}}(d+1)^2 \nu.$$

Then, with $\ell = 1 + \log_2(2 + (d+1)^2 \nu/\mu)$, one has

$$\sum_{s: N_s \leq \delta M^2} N_s \leq C \ell L^{29} \delta^{1/2080} M^2 \quad (0 < \delta \leq 1).$$

The constants and exponents are independent of d , the real frequencies, the number of scales, and all label cardinalities.

Proof. We give the cardinality argument on a common fine grid. This entails no restriction on the real data. There are finitely many labels, scales, and shifts of size at most Q . Approximate their real coordinates on one grid, taking label and frequency approximants on a fixed multiple of its step so that the divisions by 1, 2, 3 below remain on the grid. Separation remains valid with, for example, $\Sigma/2$; the annular bounds remain valid with $\mu/2$ and 2ν ; and the wide pure-shift gap retains the narrower interval $[\mu R_s/100, 100\nu R_s]$ needed after relaxing the parameters. These assertions follow by choosing the step after all the finite data, and sufficiently small relative to the least R_s . Round interval widths upward by at most one grid step. All such changes cost fixed constants, absorbed in C_{res} and C . No final estimate depends on the step. It therefore suffices to prove the grid version with these harmless fixed margins. We keep the original letters for the relaxed parameters.

A graph with many four-step paths. We first record the precise graph observation to be used. Suppose that a bipartite graph has parts of size at most M , at least ηM^2 edges, and a proper edge coloring with at most M colors, where $0 < \eta \leq 1$. Pad the second part to $m = \lceil M \rceil \leq 2M$ points, choose a uniformly random point in that part, and let Y be its neighborhood. Then

$$\mathbb{E}|Y|^2 \geq (\mathbb{E}|Y|)^2 \geq \eta^2 M^2/4.$$

Call an ordered pair in the first part bad if its codegree is less than $c_0 \eta^2 M$, with $c_0 > 0$ a sufficiently small absolute constant. Its probability of lying in Y^2 is its codegree divided by m . Summing over the at most M^2 ordered pairs gives an expected number of bad pairs at most $c_0 \eta^2 M^2$. It follows, by considering $|Y|^2 - 100$ times this number, that some Y has $|Y| \geq c\eta M$ and fewer than $|Y|^2/100$ bad ordered pairs. Delete the points with more than $|Y|/4$ bad partners. Fewer than $|Y|/25$ points are deleted. For two retained points there are at least $c|Y|$ points of Y that are good partners of

both. Each such midpoint has at least $c_0\eta^2M$ common neighbors with each endpoint. Choosing those two neighbors independently shows that any two retained points have at least

$$c\eta^5M^3$$

four-step paths between them, allowing repeated vertices. For fixed endpoints the ordered four-color tuples of these paths are distinct: the start vertex and a color determine the next edge uniquely, by proper coloring, and this reconstructs the whole path successively.

Counting blocks and difference covers. Restrict to the scales with $N_s \leq \delta M^2$. The first hypothesis gives total mass at most LM^2 . Choose parameters $\delta \leq \alpha/2$ and $0 < \kappa < 1$, whose values will be fixed at the end. Order the retained scales by descending R_s , and form consecutive blocks of mass at most αM^2 , greedily. Every block except possibly the last has mass greater than $(\alpha - \delta)M^2$. The number J_0 of blocks is therefore at most $3L/\alpha$. These are counting blocks, unrelated to the depth blocks used in the linearization.

Let R_b^+ be the largest reciprocal scale in a counting block b . With grid step h_0 , set

$$N_0 = \lceil 32B \rceil, \quad \epsilon_b = h_0 \left\lceil \frac{16\nu R_b^+}{h_0} \right\rceil,$$

taking h_0 small enough that $\epsilon_b \leq 17\nu R_b^+$ for every block. For an integer $t \geq 1$ define the finite symmetric neighborhood

$$U_b(t) = \{6n \cdot \zeta + u : n \in \mathbb{Z}^d, |n|_\infty \leq tN_0, u \in h_0\mathbb{Z}, |u| \leq t\epsilon_b\}.$$

These neighborhoods contain zero and satisfy $U_b(t) + U_b(t') \subseteq U_b(t+t')$.

Fix a core role i and a block. As long as the mass whose i -label is still uncovered exceeds κM^2 , choose two other core roles j, k . Project these occurrences to a simple graph of (i, j) pairs, keeping one scale witness in the block for each pair. There are at least $(\kappa/L)M^2$ edges. Color a pair by its k -label. The coloring is proper, because a pair in the roles (i, k) or (j, k) also determines the entire tuple. Apply the preceding observation with $\eta = \kappa/L$ to obtain a subset B_0 of role i , of size at least $c\eta M$.

For a four-step path between $a, a' \in B_0$, alternately add and subtract the four annular relations for the triple i, j, k . The intermediate i -label and both j -labels cancel. Division by $\alpha_i \in \{\pm 1, \pm 2, \pm 3\}$ gives

$$a - a' \in q(c_1, c_2, c_3, c_4) + U_b(1),$$

where q depends only on the four colors. The scalar error is at most $4\nu R_b^+$ before division. The shift has size at most $4B$ before division, and $36/\alpha_i$ is an integer multiple of 6, which explains the choice of U_b .

Choose a maximal packing in $B_0 - B_0$ whose distinct differences lie outside $U_b(2)$. Choose one endpoint pair for each packed difference. Each has at least $c\eta^5M^3$ ordered four-color tuples, and two packed differences cannot use the same tuple, since that would put their difference in $U_b(1) - U_b(1)$. There are at most M^4 color tuples in total. The packing thus has size at most $C\eta^{-5}M$, and maximality covers $B_0 - B_0$ by that many translates of $U_b(2)$. Remove B_0 and repeat. Each iteration removes at least $c\eta M$ labels, so there are at most $C\eta^{-1}$ iterations. We have proved that, except for κM^2 uncovered occurrences in each core role and block, the incident labels are covered by at most

$$K_1 = C(L/\kappa)^6$$

subsets B , each of whose difference set is covered by at most K_1M translates of that block's $U_b(2)$.

Nonfuture pairs. Discard every occurrence missing an own-block covered label in any core role. This loses at most $4J_0\kappa M^2$. A core role on a remaining occurrence is called future if its exact label belongs to a cover subset of that role in a later counting block. The sets of nonfuture allowed labels are disjoint over blocks, separately in each core role: appearance in a later cover would make the earlier one future. If $m_i^{\mathfrak{b}}$ is the number of such labels, then $\sum_{\mathfrak{b}} m_i^{\mathfrak{b}} \leq M$. The side role is always nonfuture, and in that role set $m_l^{\mathfrak{b}} = \sum_{s \in \mathfrak{b}} w_s$, which has the same sum bound.

An occurrence with fewer than three future roles has at least two nonfuture roles. Assign one such pair to it. For a fixed pair the mass in block $\mathfrak{b}$ is at most

$$\min(\alpha M^2, L m_i^{\mathfrak{b}} m_j^{\mathfrak{b}}).$$

For two cores this is pair multiplicity; for a core and the side it is the side incidence budget, with the harmless factor $L \geq 1$. The sum over blocks is at most

$$\sqrt{L\alpha} M \sum_{\mathfrak{b}} \sqrt{m_i^{\mathfrak{b}} m_j^{\mathfrak{b}}} \leq \sqrt{L\alpha} M^2.$$

There are at most six role pairs.

Every remaining occurrence has three future core roles. Assign one such triple and, for each of its labels, one future cover subset containing it, using fixed rules if there are choices. The number of resulting combinations is at most $4J_0(J_0K_1)^3$. Fix a combination with mass $\rho M^2 > 0$. It consists of a current block, a fixed triple of roles, and subsets B_1, B_2, B_3 in those roles from strictly later blocks. The indices 1, 2, 3 in the next paragraphs enumerate these roles, and do not change their progression coefficients α_i .

Four ranges with scale margins. Set

$$\Gamma = C_{\text{gap}}(d+1)^2\nu/\mu,$$

where C_{gap} is a sufficiently large absolute constant. If $\rho \leq C\delta\ell$, retain this bound and do no further work on the combination. Otherwise increase the absolute C so that all the following margins hold. Cut the ordered mass at its three quartiles and omit the individual scale bins crossing the cuts. At most three bins are lost, each of mass at most δM^2 . Each of the four remaining ordered ranges has mass at least $\rho M^2/8$, and hence at least $\rho/(8\delta)$ occupied dyadic bins. Write $R^\#$ for the largest reciprocal scale of the second range.

The first range supplies, in each B_i , a set B_{i0} of distinct anchor labels with

$$|B_{i0}| \geq c\rho M/L.$$

Indeed a fixed label is incident to at most LM occurrences, by pair determination and multiplicity with any other core role. Each anchor has an occurrence at reciprocal scale at least $R^\#$, so distinct anchors are separated modulo the reserved shifts by more than $\Sigma R^\#$.

The last range supplies a set F of distinct actual triples with

$$|F| \geq c\rho M^2/L,$$

choosing one scale witness for each. A triple has at most L occurrences. The third range contains at least $\rho/(8\delta)$ occupied dyadic bins. Thus every second-range reciprocal scale exceeds every last-range reciprocal scale by a factor at least Γ , once $\rho > C\delta\ell$. Every future block lies below the entire current block in reciprocal scale, so the same assertion holds for every future-block scale used by this combination. Greedily thin the second range so that successive selected scales have ratio at least Γ . Each selection excludes at most $1 + \lceil \log_2 \Gamma \rceil \leq C\ell$ occupied bins, and therefore retains

$$q \geq c\rho/(\delta\ell)$$

scales. Choose one actual triple t_v at each selected scale R_v . Figure 2 shows the four ranges for this fixed combination.

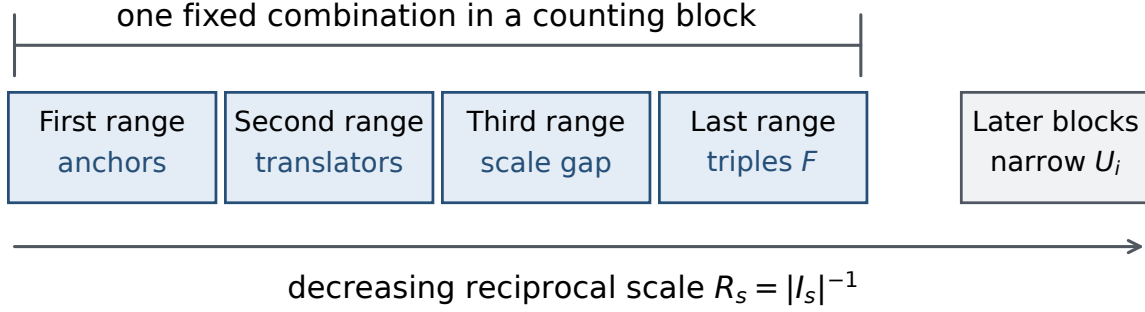


FIGURE 2. For one fixed choice of a current counting block, three core roles, and their future cover subsets, the four ranges shown remain after omitting the scale bins crossing the quartile cuts. Reciprocal scale decreases from left to right. The ranges supply anchors, separated translators, a scale gap, and the set F . The selected future subsets lie in later counting blocks and have narrower neighborhoods. These counting blocks are distinct from the consecutive K -depth blocks in Figure 1.

We will compare the $q|F|$ indexed outputs of the translated sets $t_v + F$ with a cover of a common slab. The next two steps build covers of the future sumsets; the anchors will remove one factor M from the slab cover. The scale margins will then show that no cover box contains two of these outputs.

A small-growth scale without an exponential loss. Write U_i and ϵ_i for the neighborhoods and widths of the future block associated with B_i . Subdividing each coefficient interval and the scalar interval gives, for every integer $t \geq 1$, a cover of $U_i(t)$ by at most $(2t + 1)^{d+1}$ translates of $U_i(1)$. Consequently there is an odd integer

$$1 \leq H_i \leq 100(d + 1)^2, \quad |U_i(H_i + 2)| \leq 2|U_i(H_i)|.$$

Otherwise growth by more than 2 at each odd step would exceed the displayed polynomial-in- t bound by the time t is a sufficiently large absolute multiple of $(d + 1)^2$. The exponential dependence on d of the preliminary cover is used only for this choice of H_i ; it is not paid in the estimates below.

The translates $a + U_i(H_i)$, $a \in B_{i0}$, are disjoint. An intersection would put their difference in $U_i(2H_i)$, whose shift index in the ζ lattice is $O((d + 1)^2 B)$ and whose scalar width is at most $C(d + 1)^2 \nu R^\#$. The reserves on Q and Σ exclude this. With $Z_i = B_{i0} + U_i(H_i)$, the difference cover for $B_i - B_i$ gives

$$\begin{aligned} |Z_i - B_i| &\leq K_1 M |U_i(H_i + 2)| \leq 2K_1 M |U_i(H_i)| \\ &\leq C(K_1 L / \rho) |Z_i|. \end{aligned}$$

The minimal-growth subset argument. Choose a nonempty $X_i \subseteq Z_i$ minimizing $r_i = |X_i - B_i| / |X_i|$. Then $r_i \leq CK_1 L / \rho$. We include the iteration argument, in the form used in [10]. For a finite family of translates $C_0 = \{c_1, \dots, c_q\}$ let $X_{i,j}$ consist of the points $x \in X_i$ whose translate $x + c_j$ is not in any earlier $X_i + c_v$. Thus $|X_i + C_0| = \sum_j |X_{i,j}|$. Every point of $(X_i \setminus X_{i,j}) - B_i + c_j$ already belongs to an earlier translate of $X_i - B_i$. The new points contributed at step j are therefore at most

$$\begin{aligned} |X_i - B_i| - |(X_i \setminus X_{i,j}) - B_i| &\leq r_i |X_i| - r_i |X_i \setminus X_{i,j}| \\ &= r_i |X_{i,j}|, \end{aligned}$$

by minimality; the empty-complement case is included. Summing and then iterating with $C_0 = -(m-1)B_i$ proves

$$|X_i - mB_i| \leq r_i^m |X_i| \quad (m \geq 1),$$

where mB_i denotes the m -fold sumset.

One anchor translate contains a portion Y_i of X_i of size at least $|X_i|/M$, since there are at most M anchors. Its difference set is contained in $U_i(2H_i)$. If $\mathcal{P}$ is a packing of mB_i with distinct differences outside $U_i(2H_i)$, the translates $Y_i - p$, $p \in \mathcal{P}$, are disjoint and belong to $X_i - mB_i$. Hence

$$|\mathcal{P}| \leq r_i^m M.$$

Maximal packings give covers by the same neighborhood. We need only $m = 2, 3$.

A slab cover with only two label-count factors. Put $r_0 = CK_1 L/\rho$, enlarging the absolute constant so that $r_i \leq r_0$ for every role. Consider the set of outputs in $2B_1 \times 2B_2 \times 2B_3$ satisfying

$$\left| \sum_i \alpha_i x_i + 36n \cdot \zeta \right| \leq 2\nu R^\# \quad \text{for some } n \in \mathbb{Z}^d, \quad |n|_\infty \leq 2B.$$

Cover the first two sumsets by at most $r_0^2 M$ translates each of $U_i(2H_i)$. Above a fixed pair of boxes, the possible third coordinate lies in $2B_3$ intersected with a set of the form

$$c + \{n \cdot \zeta : |n|_\infty \leq C(d+1)^2 B\} + [-C(d+1)^2 \nu R^\#, C(d+1)^2 \nu R^\#].$$

Indeed substitute the first two coordinate boxes into the slab relation and divide by $\alpha_3 \in \{\pm 1, \pm 2, \pm 3\}$. The box shifts are $6n \cdot \zeta$ and the slab shifts are $36n \cdot \zeta$, so the divided coefficients remain integral. Their sizes have the stated bound.

Let $\mathcal{P}$ be a packing in this local third-coordinate set outside $U_3(2H_3)$. Then $B_{30} + \mathcal{P} \subseteq 3B_3$ is still such a packing. For an equal anchor this is the definition of $\mathcal{P}$. For distinct anchors, a collision would express their difference, modulo a reserved shift, with absolute value at most $C(d+1)^2 \nu R^\#$, contradicting their separation. The global $3B_3$ packing bound gives

$$|B_{30}| |\mathcal{P}| \leq r_0^3 M, \quad |\mathcal{P}| \leq Cr_0^3 L/\rho.$$

Covering the local third-coordinate set by a maximal packing shows that the whole slab is covered by at most

$$Cr_0^7 (L/\rho) M^2 \leq CK_1^7 (L/\rho)^8 M^2$$

translates of $\prod_{i=1}^3 U_i(2H_i)$. The use of all the separated anchors is what removes a third factor M . All translated last-range triples are separated. For each chosen second-range triple t_v , translate F by t_v . All outputs belong to the preceding slab, by adding their annular relations. A cover box contains at most one output. First, within one translation, distinct members of F differ in some label coordinate. Their last-range witnesses lie above every future scale, so hereditary separation is stronger than membership of that coordinate difference in $U_i(4H_i)$.

Next consider different translations, and let R_{big} be the larger of their two second-range scales. The annular residual of that triple has magnitude between μR_{big} and νR_{big} . The sum of the magnitudes of the other second-range residual and the two last-range residuals is at most $\mu R_{\text{big}}/10$, by the factor- Γ margins. Thus the linear form of the output difference, after a shift of index $O(B)$, has absolute value between $9\mu R_{\text{big}}/10$ and $2\nu R_{\text{big}}$. If the outputs lay in the same box, their coordinate differences would be in $U_i(4H_i)$. The total scalar error in the linear form would then be at most

$$C(d+1)^2 \nu \max_i R_{\text{future},i}^+ \leq \mu R_{\text{big}}/10.$$

All shifts still have index $O((d+1)^2 B) \leq Q$. We would obtain a pure shift of magnitude between $4\mu R_{\text{big}}/5$ and $3\nu R_{\text{big}}$, inside the forbidden gap. This proves the required separation.

The number $q|F|$ of outputs is consequently bounded by the slab cover. Cancelling M^2 gives

$$c \frac{\rho}{\delta \ell} \frac{\rho}{L} \leq CK_1^7 (L/\rho)^8, \quad \rho^{10} \leq C\delta \ell K_1^7 L^9.$$

Including the previously omitted case $\rho \leq C\delta \ell$, a convenient uniform consequence is

$$\rho \leq C\ell K_1 L \delta^{1/10}.$$

Choice of the auxiliary powers. There are at most $4J_0^4 K_1^3$ combinations. Combining their mass, the uncovered mass, and the nonfuture-pair mass, we obtain

$$\begin{aligned} M^{-2} \sum_{s: N_s \leq \delta M^2} N_s &\leq C \left(J_0 \kappa + \sqrt{L\alpha} + \ell L J_0^4 K_1^4 \delta^{1/10} \right) \\ &\leq C \left(\frac{L\kappa}{\alpha} + \sqrt{L\alpha} + \ell L^{29} \alpha^{-4} \kappa^{-24} \delta^{1/10} \right). \end{aligned}$$

For sufficiently small absolute δ , take $\alpha = \delta^{1/1040}$ and $\kappa = \alpha^2$. Then $\delta \leq \alpha/2$, and the three powers of δ on the right are $1/1040$, $1/2080$, and $1/10 - 52/1040 = 1/20$. Since $L, \ell \geq 1$, this proves the asserted estimate. For the remaining δ , enlarge the absolute constant and use the total-mass bound LM^2 . $\square$

9.4. Application to the progression tests. Fix a retained path point x_* , the height, all role types, the frequency bands, and all amplitude bins in (33). In each group define

$$M_i^G = |\mathcal{A}_i^G| \quad \text{for a core role,} \quad M_l^G = \sigma_l^{-2} \sum_{s: G \text{ used}} \|S_{l,G,s}\|_{2,I_s}^2 \quad \text{for a side role,}$$

and put $M' = \max_i M_i^G$. We omit groups with no nonzero counts; then $M' \geq 1$. In the side case this also follows directly from $b_\Lambda \geq \sigma_l$ and Parseval at a nonzero use.

Proposition 9.2 (Counts for the tested core tuples). *There is an absolute $p_0 \in (0, 1)$ such that, for the preceding data and some $L_0 \leq P^C(1 + \log(\nu/\mu))^C$, one has*

$$\begin{aligned} \sum_s N_s^G &\leq L_0 \min_{i \neq j} M_i^G M_j^G, \\ \sum_s (N_s^G)^{p_0} &\leq P^C(1 + \nu + \mu^{-1} + \Lambda)^C (M')^{2p_0}. \end{aligned} \tag{34}$$

All exponents are absolute. They do not depend on the path, the group, the horizontal dimension, or the number of depths.

Proof. Identify a label a with its fixed value $\omega_a(x_*)$. Different labels in one fixed role and type have distinct values by (25) with zero shift. We first establish pair determination. In the four-core case, subtract the two frequency tests for two incident tuples that have a given pair of labels in common. At a cost $O(\nu \max(R_s, R_t))$, the two remaining frequencies solve the linear equations for Q and T , modulo the bounded shifts. The inverse determinant divides by an integer of magnitude at most 3. Separation (25) forces the two alternative labels in each remaining role to agree. In the three-core case, the single equation for T_l similarly determines the third core label from the other two, with a divisor of magnitude at most 3. These conclusions hold across different scales, because the separation threshold uses the larger reciprocal scale. All shift differences and divisions fit the height reserve.

In addition, for any triple of core roles let l' be the excluded fourth role and put $\alpha'_i = i - l'$ on that triple. The tests imply

$$c\mu R_s \leq \left| \sum_{i \text{ in the triple}} \alpha'_i a_i + n \cdot \xi \right| \leq C\nu R_s, \quad n \in \mathbb{Z}^{\dim \theta_z}, \quad |n|_\infty \leq CB'_h. \tag{35}$$

For four cores this is $t' - l'q'$, using both bounds in (31); for three cores it is precisely the nonzero T_l test. Fixing a pair fixes the whole core tuple, and hence the label values in this annulus. The height-bin observation, applied to those fixed values together with the coordinates of ξ , bounds the number of scale occurrences by $P^C(1 + \log(\nu/\mu))^C$. We use $L_0 \geq 1$ for a bound of this form. This proves the first count for any pair of core roles.

For a pair consisting of a core role and the side, fix the core label and a scale. The windows in (32) associated with different incident triples are disjoint. If two windows overlapped, their Q values would be close modulo bounded shifts, with error $O(\Lambda R_s)$. Combining this with the two T_l relations determines the two remaining core labels, again by a two-by-two system with divisor at most 3. Since $\nu, \Lambda \leq CU_0 \ll \Xi$, separation forces those labels to be identical. Couple the fresh offset in all these windows. Parseval for each offset, followed by expectation, shows that the number of incident triples in the chosen side bin is at most

$$w_s = \sigma_l^{-2} \|S_{l,G,s}\|_{2,I_s}^2.$$

This remains valid on any subset of scales after summing. Summing over the fixed core labels proves the first inequality of (34) for a core/side pair as well.

We now apply Lemma 9.1 with $M = M'$, $L = L_0$, $d = \dim \theta_z$, and

$$\zeta = \xi/36.$$

The fixed constants in (35) may be absorbed by decreasing μ and increasing ν by absolute factors. The pair and side-budget hypotheses were just verified. For separation, note that

$$\frac{\xi}{36} = 6^{\Delta-2} \frac{\vartheta}{L_h}.$$

Thus the shifts in the finite lemma are among those of (25), with $Q = \lfloor E_h/6^{\Delta-2} \rfloor$ and $\Sigma = \Xi/2$. The choice of A_0 and $B'_h = D_*^5 E_{h-\Delta}$ ensures $Q \geq C_{\text{res}}(d+1)^2 CB'_h$. Likewise $\Xi = D_*^{1000}$ and $\nu \leq CD_*^{250}$ ensure the separation reserve, with room for the fixed annular constants.

Finally, any successful component contains a nonexceptional primary, even when its representative is itself exceptional. For that primary the pure-horizontal cases of (23) apply to every shift in the finite lemma: their rational heights are smaller than $\mathcal{H}$. They are either at most $\mathcal{T}^{-1}/r_{v_b}$ or at least $\mathcal{T}/r_{t_b}$. Since $U_0^{-1} \leq \mu \leq 1 \leq \nu \leq CU_0$ and the test scale lies between these depth scales, the prescribed choice of $\mathcal{T}$ excludes the entire intermediate interval required by the pure-shift hypothesis. This checks every hypothesis of the finite lemma, uniformly over the current group.

It follows that, for an absolute $c' > 0$,

$$\sum_{s: N_s^G \leq \delta(M')^2} N_s^G \leq P^C(1 + \nu + \mu^{-1} + \Lambda)^C \delta^{c'} (M')^2 \quad (0 < \delta \leq 1). \quad (36)$$

For example, $c' = 1/2080$ is admissible after changing the absolute constants. Indeed Lemma 9.1 gives the stronger coefficient $C\ell L_0^{29}$; the dimension and the logarithms in this expression fit the displayed budget.

Pair determination also gives $N_s^G \leq (M')^2$. Choose $p_0 \in (0, 1)$ with $1 - p_0 < c'$, for instance $p_0 = 1 - c'/2$. On the set of scales with

$$2^{-j-1}(M')^2 < N_s^G \leq 2^{-j}(M')^2$$

we may bound $(N_s^G)^{p_0}$ by $[2^{j+1}/(M')^2]^{1-p_0} N_s^G$. Sum this inequality and use (36) with $\delta = 2^{-j}$. The remaining geometric series has ratio $2^{1-p_0-c'} < 1$, and proves the second inequality of (34). $\square$

9.5. Summing groups and amplitude bins.

Proposition 9.3 (Powered estimate for the core terms). *There are absolute $\varepsilon > 0$ and $C_4 < \infty$, with $1 - \varepsilon > 2/3$, such that the terms containing at least three core roles in the localized expansion of (11) have total contribution at most*

$$C_q \eta^\varepsilon P^{C_4} |R| + O(e^{-k} |R|). \quad (37)$$

Here the original cell minimum is applied before the subadditive estimates, and all numerical errors are estimated absolutely. The constants are uniform in the finite depth range and the retained attempt. In particular ε and C_4 can be fixed before choosing how close q is to 2.

Proof. Set

$$p = 1 - \varepsilon = (1 - \theta') + \theta' p_0,$$

where $\theta' > 0$ is a sufficiently small absolute number, to be fixed below. Hölder interpolation between the two inequalities in (34) gives, for any distinct roles i, j ,

$$\sum_s (N_s^G)^p \leq (L_0 M_i^G M_j^G)^{1-\theta'} [P^C (1 + \nu + \mu^{-1} + \Lambda)^C (M')^{2p_0}]^{\theta'}.$$

Order the amplitude bins from largest to smallest, temporarily calling their roles 1, 2, 3, 4, so that $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \sigma_4$. Use the pair 1, 2 in the first factor. At least one of this pair is a core role, because there is at most one side.

The path bounds (28) and the side square budget give

$$\sum_G M_i^G \leq C P^C \sigma_i^{-2} \quad \text{in every role,} \quad \sum_G M_i^G \leq D_* \quad \text{in a core role.}$$

Expand the maximum M' into at most four choices of the role attaining it. The total exponent of the resulting product of M_i^G is $2p > 1$. For nonnegative sequences m_i^G and exponents $\beta_i \geq 0$ with $\sum_i \beta_i \geq 1$, one has

$$\sum_G \prod_i (m_i^G)^{\beta_i} \leq \prod_i \left(\sum_G m_i^G \right)^{\beta_i}.$$

Indeed normalize each nonzero sum to one, apply Hölder with exponents $(\sum_i \beta_i)/\beta_i$, and use that the normalized sequences have $\ell^{\sum \beta_i}$ norm at most one. This proves the needed group summation.

The largest size-to-count factor is σ_4^{-2} . In the core member of the first pair we may also use the count cap D_* . If this member is role 2, its factor $\min(1, D_* \sigma_2^2)$ is bounded by the corresponding factor with σ_1 . Multiplying by $(\prod_i \sigma_i)^p$ from (33), the amplitude dependence is consequently at most a P^C factor times

$$\min(1, D_* \sigma_1^2)^{1-\theta'} \sigma_1^{p-2(1-\theta')} \sigma_2^{p-2(1-\theta')} \sigma_3^p \sigma_4^{p-4\theta' p_0}.$$

Let the four displayed power exponents be e_1, e_2, e_3, e_4 . Their sum is zero, and their three preceding partial sums are

$$-1 + \theta'(1 + p_0), \quad -2 + 2\theta'(1 + p_0), \quad -1 + \theta'(1 + 3p_0).$$

All are strictly negative when θ' is sufficiently small. Writing $g_j = \log_2(\sigma_j/\sigma_{j+1}) \geq 0$, the remaining power factor is $2^{\sum_{j=1}^3 g_j(e_1 + \dots + e_j)}$. Thus all three amplitude-gap sums are convergent geometric series.

Only a common shift of all the bins remains. The retained upper amplitude bound is P^C . Below $\sigma_1 = D_*^{-1/2}$ the prefactor $\min(1, D_* \sigma_1^2)^{1-\theta'}$ decays geometrically; above it there are only $O(1 + \log D_* + \log P)$ possible dyadic bins up to that upper bound. The common-shift cost is therefore polynomial in the allowed logarithmic budgets. There are only 24 amplitude orderings, so this completes the amplitude summation.

It remains to sum the frequency bands. The first count loses only powers of $1 + \log(\nu/\mu)$. The second count, after interpolation, loses at most $\mu^{-C\theta'} \nu^{C\theta'} \Lambda^{C\theta'}$. The powered kernel factor in (33)

is $\mu^p \nu^{-Lp} \Lambda^{-Lp}$. First take θ' small enough that $p - C\theta' > 0$, and then take the fixed derivative order L sufficiently large. The sums over dyadic $0 < \mu \leq 1$ and $1 \leq \nu, \Lambda < \infty$ converge, including the logarithmic losses. Extending the actual truncated ranges to these ranges only increases the bound. The number of heights, types, role patterns, and retained leaves has an allowed polynomial cost. We have proved a bound $C_q P^{C_4}$ for the path sum of the p th powers of all the principal core contributions, keeping each N_s^G inside its power.

To finish, let $\mathcal{C}_I$ be the sum of their absolute principal contributions at a cell. Subadditivity for $p < 1$ and $\min(\eta, t) \leq \eta^{1-p} t^p$ give

$$\min(\eta, \mathcal{C}_I) \leq \eta^\varepsilon \mathcal{C}_I^p \leq \eta^\varepsilon \sum_{\text{configurations}} (\text{principal contribution at } I)^p.$$

Integrating the path estimate uses the exact identity $\sum_I |I| F_I = \int_R \sum_{I \ni x_*} F_I dx_*$ for nonnegative cell data. It gives $C_q \eta^\varepsilon P^{C_4} |R|$. The previously separated absolute errors contribute $O(e^{-k} |R|)$, which proves (37). All polynomial degrees in the finite counting lemma, and hence $p_0, \theta', \varepsilon$, and C_4 , are absolute. Taking θ' smaller if necessary gives $p > 2/3$ as required by the side estimates. $\square$

10. PARAMETER CHOICE AND COMPLETION OF THE PROOF

We finish Proposition 2.3. Throughout this section all collections of summation cells are finite and lie in the original N depths. Thus the defining constant $C_N(q)$ is finite, and every application of it in Section 7 concerns precisely this finite constant.

10.1. One attempt and its restarts. Fix one of the normalized base segments from Section 5, with root R , a tuple of increment levels, and one active pattern. For the noninitial levels, its remaining quantity is

$$\sum_{I \text{ active}} |I| \min\{\eta, |H_I(z)|\}, \quad \eta = e^{-k\theta},$$

where k is the later accuracy log and $\theta > 0$ is fixed. This is Equation (11). We explain explicitly how the stopping constructions preserve this quantity.

Start an attempt at R . Construct the main catalogs, color histories, and ball filters, making the small marking-failure cuts from Section 6. The objects needed on an unused branch can be continued by inheritance and local refitting. Next construct the fixed first-match mode expansions and the linear side states of Section 8. Choose the branch offsets by the small integrated error bound over all the pre-existing actual uses. Finally close the helper catalogs on the entire side states and the specified histories. A helper base is one such state, not an individual Fourier mode or a choice of heights and amplitude bins.

All subsequent size and counting stops are imposed on these fixed objects. A stop removes original summands from the attempt; it does not truncate or redefine a previously fixed stack function. Partial path counts and ancestor suprema are accumulated in increasing depth, and a cell is stopped at their first threshold crossing. Hence the retained summands see only retained partial counts. For the hull estimates of Section 7, include all potentially needed ancestors within a block before imposing the common cuts. The maximal functions remain defined on the full retained hull cells, including portions below a cut. Their joint caps are consequently valid on the sparse remainders used in the proof of Equation (19). Large-frozen-complement flags are charged as powered errors within the current attempt; only a first crossing of their accumulated power-cost budget produces a restart root.

The integral bounds proved in the preceding sections permit each class of stops to have arbitrarily small fixed relative root length by increasing its threshold by a fixed factor. The number of configurations requiring the strong caps is a small overhead, which can be included in those thresholds. Polynomially many configurations in P, k occur only in the estimates already permitted

a polynomial cost. Choose these thresholds together so that the union of first stopped cells has total length at most, say, $|R|/4$.

At every stopped cell send the original base summands there and below to a fresh attempt. The original basic arguments and their inherited lists remain the same. Their normalized sizes hold on the new root; the rolling basic projection bounds and the other local hypotheses hold there as well. Thus each attempt has the same estimates, uniformly in its root and in the remaining number of depths. The total length of all roots in the resulting recursion is at most

$$|R| \sum_{n \geq 0} 4^{-n} = \frac{4}{3} |R|.$$

The construction terminates on a finite depth tree. Equivalently, one can prove the required bound by induction on the remaining depth, with the displayed geometric root-length estimate.

10.2. The estimate on retained cells. Proposition 6.8 first gives the group decomposition in Equation (14). In each coarse role the decomposition of Section 8 is a core plus a rolling and a linear side, with a separately estimated physical error. The basic rolling U role, if present, is always a side.

Terms with at least three core roles are estimated by Proposition 9.3. Terms with at least three side factors have three square-size budgets, so they are summable at every fixed power $p > 2/3$ sufficiently close to one, with a polynomial cost in P, k . In a term with exactly two sides, replace its two core complements by their whole current localized arguments. The correction terms have at least three sides. Apply Lemma 6.9 to the two whole complements. After transfer, replacing their basic coarse parts by the full Y arguments adds a third basic side; the remaining term has the full complementary pair for which Section 5 proves smallness. This is the pair-small main term in Proposition 7.5.

The near-band defects are retained as their coupled band sums and estimated at power one by Equation (19). Numerical errors are also estimated at power one. For all the other terms we use

$$\min\{\eta, a\} \leq \eta^{1-p} a^p \quad (a \geq 0).$$

The minimum is first applied to the sum at the original cell and then split by subadditivity. In particular no separate pair-smallness assertion for an individual group is being used. Put $p = 1 - \varepsilon$, where $\varepsilon > 0$ is fixed small enough for both the three-side estimate and the interpolation in Section 9.

The first term below collects the estimates taken at power $p = 1 - \varepsilon$ and the smaller pair-small main term; the second is the coupled near-band estimate at power one. Combining Equations (19) and (37) with the other side estimates, and increasing one fixed exponent C_4 to include all powered costs, gives on retained cells

$$C_q \left[\eta^\varepsilon P^{C_4} + (1 + C_N) P^{-1/2} G_0(K + 2)^{C_{19}(q-2)} \right] |R| + O(e^{-k} |R|).$$

Fixed polynomial factors in k can be retained here and absorbed into the first term's power of P once the parameters below are fixed. The main term after transfer, of size $\eta^{PC} |R|$, is also covered after increasing C_4 and decreasing ε if necessary. The constant C_{19} is absolute. Restarting as above changes this bound only by a fixed factor.

10.3. Order of the parameters. For clarity, the two types of complexity used in the proof are listed separately:

Quantity	Allowed size	Role in the argument
$D_{\text{base}}, D_{\text{m}}$	$\log D \leq C_q k^C$	Main columns, colors, helpers and their projection products.
D_* and the shift heights	$\log D_* \leq C_q(1 + k + P)^C$	Fixed linear modes, Fourier truncations and graph tests.
K	$K \leq C_q(1 + k + P)^{C_5}$	Depth-block length after the mode and gap choices.
G_0	$\exp(C_q(1 + \log(2k))^C)$	Strong energy and stack overhead in Equations (17)–(19).

Every exponent denoted by C, C_4, C_5, C_{19} in this hierarchy has an absolute upper bound before q is chosen. Constants such as C_q may depend on that later fixed choice.

Here is why the ordering is not circular. Detection changes a power of an accuracy log by an absolute amount; the dependence on q from clipping multiplies that log by a q -dependent constant. Thus choose the stagger exponent A in Section 5 and the small quality exponent θ independently of q . The main/helper catalogs have polynomially many ranks and multiplicative count increments $\exp(k^{C_2})$. Their logarithms remain polynomial in k . The number of masks and histories enters this recurrence logarithmically, including the P pad choices, because $\log P = O(k)$. The tolerance in Equation (13) includes $n_{r-1}^{-1/2}$, so summing a lower-rank source does not require its full count as an additional error loss. Choose the mismatch accuracy and the edge height after these count bounds. Color recourse bounds are uniform in that edge height.

The short-path construction then produces the larger mode budget. Its Fourier cutoffs precede the height range, shift radii and horizontal gap, and K is chosen last among these quantities. All their logarithms are still polynomial in P, k . The strong side estimate compresses only main projection families; mode projections have already been incorporated in Bessel rows. The bad-density argument removes the large height factor from its chain count. Thus G_0 has the smaller size in the table, whereas the top counts and core estimates may pay powers of P . No later mode count is fed back into main/helper closure.

Now set $P = \lceil \eta^{-\alpha} \rceil$ for a fixed $\alpha > 0$ to be chosen. For every such fixed choice and sufficiently large k , P dominates each fixed power of k and $G_0 = P^{o(1)}$. First choose $2 < q < 3$ so close to two that

$$C_{19}C_5(q-2) < \frac{1}{8}.$$

After the finite overheads are combined, increasing the starting log makes them at most $P^{1/8}$. Hence

$$P^{-1/2}G_0(K+2)^{C_{19}(q-2)} \leq C_q P^{-1/4}.$$

Next choose $\alpha > 0$ small enough that $\alpha C_4 < \varepsilon/2$, and impose the finitely many analogous upper bounds coming from the other powered side costs. Then

$$\eta^\varepsilon P^{C_4} \lesssim \eta^{\varepsilon/2}, \quad P^{-1/4} \leq \eta^{\alpha/4}.$$

Finally increase the starting log so that all earlier accuracy and small-overhead requirements hold. Since $\theta < 1$, numerical errors e^{-k} are smaller than the resulting quality gain for large k . We have therefore proved, for some fixed $c_* > 0$, the bound

$$\sum_{I \text{ active}} |I| \min\{\eta, |H_I(z)|\} \leq C_q \eta^{c_*} (1 + C_N) |R|$$

on every base segment and pattern.

10.4. Summing the base expansion. For later level t , there are $O(t)$ choices of the earlier increment level and only a bounded number of slot patterns. Each tuple uses intersections of a bounded number of inheritance segments. Their roots have uniformly bounded total length by the reset packing in Section 5; the further attempts just considered preserve this property. Thus the total coefficient of $1 + C_N$ from the noninitial tuples is bounded by a fixed multiple of

$$\sum_{t \geq 12} t \exp(-c_* k_t^\theta), \quad k_{t+1} = k_t^A.$$

This series converges and tends to zero as k_1 tends to infinity. The absolute errors $O(e^{-k_t})$ are summable with the same multiplicity. The finitely many initial levels are bounded by the active-scale count in Equation (10); their cost is a finite constant depending on the chosen starting logs but independent of N .

Given the small c in Proposition 2.3, take k_1 large enough that the preceding coefficient of C_N is at most c . All remaining constants are included in A . The base expansion is first taken at a finite accuracy cutoff; on a fixed finite tree its residual terms tend to zero by the staggered pair comparisons of Section 5. The uniform bounds just proved permit passage to that limit. We obtain the assertion of Proposition 2.3. Lemma 2.4 now proves Theorems 2.2 and 1.1.

REFERENCES

1. David V. Cruz-Uribe, José María Martell, and Carlos Pérez, *Weights, extrapolation and the theory of Rubio de Francia*, Operator Theory: Advances and Applications, vol. 215, Birkhäuser, Basel, 2011.
2. W. T. Gowers and Julia Wolf, *Linear forms and quadratic uniformity for functions on $\mathbb{Z}_N$* , arXiv:1002.2210, 2010.
3. Martin Henk, *Successive minima and lattice points*, arXiv:math/0204158, 2002.
4. Bingyang Hu and Victor Lie, *On the curved trilinear Hilbert transform*, arXiv:2308.10706, 2023.
5. ———, *On the boundedness of the curved trilinear Hilbert transform and the curved n -linear maximal operator in the quasi-Banach regime*, arXiv:2511.07373v1, 2025.
6. Michael Lacey and Christoph Thiele, *L^p estimates on the bilinear Hilbert transform for $2 < p < \infty$* , Annals of Mathematics **146** (1997), no. 3, 693–724.
7. ———, *On Calderón’s conjecture*, Annals of Mathematics **149** (1999), no. 2, 475–496.
8. James Leng, Ashwin Sah, and Mehtaab Sawhney, *Quasipolynomial bounds on the inverse theorem for the Gowers $U^{s+1}[N]$ -norm*, arXiv:2402.17994v3, 2024.
9. Vladimir A. Mikhailets and Aleksandr A. Murach, *General forms of the Menshov–Rademacher, Orlicz, and Tandori theorems on orthogonal series*, arXiv:1110.4253v2, 2011.
10. Giorgis Petridis, *New proofs of Plünnecke-type estimates for product sets in groups*, Combinatorica **32** (2012), no. 6, 721–733.
11. Terence Tao, *Cancellation for the multilinear Hilbert transform*, arXiv:1505.06479v3, 2015.